PI User Training

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DAY 2

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- Current Value
- Archive Value
- Sampled Data
- Compressed Data
- Calculated Data
- Timed Value
- Trend

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1. PI System Overview

WHAT IS THE PI SYSTEM?

The PI System is a suite of software products that ...

Collect

PI Interface
- 400+ protocols
- OPC, Modbus, ODBC
- Data buffering
- Failover

Historize

PI Data Archive
- Time-series database
- 24 mil points supported
- Robust & compact

Asset

PI Asset Framework
- Asset Hierarchy
- Site, Area, Process
- Unit, Equipment
- Controller

Analyse

PI PE
- Real-time calculation
- Steam table functions

PI ACE
- .NET Framework

Notify

PI Notifications
- Email notifications
- Acknowledge
- Escalation

Visualize

PI ProcessBook
- Graphic display

PI DataLink
- Excel add-in

PI Coresight
- Web Application
- Mobile Application

Data Provider

PI Data Access
- Data gateway
- ODBC, OLEDB, JDBC
- Web Service, WebAPI
- OPC DA & HDA, SDK

PI SYSTEM CONNECTIVITY

Microsoft
esri
mro SOFTWARE
SAP
ORACLE
iOS

ERP | GIS | CMMS | LIMS | Asset Mgmt. | Other Apps.

OSIsoft

Honeywell Plantscape
YOKOGAWA
Bailey
Honeywell PHD
EMERSON
invensys Wonderware
HACH
invensys FOXBORO
DeltaV
ROSEMOUNT Analytical
GE FANUC
FISHER
ABB
TELVENT
ROCKWELL SOFTWARE
Honeywell
Rockwell Automation RSView





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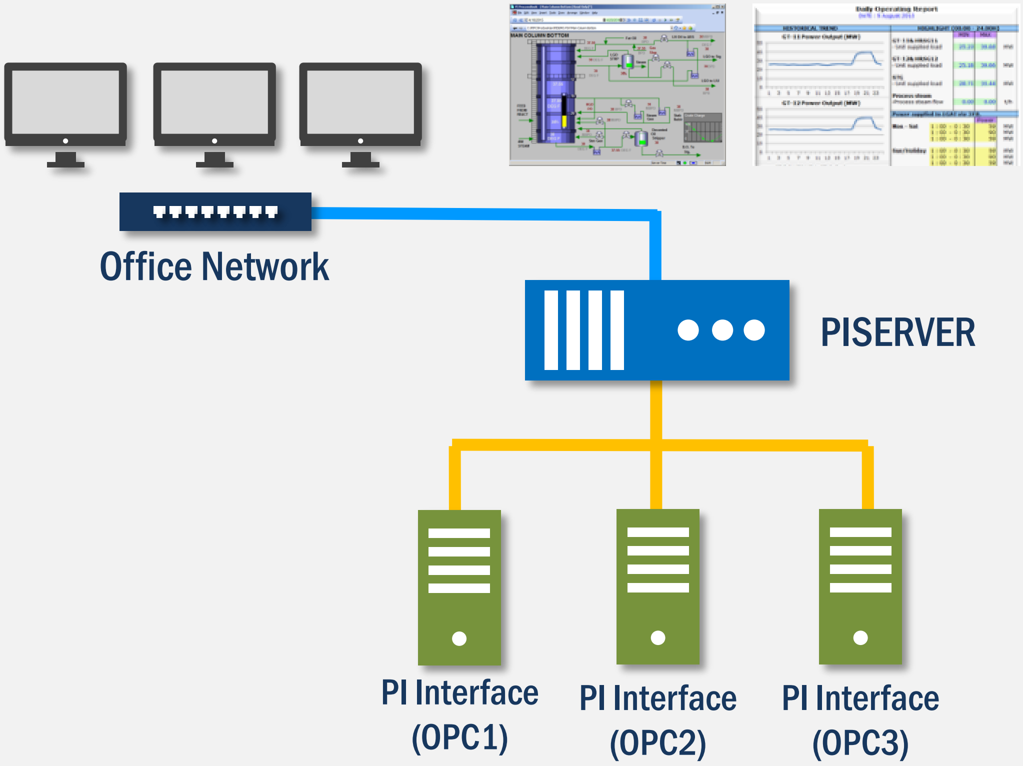
SYSTEM ARCHITECTURE



Why we need the PI System ?

OUR CUSTOMER'S CHALLENGE

COO

What is the forecast of productivity?

Engineer

The turbine is not working – what's the problem?

Plant Manager

I need to combine data from 3 sources

Maintenance Technician

I need to know the *moment* it goes out of tune

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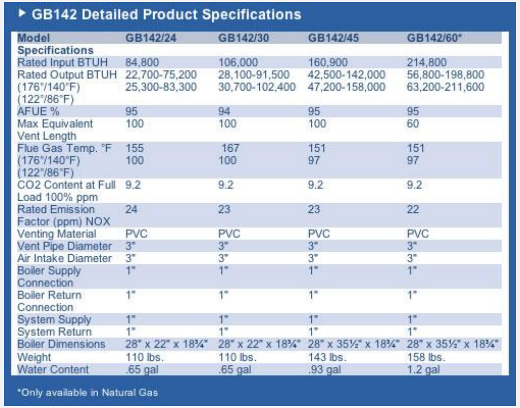
MORE CHALLENGE



GB142 Detailed Product Specifications

Model	GB142/24	GB142/30	GB142/45	GB142/60*
Specifications				
Rated Input BTUH	84,800	106,000	160,900	214,800
Rated Output BTUH (176°/140°F) (122°/86°F)	22,700-75,200 25,300-83,300	28,100-91,500 30,700-102,400	42,500-142,000 47,200-158,000	56,800-198,800 63,200-211,600
AFUE %	95	94	95	95
Max Equivalent Vent Length	100	100	100	60
Flue Gas Temp. °F (176°/140°F) (122°/86°F)	155 100	167 100	151 97	151 97
CO2 Content at Full Load 100% ppm	9.2	9.2	9.2	9.2
Rated Emission Factor (ppm) NOX	24	23	23	22
Venting Material	PVC	PVC	PVC	PVC
Vent Pipe Diameter	3"	3"	3"	3"
Air Intake Diameter	3"	3"	3"	3"
Boiler Supply Connection	1"	1"	1"	1"
Boiler Return Connection	1"	1"	1"	1"
System Supply	1"	1"	1"	1"
System Return	1"	1"	1"	1"
Boiler Dimensions	28" x 22" x 18¾"	28" x 22" x 18¾"	28" x 35½" x 18¾"	28" x 35½" x 18¾"
Weight	110 lbs.	110 lbs.	143 lbs.	158 lbs.
Water Content	.65 gal	.65 gal	.93 gal	1.2 gal

*Only available in Natural Gas

Plant status outside control room ?

Fastest Information to Top Management ?

Actual performance vs. design ?

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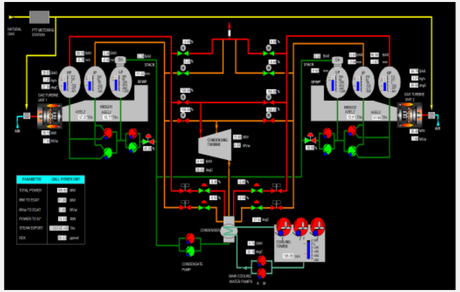
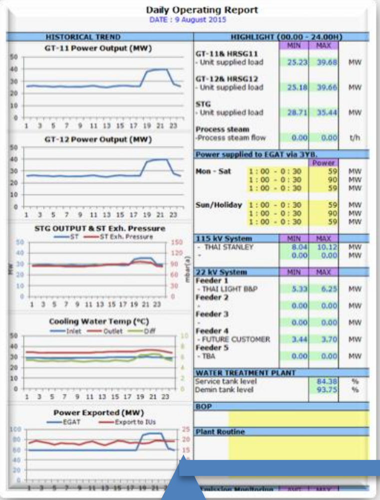
WHAT'S A SOLUTION?





Operation

Performance

Plant Mgmt.

Executive Mgmt.

PI SDK Utility

WHAT IS PI SDK UTILITY?

PI SDK Utility is a tool for

- Manage & Test Connection
- Change Password

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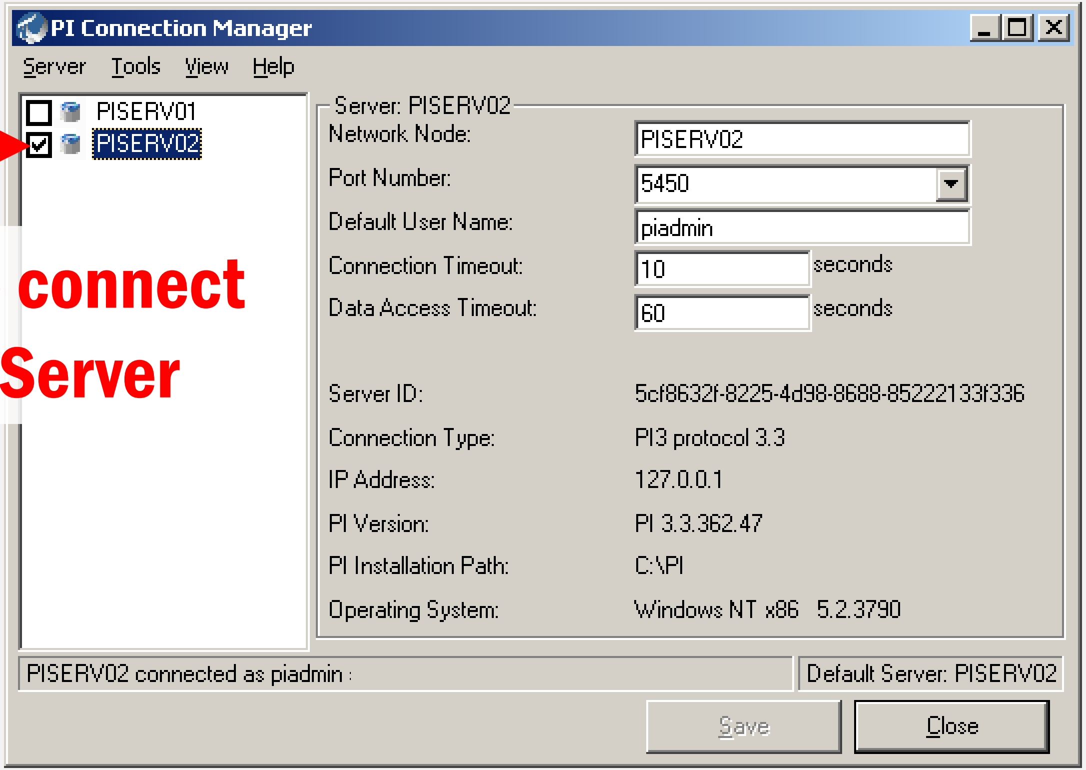
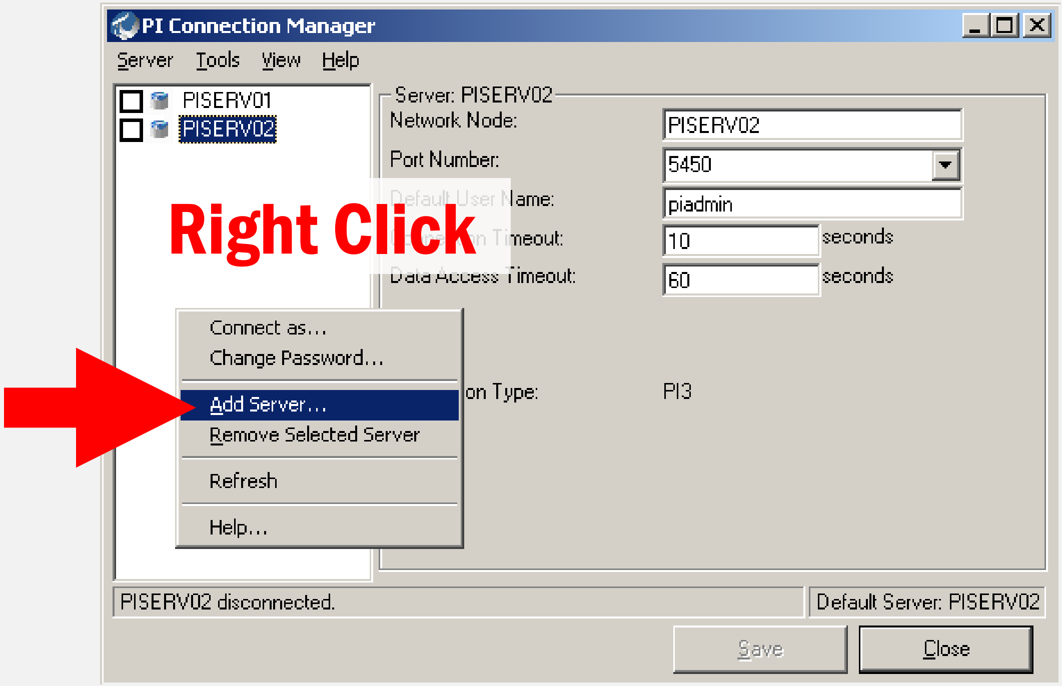
PI SERVER CONNECTION

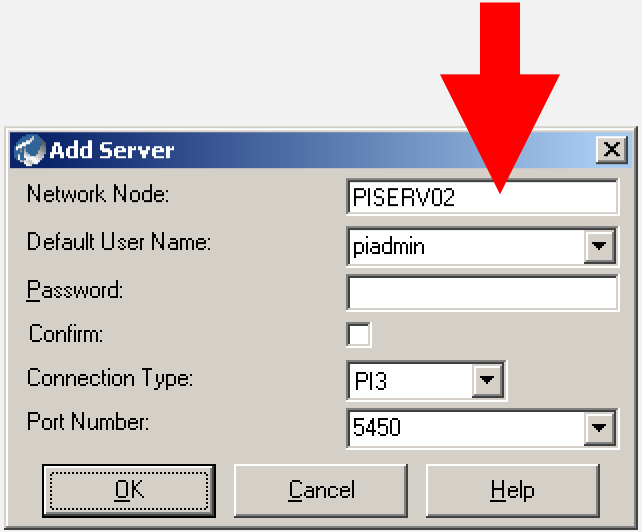


Check to connect the PI Server

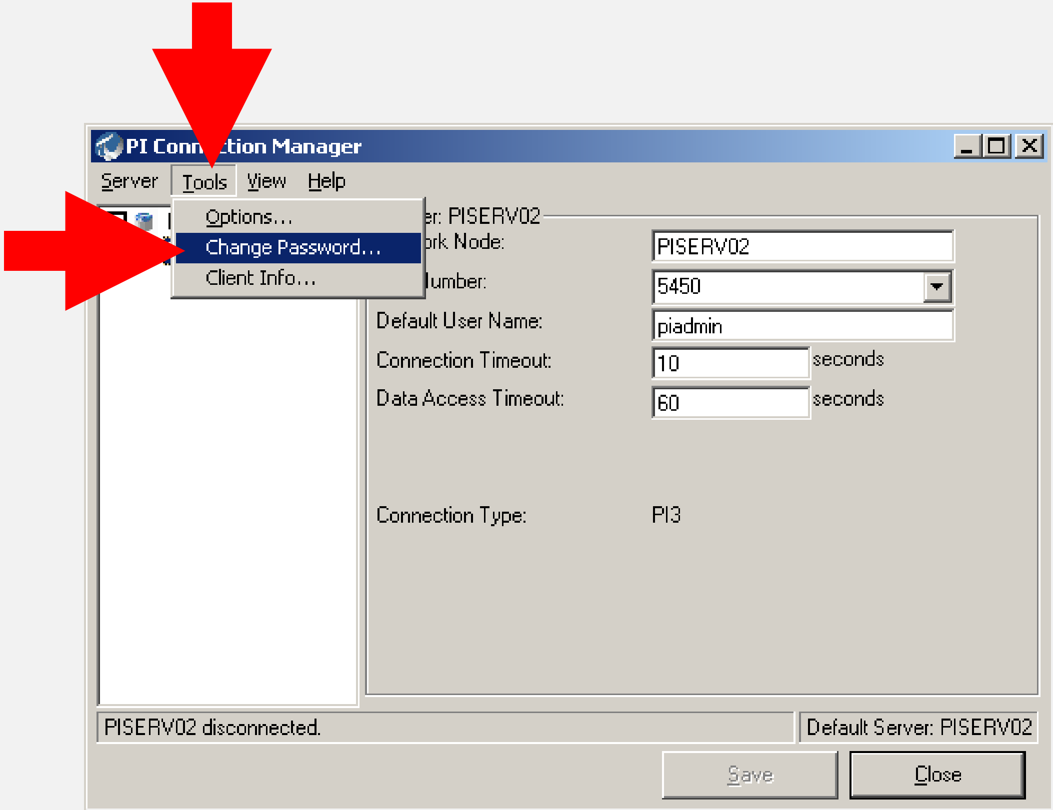
ADDING A PI SERVER CONNECTION

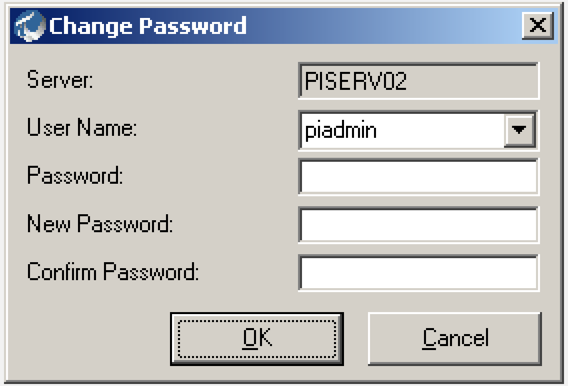


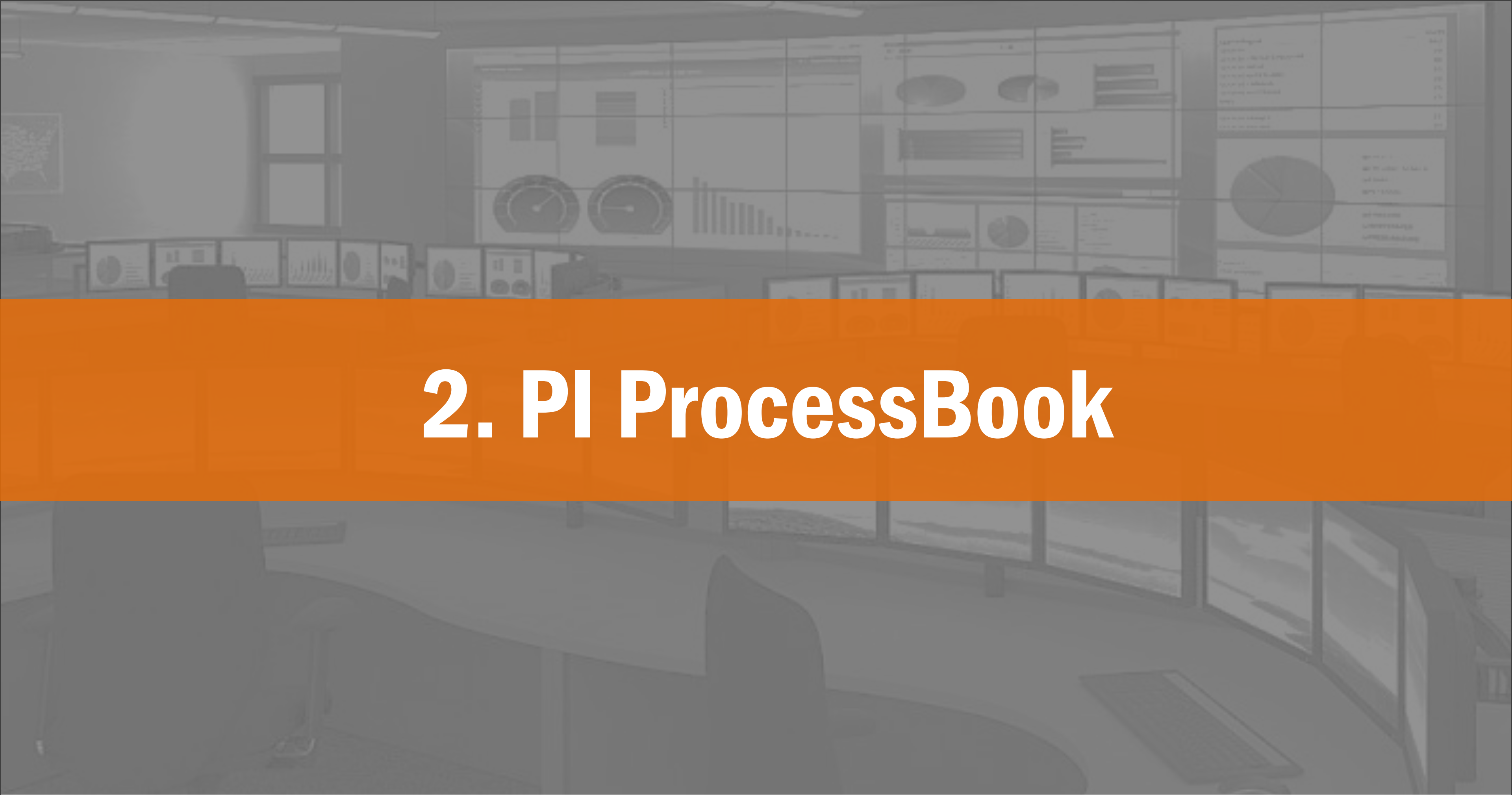
PISERV01
PISERV02



CHANGING PASSWORD







2. PI ProcessBook

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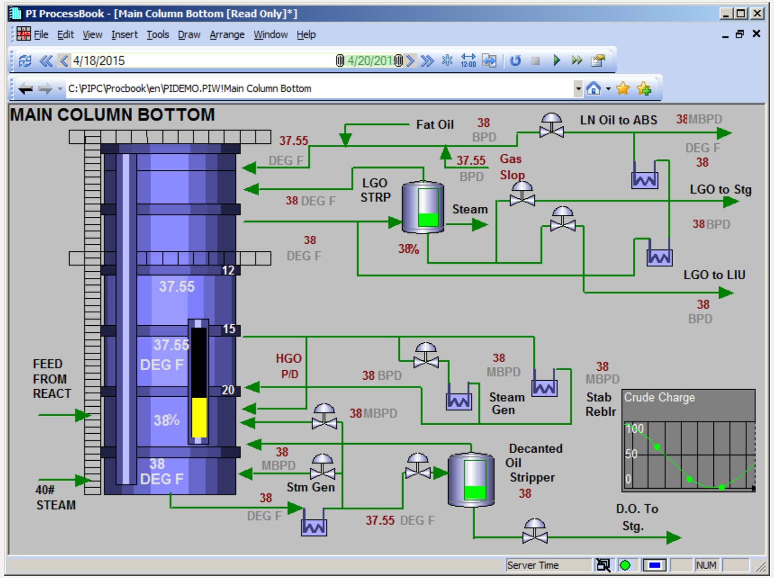
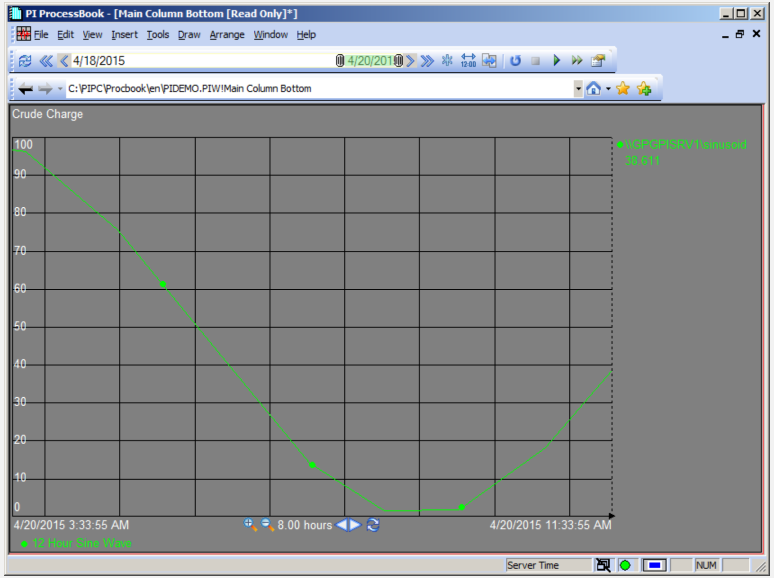
WHAT IS PI PROCESSBOOK?

- Access and visualize real-time and historical PI System data through interactive, graphical, displays.

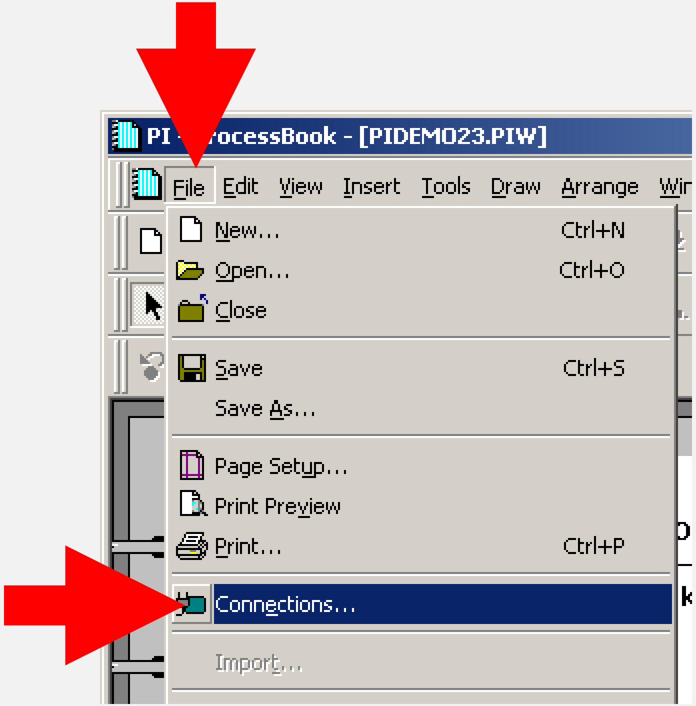


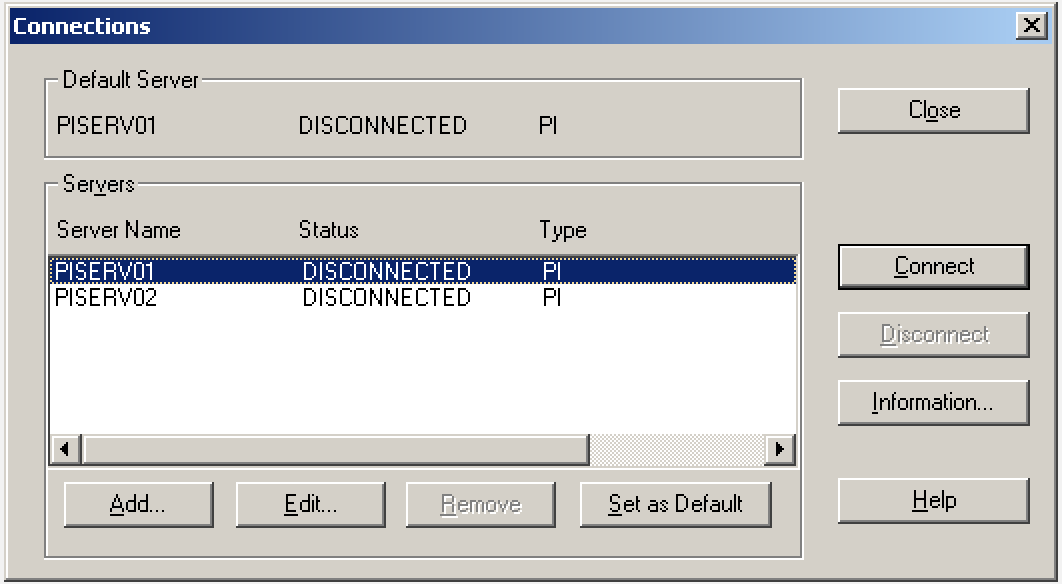


PI PROCESSBOOK COMPONENTS

- Build Visualization Display with
  - Shapes (Line, Rectangle, Text, Ellipse, Arc, Polyline, Polygon)
  - Images (jpg, bmp, png, tiff, ...)
  - Symbols (Tank, Valve, Pipe, Pump, ...)
  - Real Time Value and Bar
  - PI Trend (Value vs Time, Value vs Value)
  - PI Calculation
  - ActiveX Controls (VBA)

PI SERVER CONNECTION





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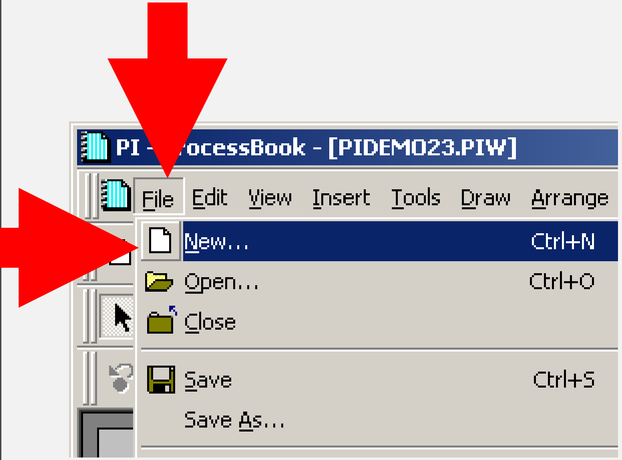
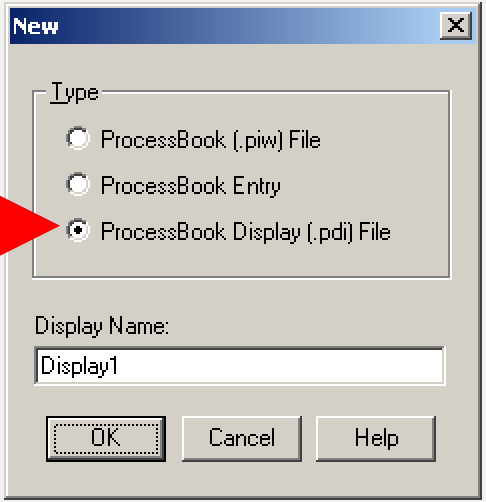
CREATE A NEW DISPLAY

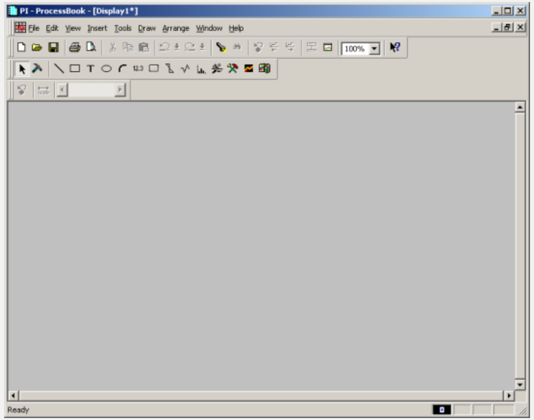
File > New



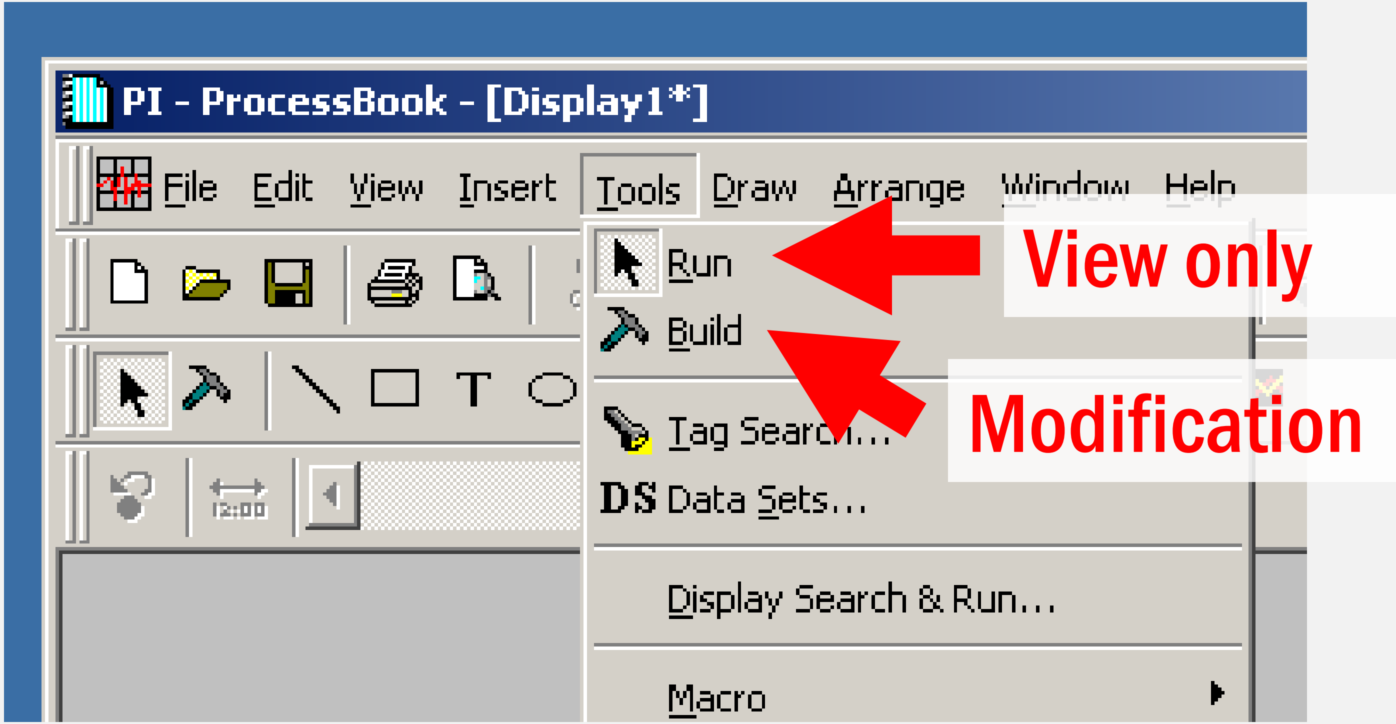
Choose display type



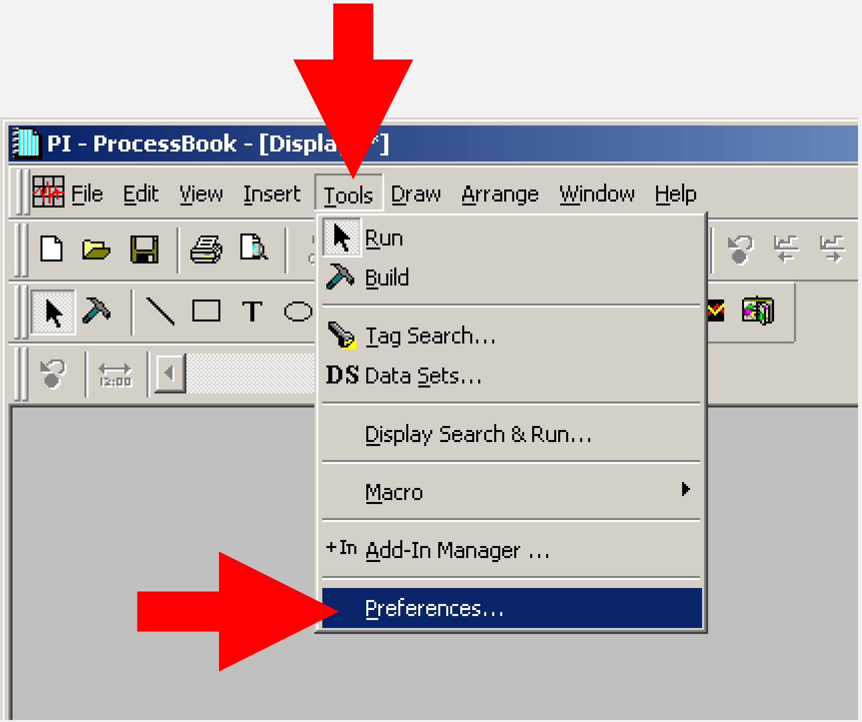
New display

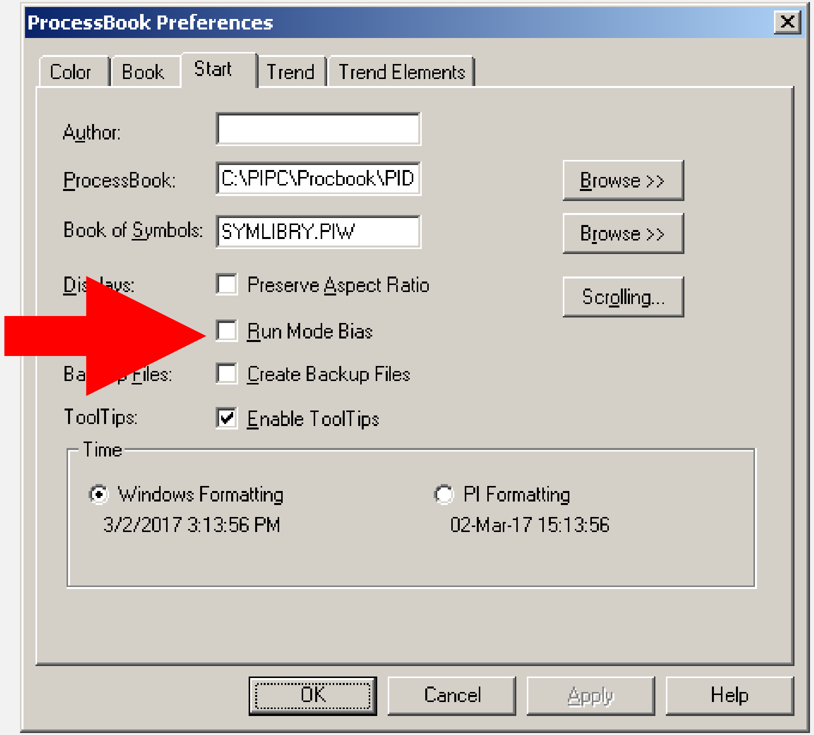


“RUN MODE” VS “BUILD MODE”



DISABLE AUTO RUN MODE





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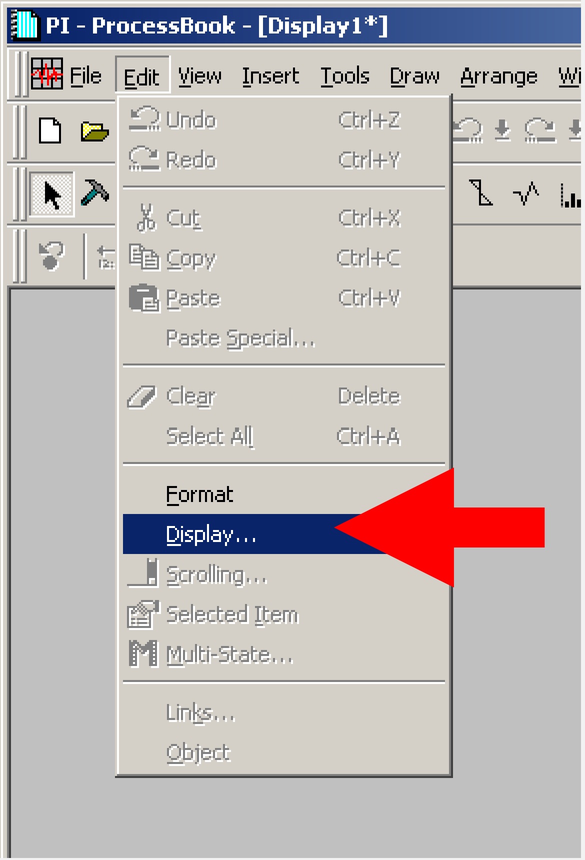
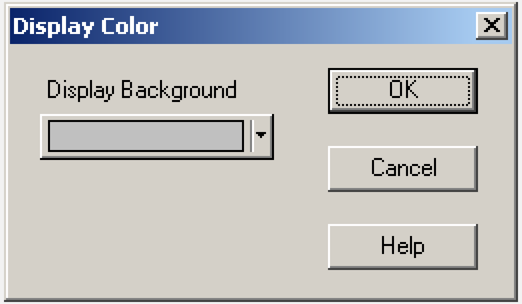
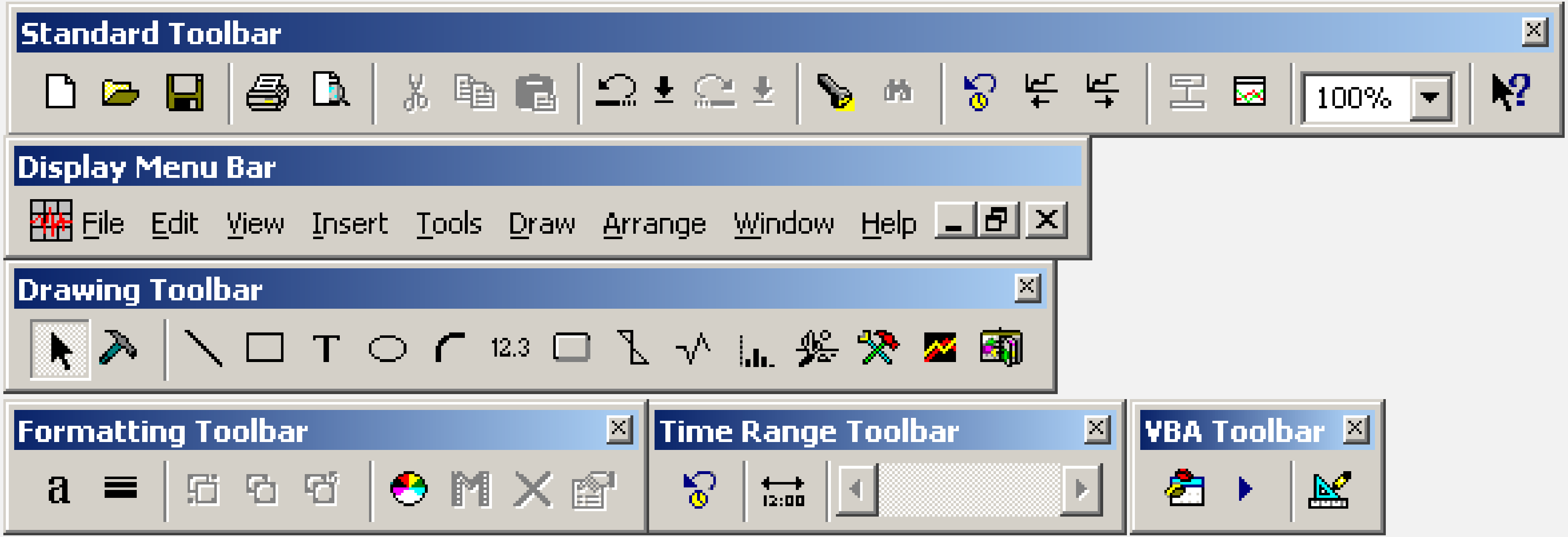
CHANGE BACKGROUND COLOR

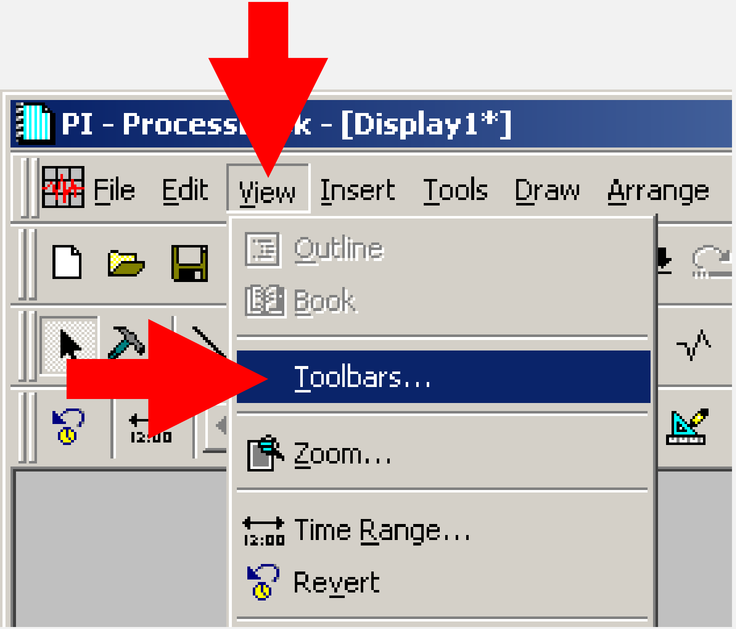


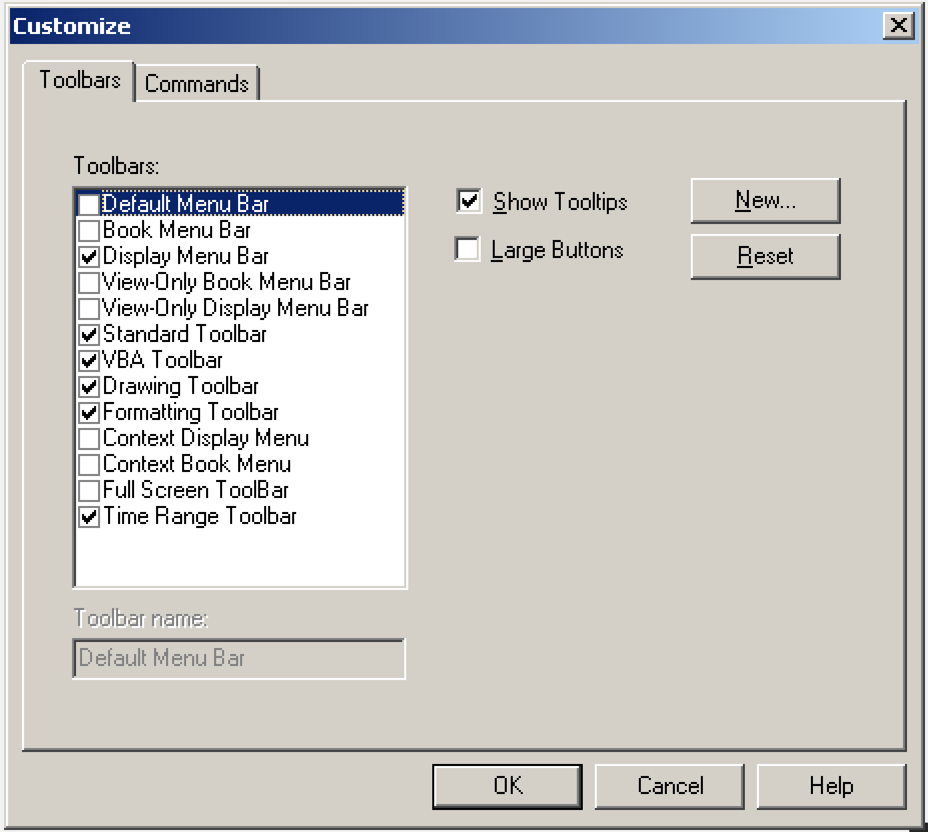


PI PROCESSBOOK TOOLBARS

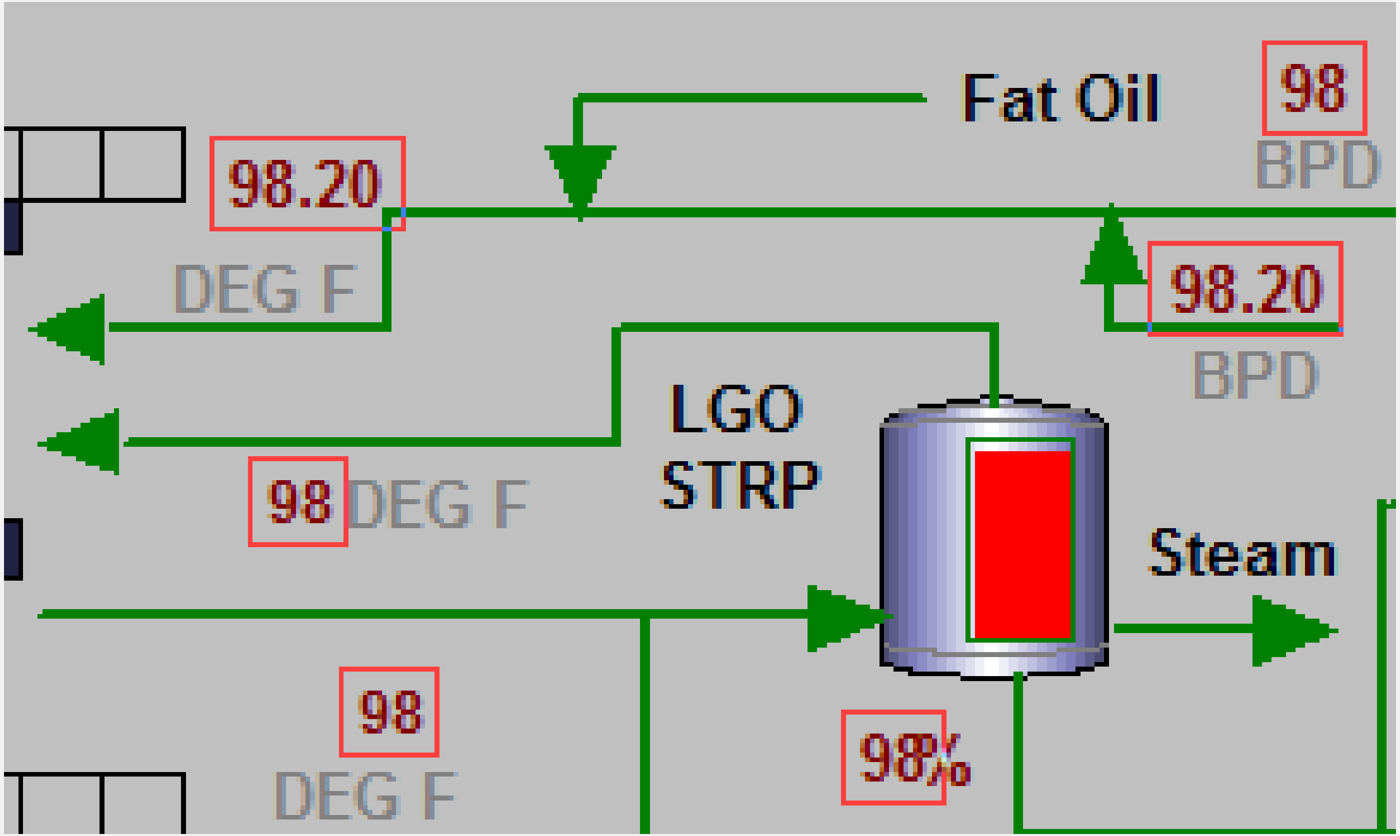


MANAGE TOOLBARS

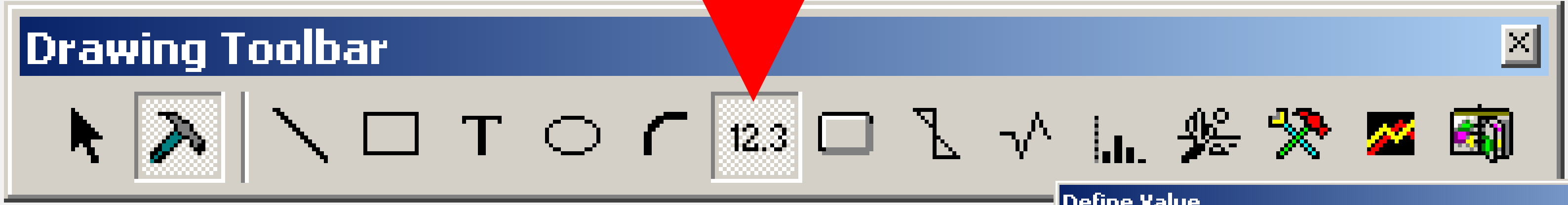


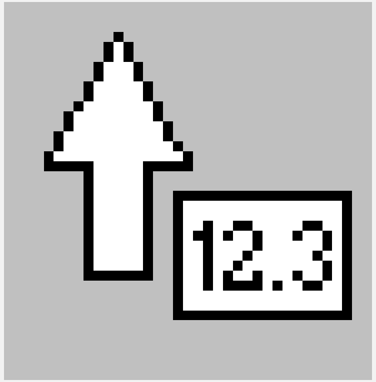


VALUE OBJECT

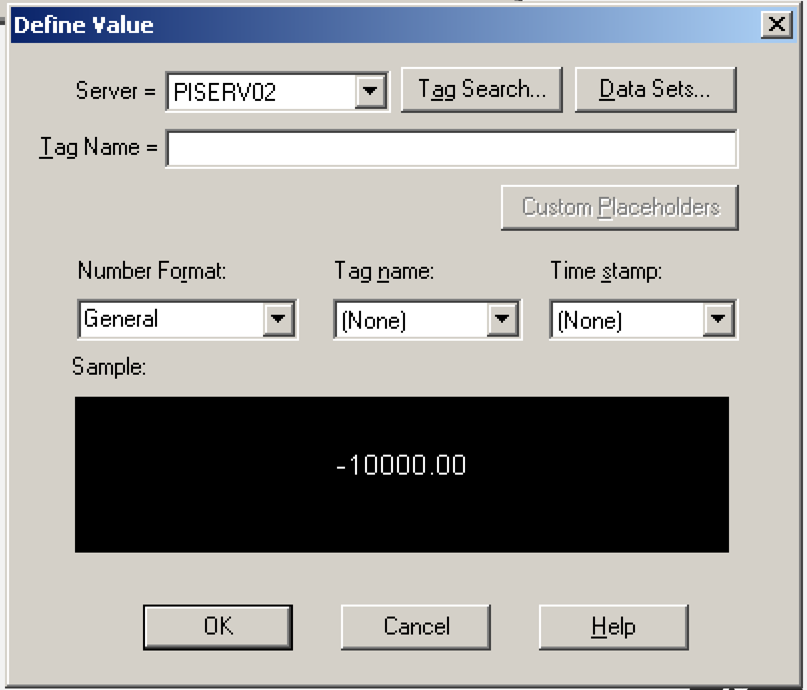


ADDING A VALUE OBJECT



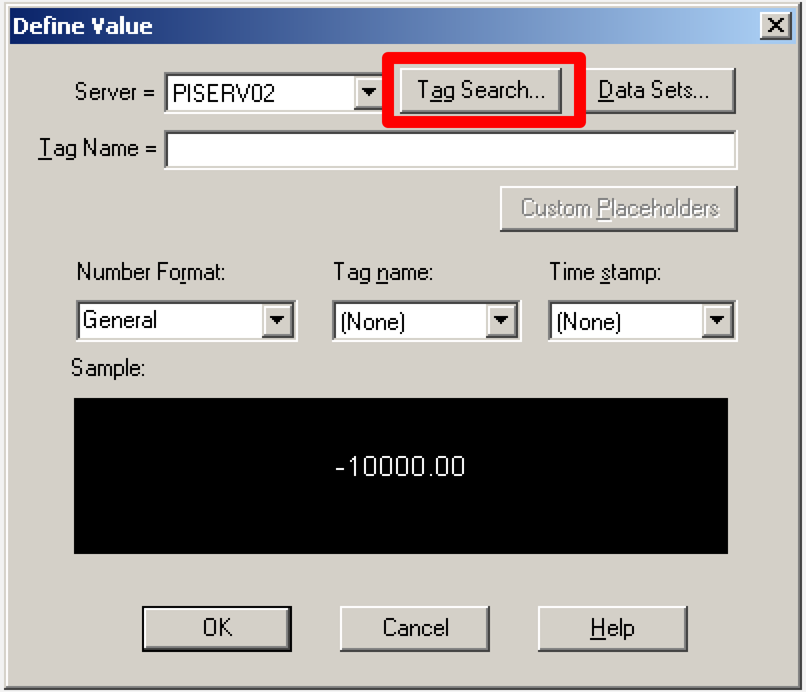


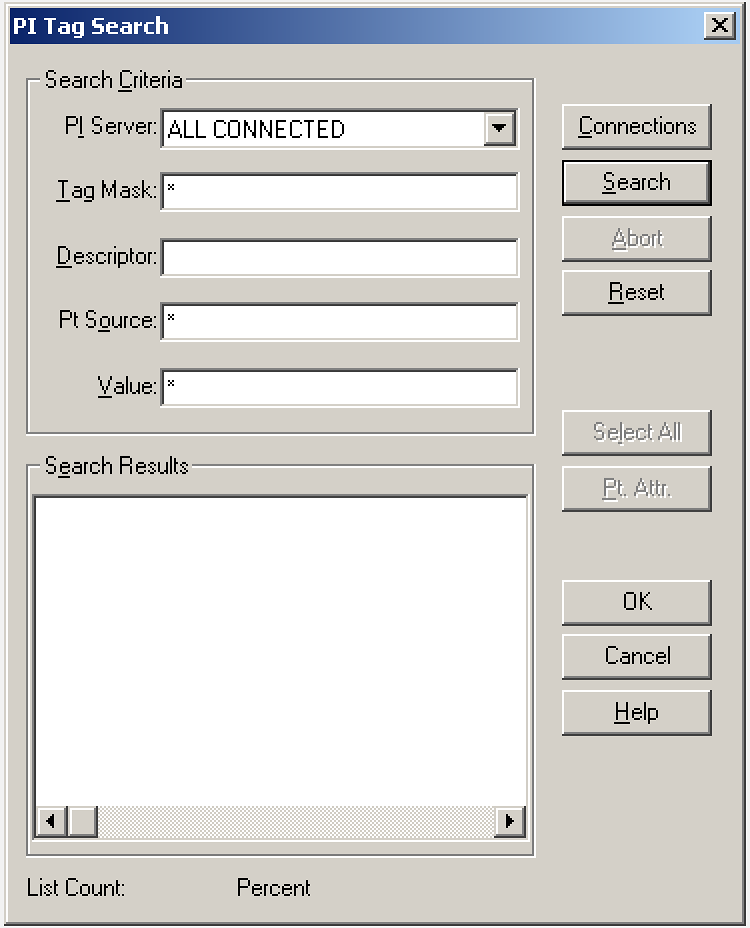
Click to the desired location



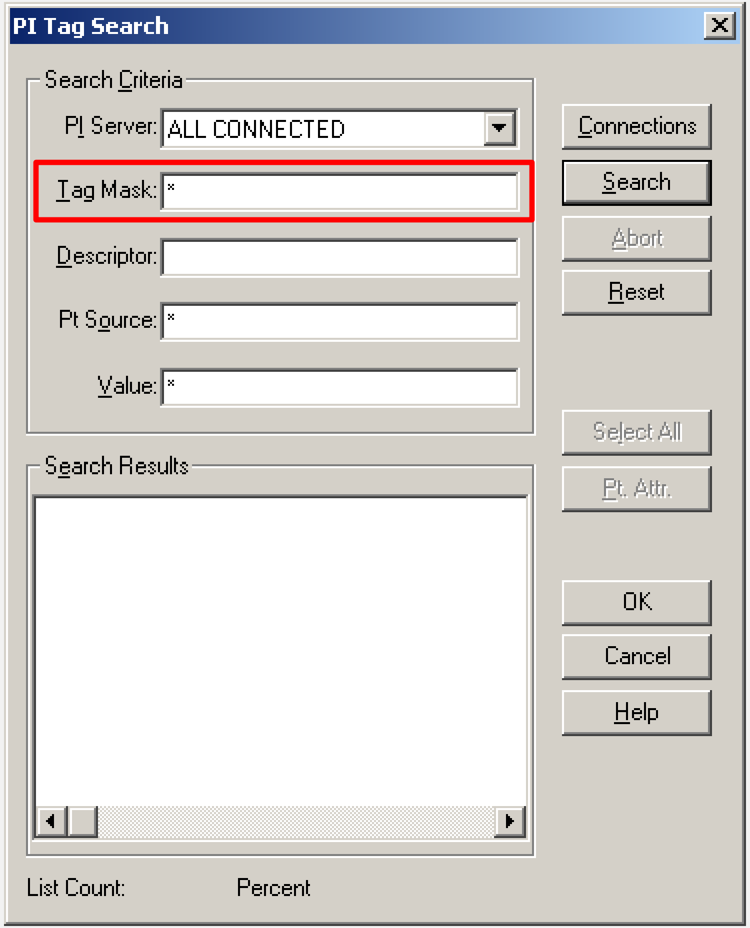
Value configuration pop-up

ADDING A VALUE OBJECT



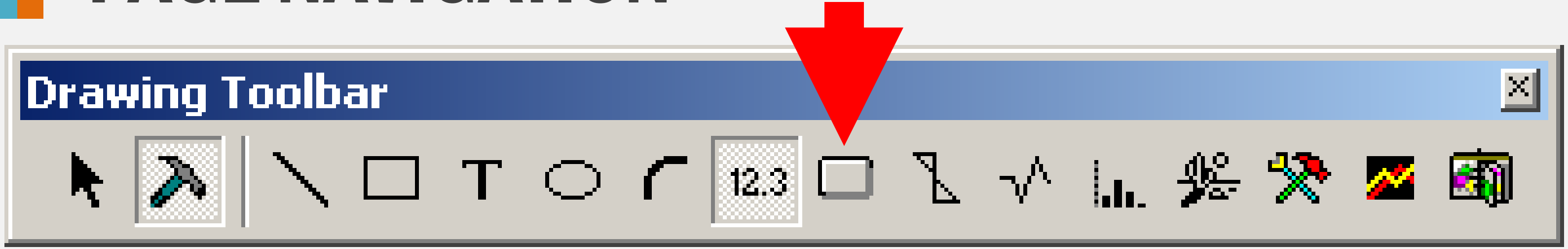


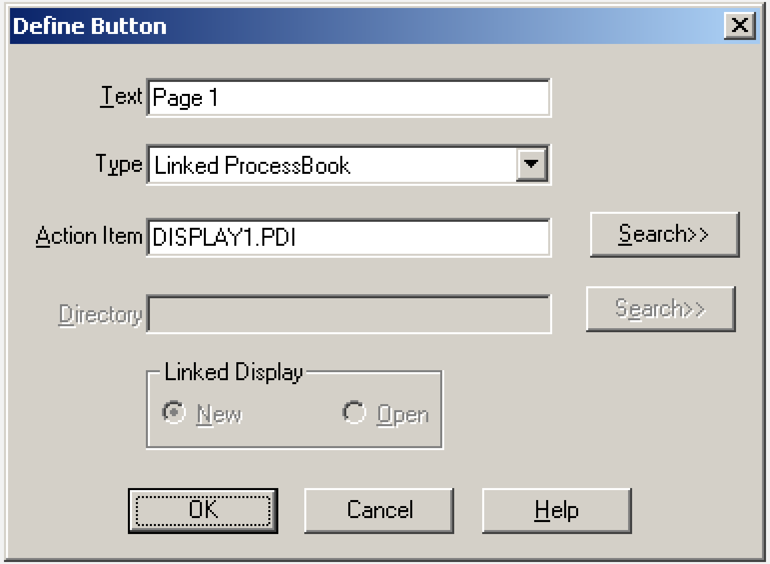
TAG SEARCH



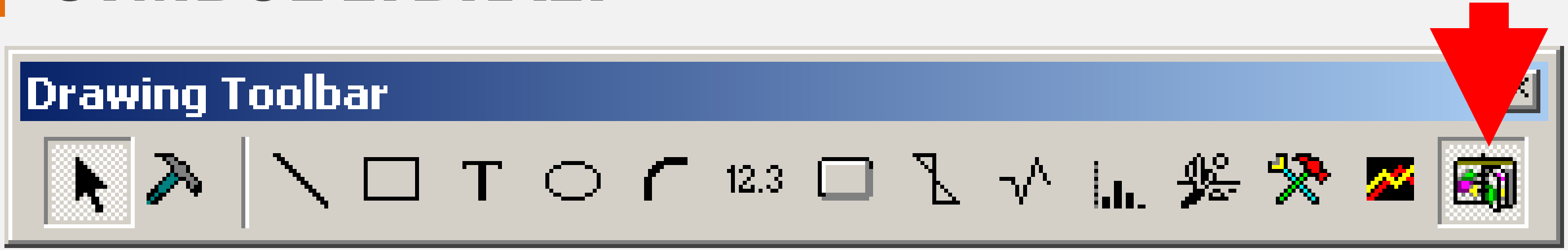
Expression	Meaning
FI	(Any) + FI + (Any) ▪ 10FI001.PV ▪ FI555.PV ▪ 555.FI
FI*	FI + (Any) ▪ FI123.PV
*.PV	(Any) + .PV ▪ 10TI003.PV
LIC.MV	(Any) + LIC + (Any) + .MV ▪ 10LIC01.MV

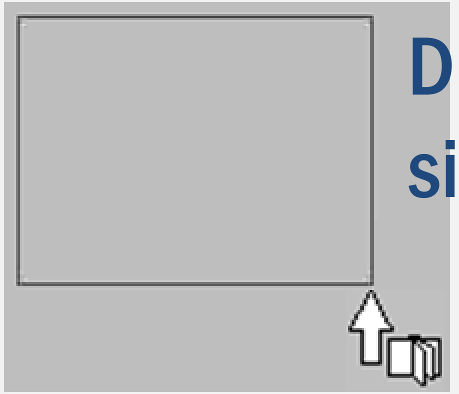
PAGE NAVIGATION





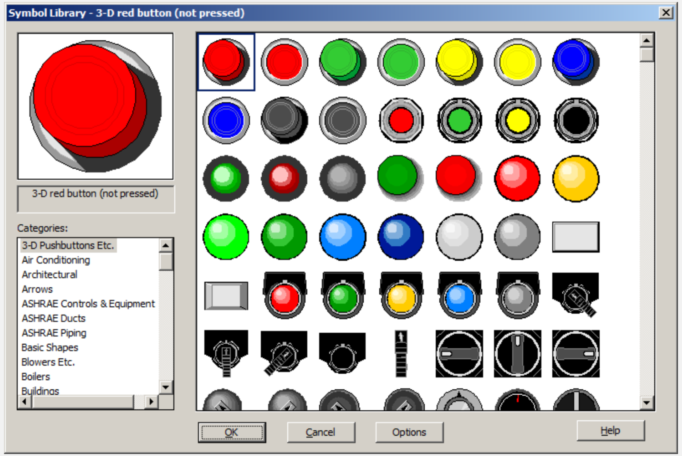
SYMBOL LYBRALY





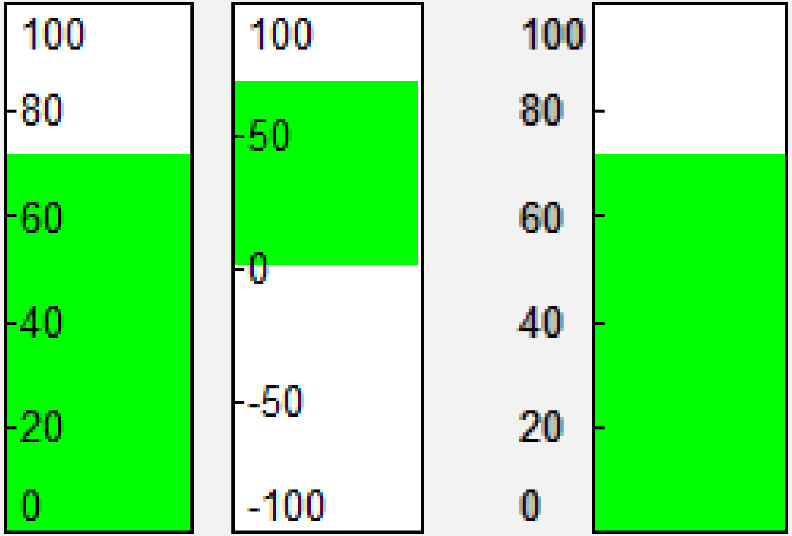
Draw the location and size of the symbol

Symbol selection pop-up

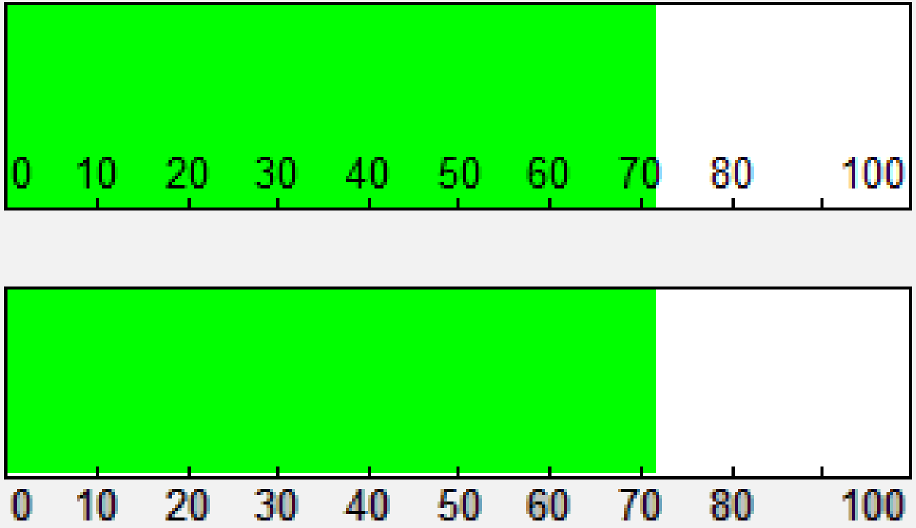


BAR OBJECT

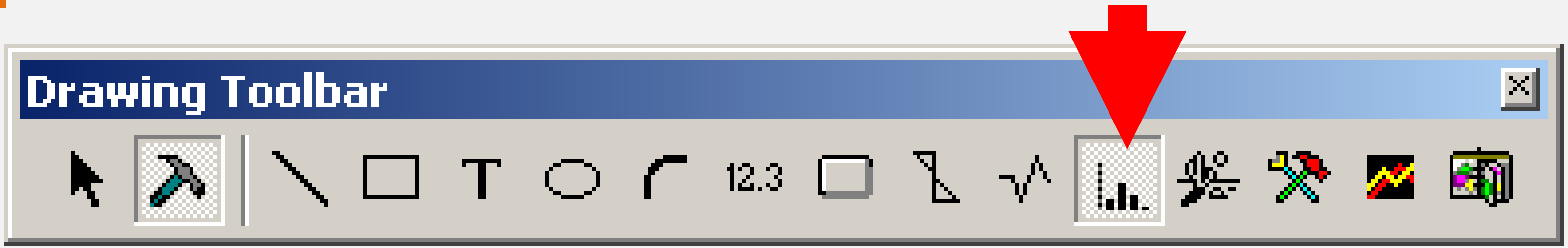
Vertical Bar

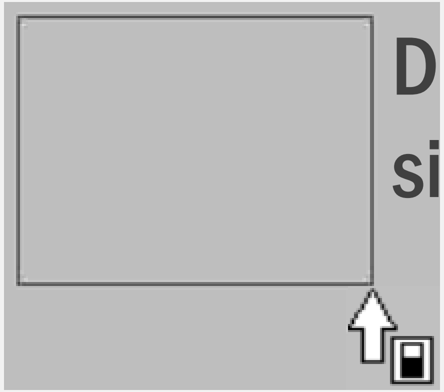


Horizontal Bar



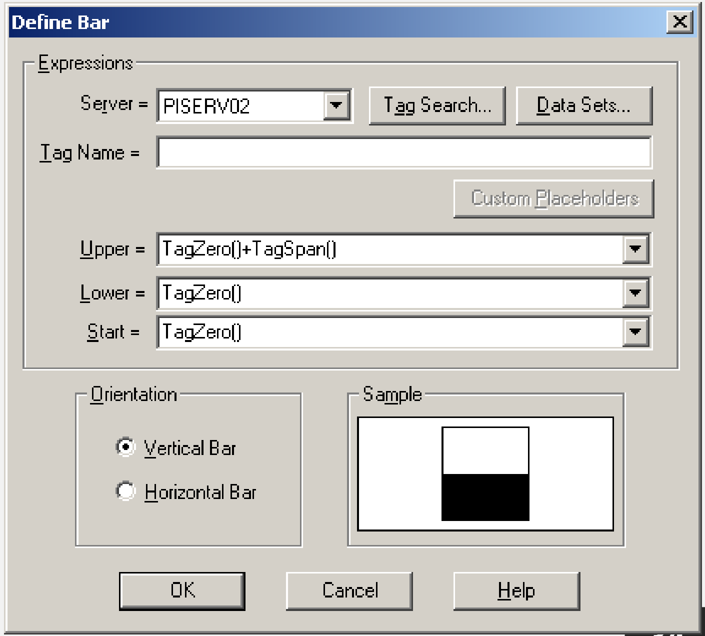
INSERT A BAR OBJECT





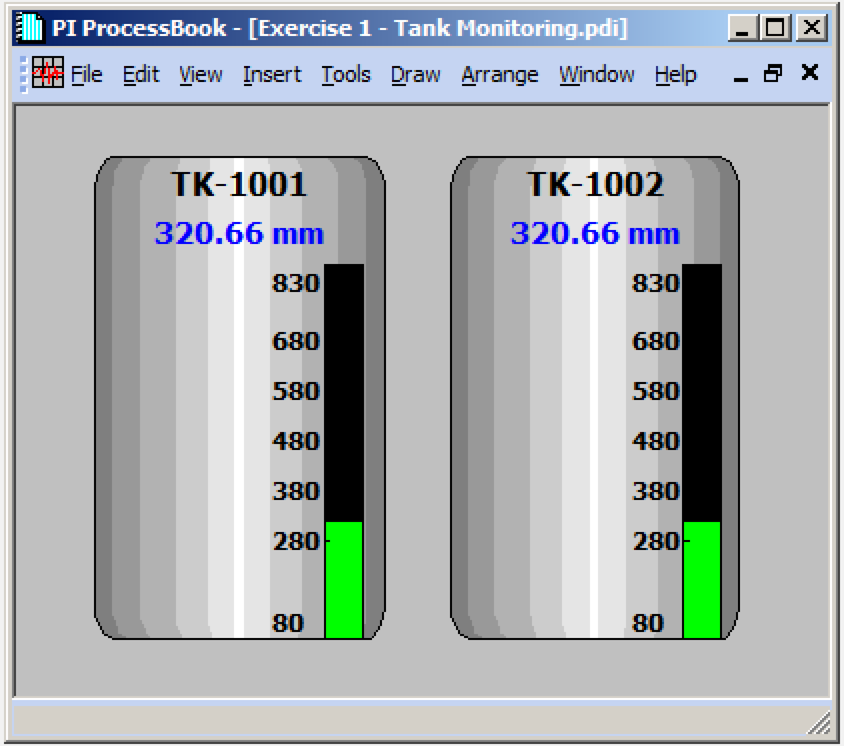
Draw the location and size of the bar

Bar configuration pop-up



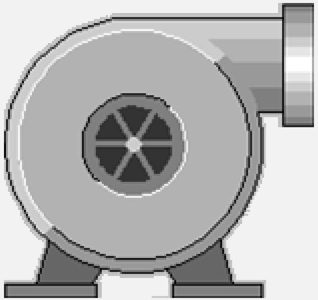
EXERCISE 2.1 : TANK MONITORING

- Build a simple display with
  - Basic Shapes
  - Use Symbols
  - Use Text
  - Use Value
  - Use Bar

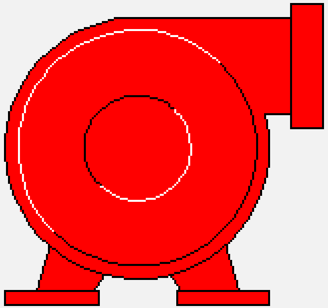


MULTI-STATE

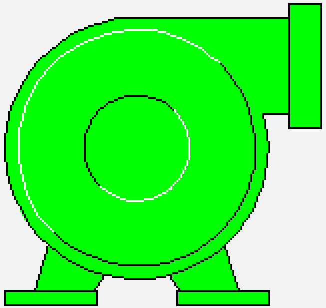
- Change the symbols color upon value



No Multi-State



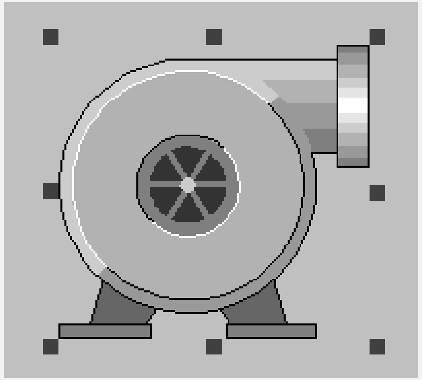
Running



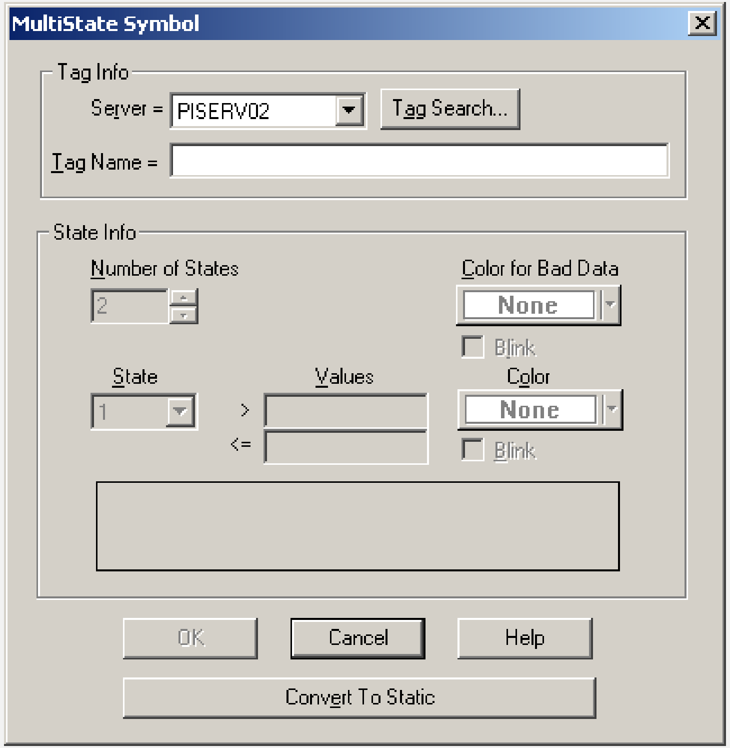
Stopped

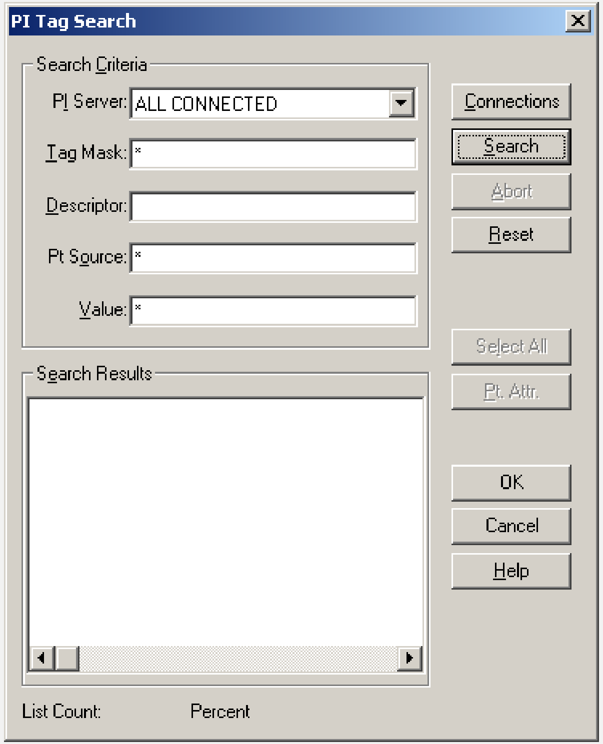
MULTI-STATE

Select the target symbol



MULTI-STATE CONFIGURATION





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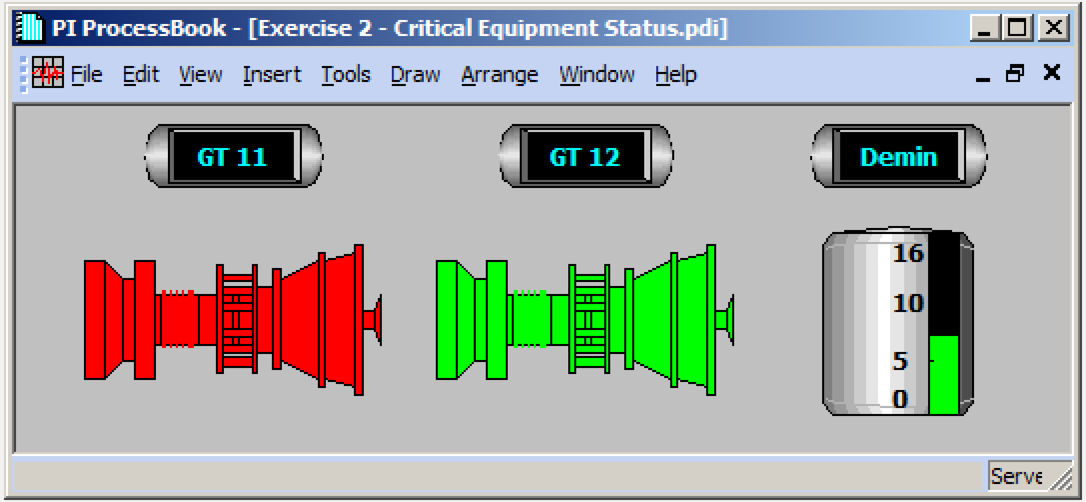
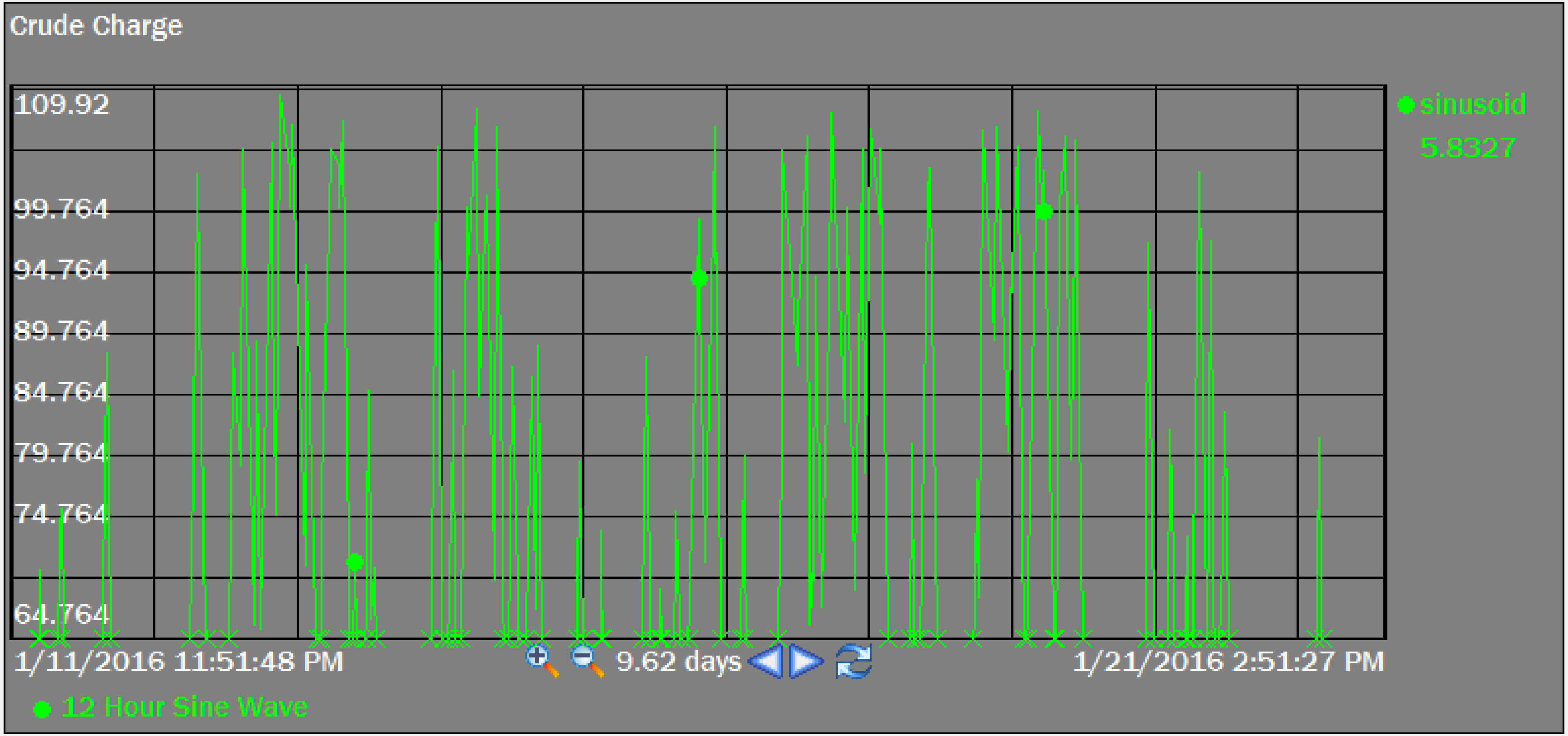
EXERCISE 2.2 : CRITICAL EQUIPMENTS STATUS

- Build a display with symbols that change the color following measured value.

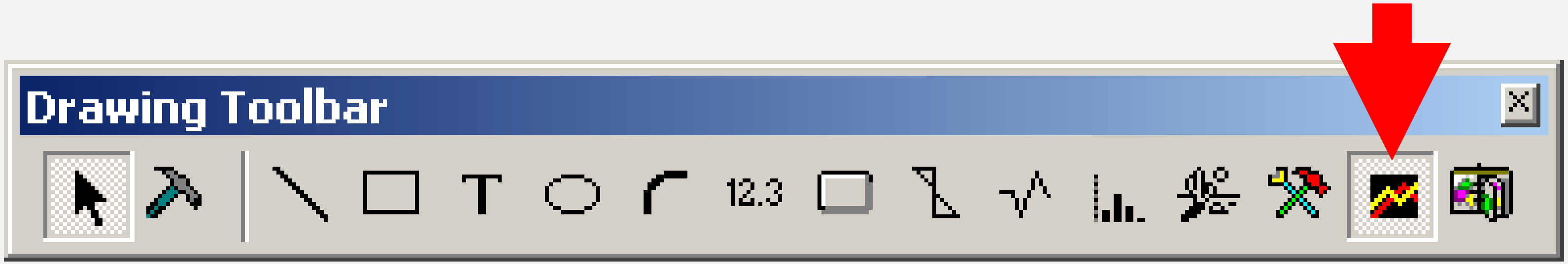


TREND

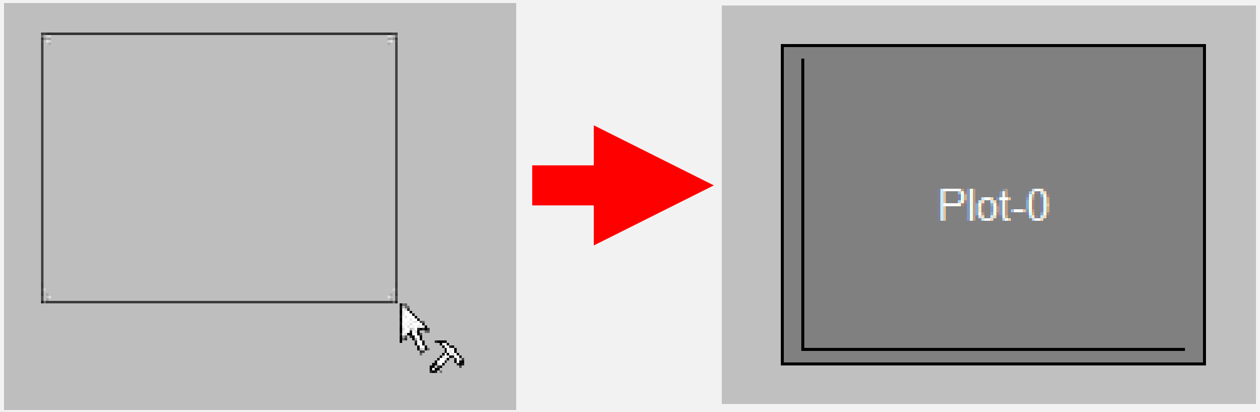


Crude Charge
109.92
99.764
94.764
89.764
84.764
79.764
74.764
64.764
sinusoid
5.8327
1/11/2016 11:51:48 PM
9.62 days
1/21/2016 2:51:27 PM
12 Hour Sine Wave

ADDING A TREND OBJECT

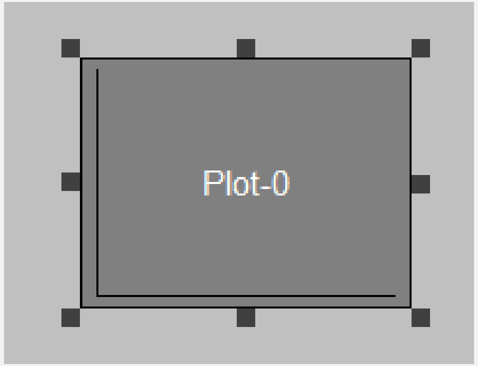


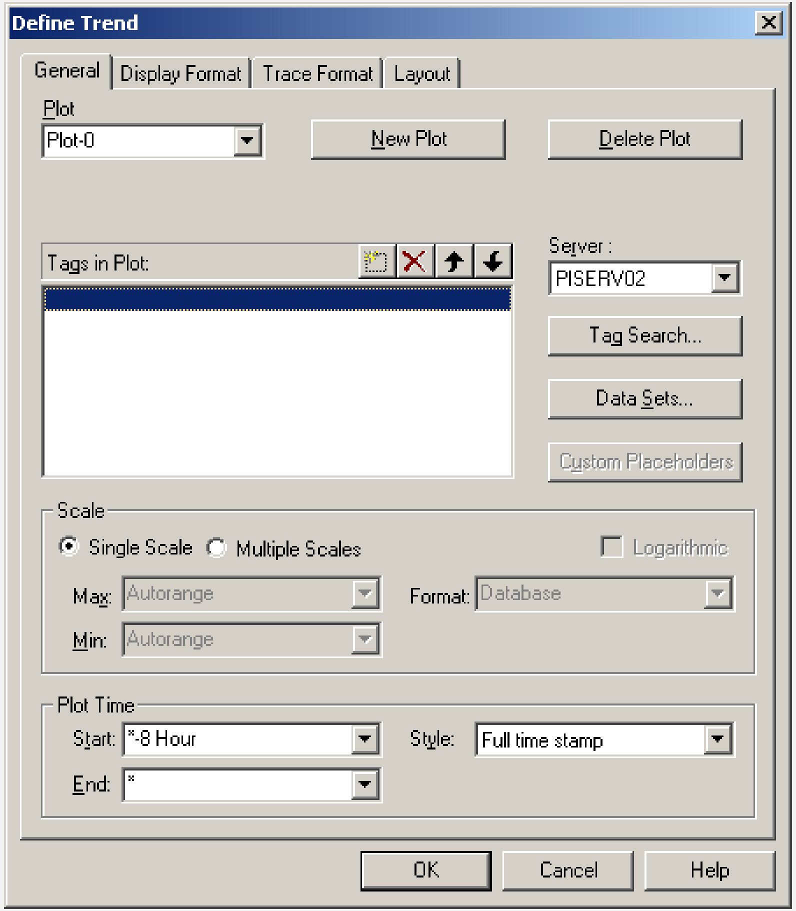
Drag a trend location and size



ADDING A TREND OBJECT

Double click to the trend to open the trend configuration window





PI TIME FORMAT

Expression	Meaning
dd-MMM-yy HH:mm:ss	dd-MMM-yy (Any missing part mean *CURRENT*) HH:mm:ss (Any missing part mean *ZERO*)
12-Apr-15 9:30:25	12 April 2015 9:30:25
12	12 - (Current Month) - (Current Year) at 00:00:00
9:	(Current Date) 9:00:00
12 9:	12 - (Current Month) - (Current Year) at 9:00:00
*	Now
t	Today at 00:00:00
y	Yesterday
Sun, Mon, Tue, ... , Sat	Sun = Last Sunday at 00:00:00

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PI TIME SHIFTING

Expression	Meaning
1s, 1m, 1h 1d, 1w, 1mo, 1y	s = second, m = minute, h = hour d = day, w = week, mo = month, y = year
* - 8h	Now – 8 Hours
* - 1d	Now – 1 Day
t + 8h	Today + 8 Hours
t – 1mo	Today – 1 Month
y – 1y	Yesterday – 1 Year

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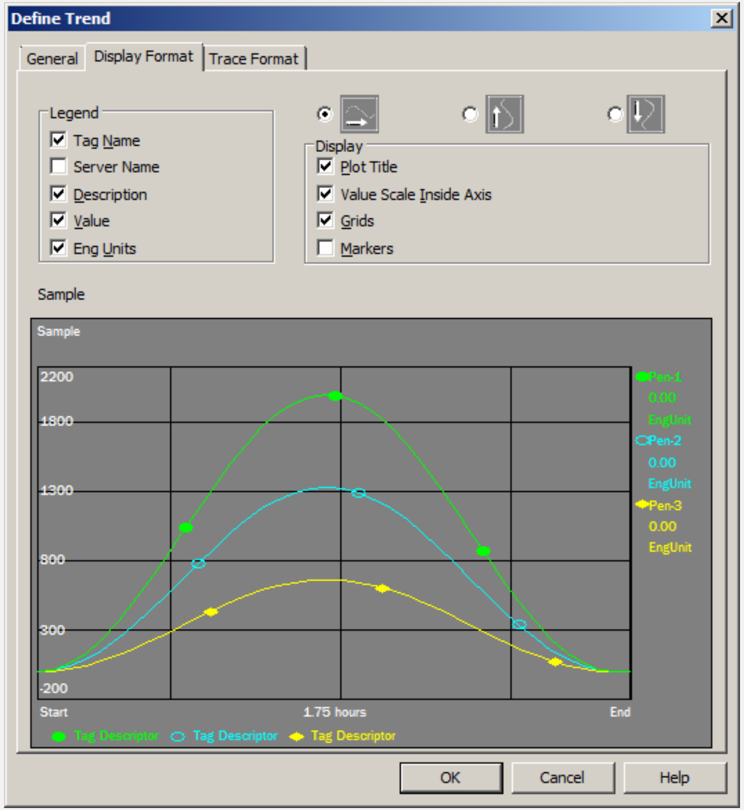
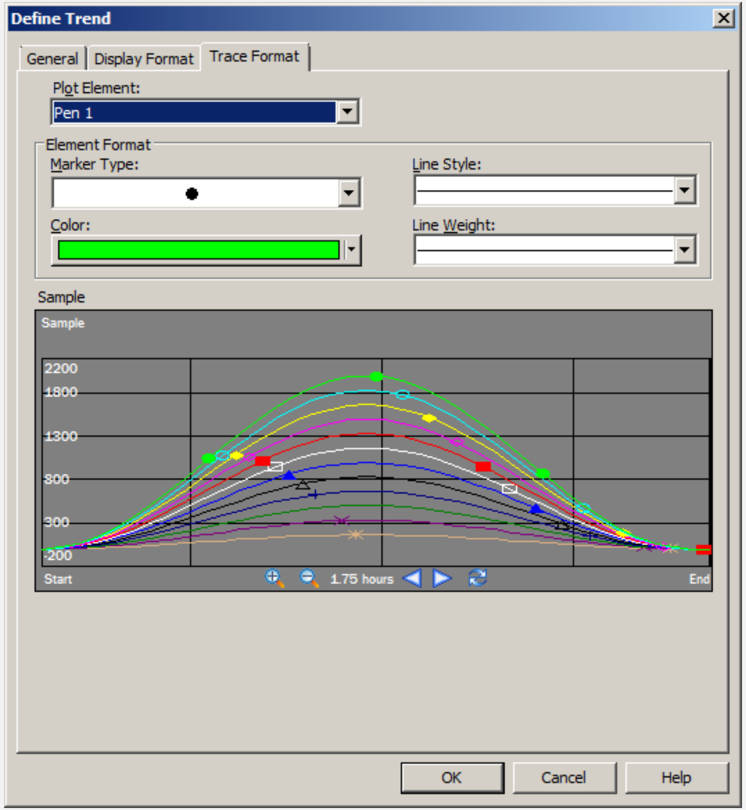
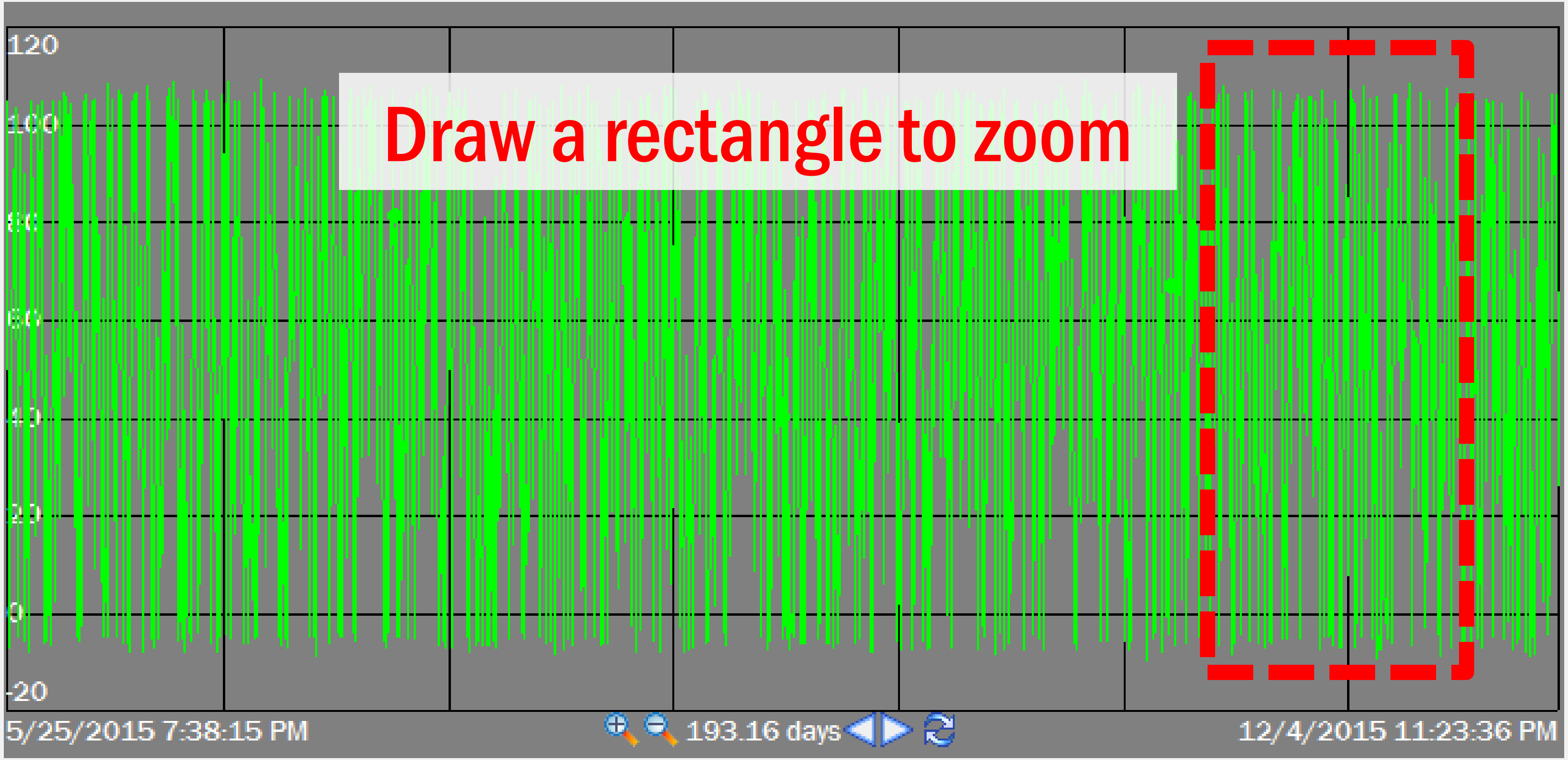
TREND FORMAT

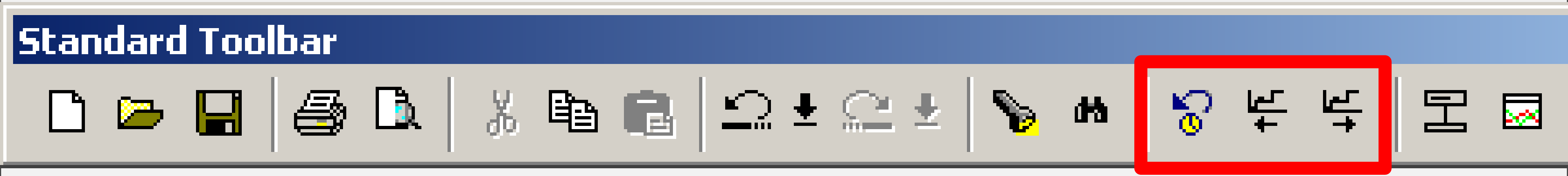




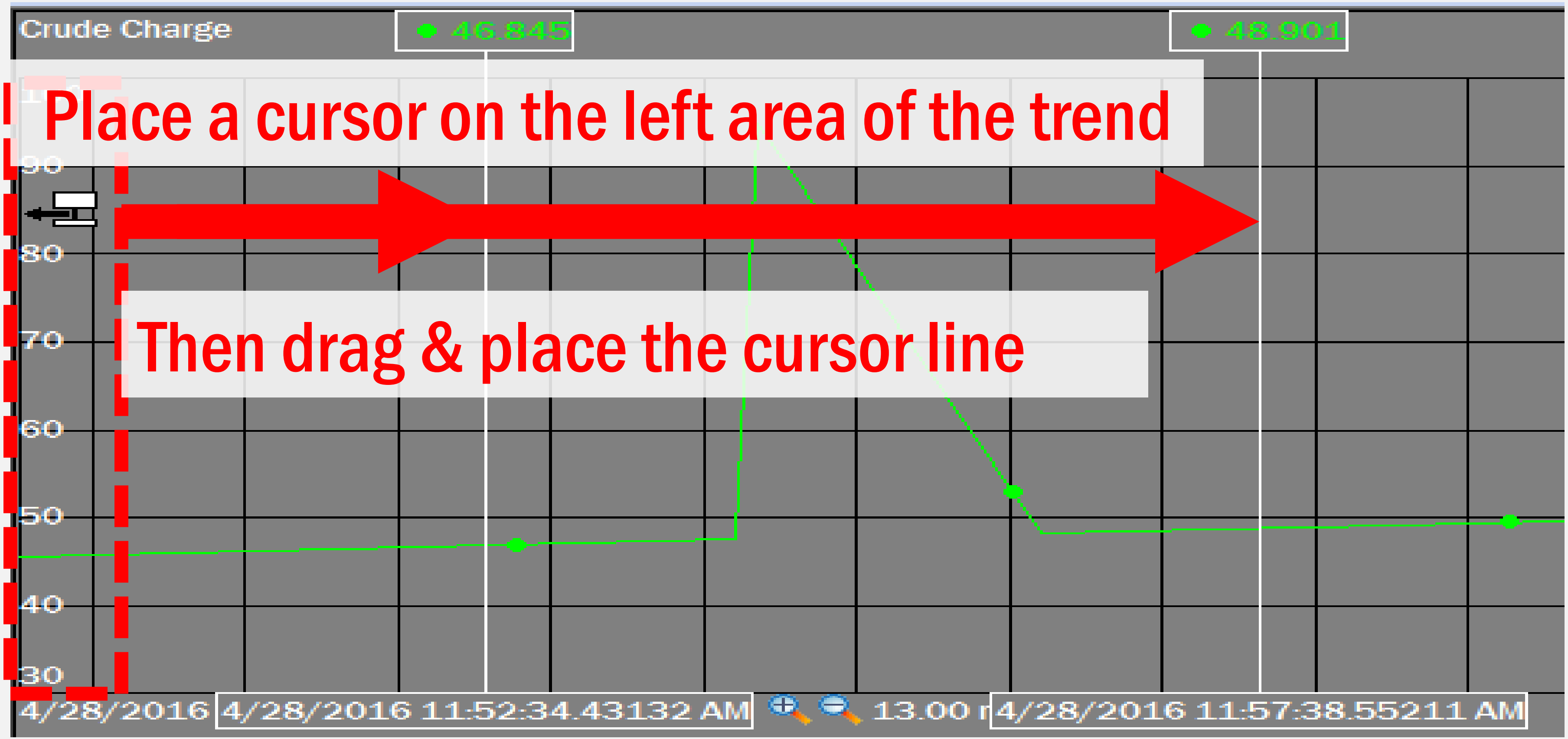
TREND ZOOMING



SCROLLING TREND TIME RANGE

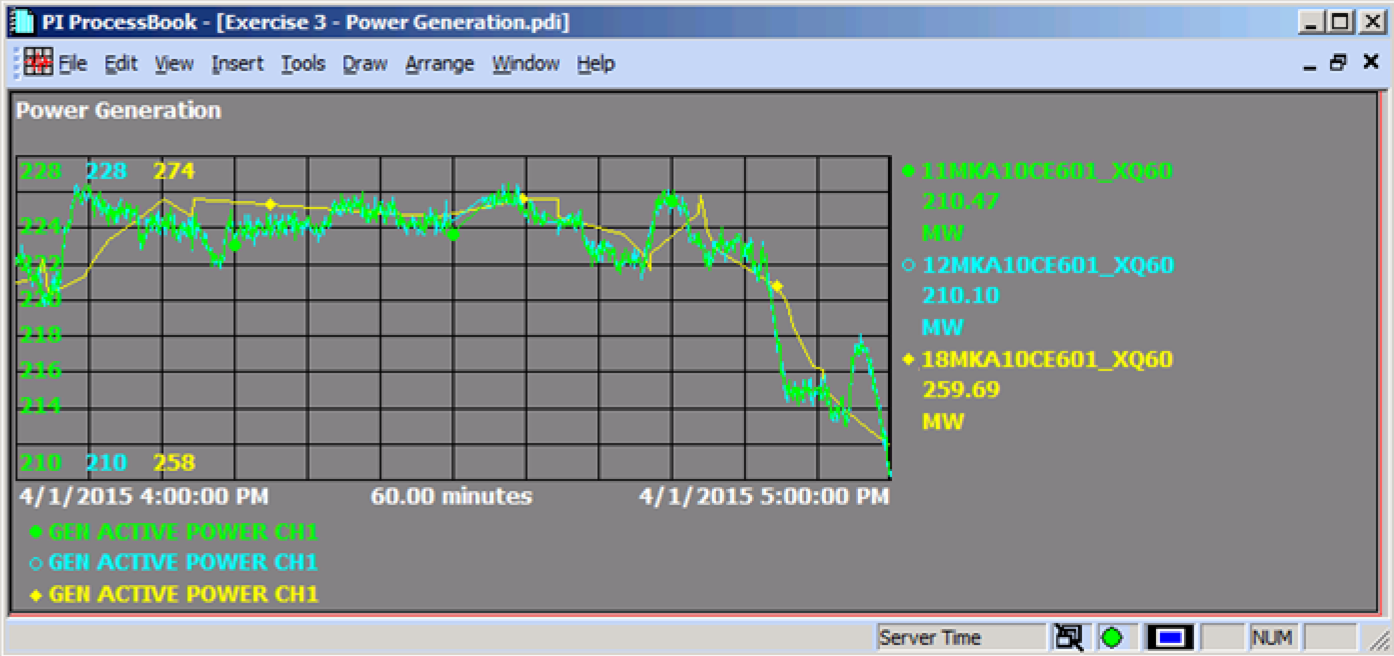


TREND CURSOR



EXERCISE 2.3 : POWER GENERATION TREND

- Build a display with trends



SAT Solutions

CALCULATION ON PI PROCESSBOOK

- PI Calculation Data Sets allow the user to enter an equation in PI Expression syntax, or a PI point. Data sets will automatically calculate (on-demand) the expression itself, statistics on the expression, and statistics on a single point.

CALCULATION SYNTAX

<u>Tagname</u> <u>Timestamp</u>

‘TAG’ ‘TIME’

SAT Solutions

CALCULATION SYNTAX

Number	String
20	“ON”

CALCULATION SYNTAX

- Numbers

  0 0.125 34.56

- Tag Name - Single Quotes

  'CDT158' 'temp_tank_1' 'FIC5821.PV'

- Timestamps - Single Quotes

  '*' '14-NOV-04' 'T-8h'

- Strings - Double Quotes

  "This is a string"

SAT Solutions

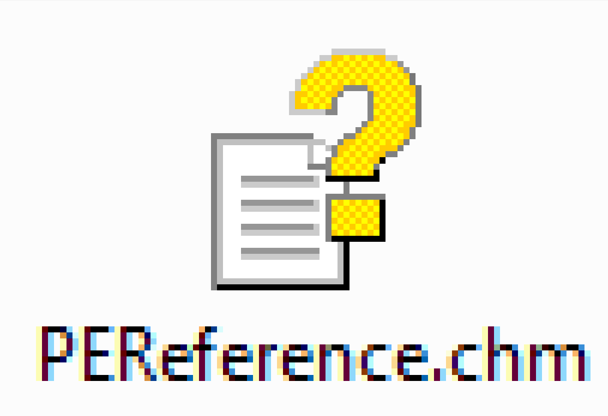
SAMPLE FUNCTIONS

Mathematical operators	Mathematical functions	Mathematical functions	Aggregate Functions
+	cos()	exp()	StDev()
-	sin()	abs()	TagAvg()
*	tan()	int()	TagMax()
/	log()	atn()	TagMin ()
^	log10()	sqr()	TagTot()
		sgn()	

FUNCTION REFERENCE

See all available functions in

.\PIPC\Help\PEReference.chm



AGGREGATION FUNCTIONS

PctGood()	• % of time tag has good values
Range()	• Range of min to max
StDev()	• Time-weighted standard deviation
TagAvg()	• Time-weighted average
TagMean()	• Event-weighted average
TagMax()	• Maximum value in period
TagMin ()	• Minimum value in period
TagTot()	• Time integral over a period

SAT Solutions

IF-THEN-ELSE SYNTAX

IF	CHECK_EXPR
THEN	WHEN_TRUE
ELSE	WHEN_FALSE

IF-THEN-ELSE EXAMPLE

IF	**‘Tag1’ > 80**
THEN	**“Alarm Hi”**
ELSE	**“Normal”**

IF-THEN-ELSE (NOOUTPUT)

Must have all the **IF, THEN and ELSE**

Use **NoOutput()** when no value returned

IF ‘Tag1’ >= 85

THEN “Alarm”

ELSE NoOutput()

SAT Solutions

IF-THEN-ELSE (NESTED)

```
IF      'Tag1'  >=  50
THEN
    IF      'TAG1' >= 75
    THEN  "HIHI"
    ELSE  "HI"
ELSE  "NORMAL"
```

SAT Solutions

CONVERSION FACTOR

- For TOTAL calculation only
- TagTot('tag', 'start', 'end') x ConversionFactor

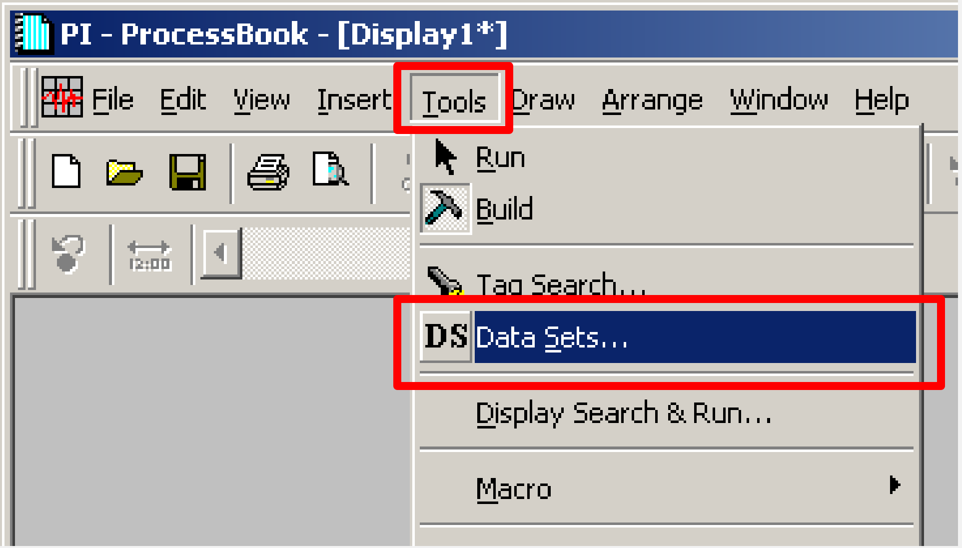
EngUnits	Conversion Factor
Units / day	1
Units / hour	24
Units / minute	1,440
Units / second	86,400

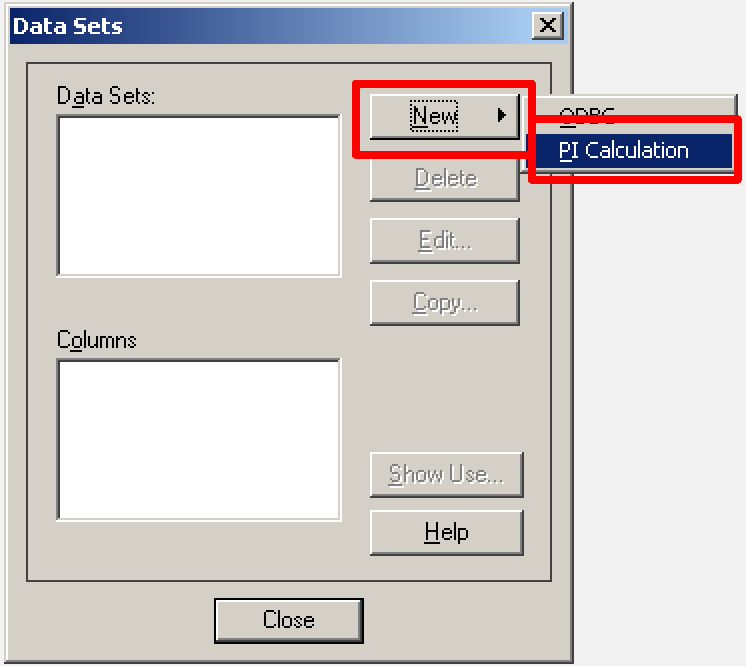
CONVERSION FACTOR

- Example
  - The units of WATER_FLOW is m^3/h
  - The conversion factor will be 24
- Yesterday Water Usage (m^3) will be

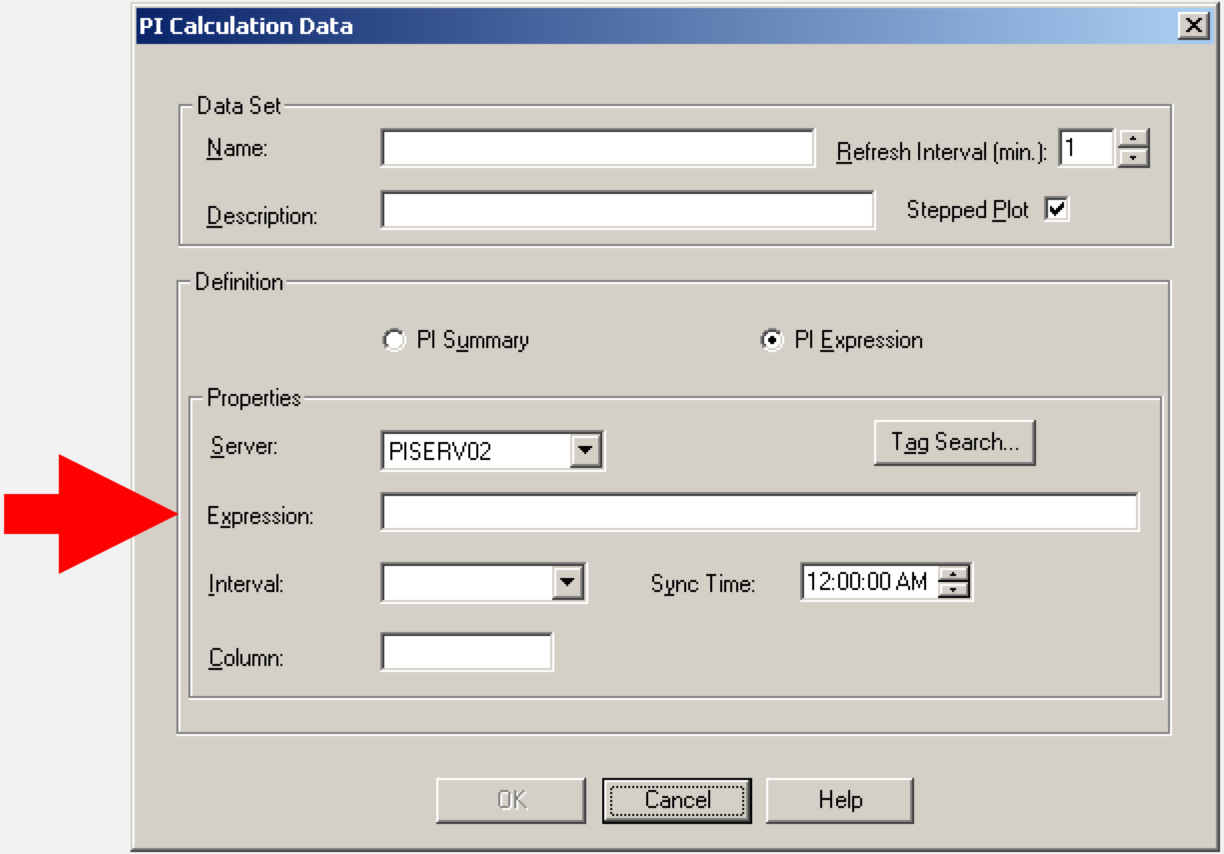
  TagTot('WATER_FLOW', 'Y', 'T') * 24

ADDING A CALCULATION





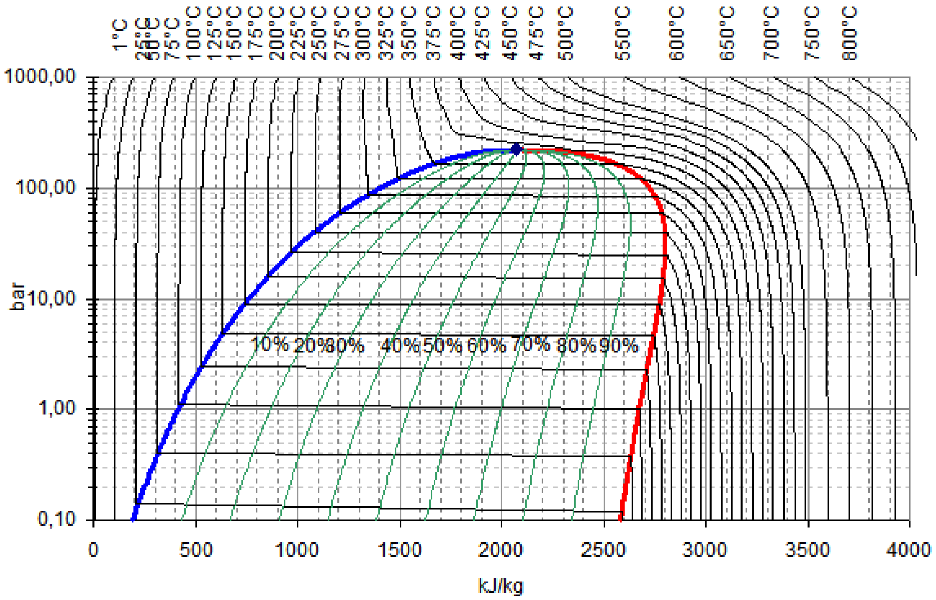
CALCULATION EXPRESSION



STEAM TABLE

See Steam Table functions in

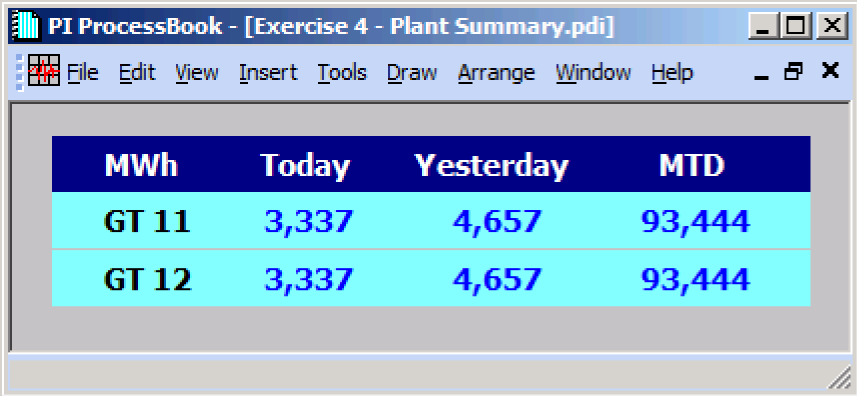
PISteamTable.chm

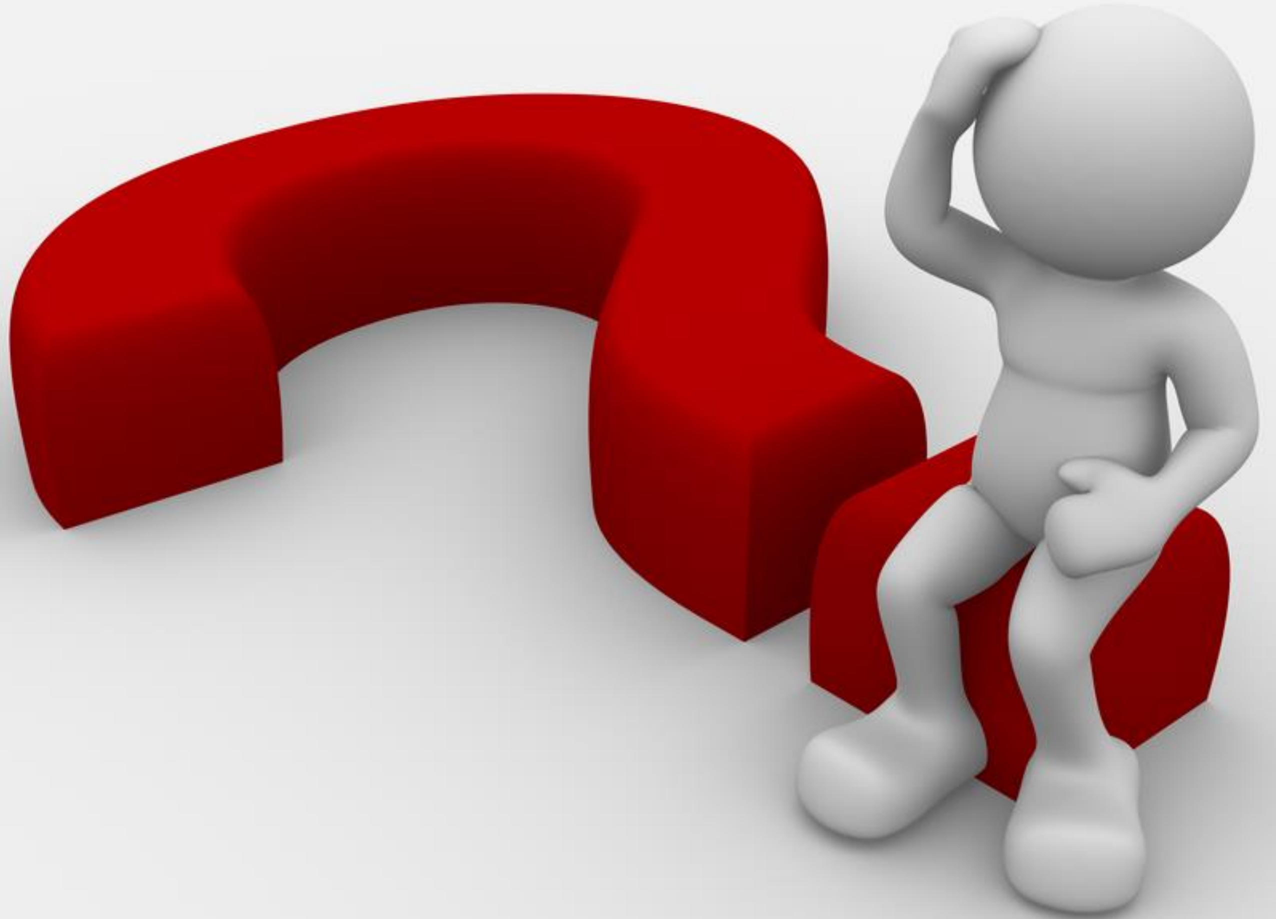


Function	Output	Input
TsatP	Temp.	Pressure
HsatP	Enthalpy	Pressure
SsatP	Entropy	Pressure
VsatP	Volume	Pressure
PsatP	Pressure	Pressure
HsatP	Enthalpy	Pressure
SsatP	Entropy	Pressure
VsatP	Volume	Pressure
VPT, VPTL	Volume	Pressure, Temp.
VPH	Volume	Pressure, Enthalpy
VPS	Volume	Pressure, Entropy
HPT, HPTL	Enthalpy	Pressure, Temp
HPS	Enthalpy	Pressure, Entropy
SPT, SPTL	Entropy	Pressure, Temp.
SPH	Entropy	Pressure, Enthalpy
TPH	Temp.	Pressure, Enthalpy
TPS	Temp.	Pressure, Entropy
XPH	Quality	Pressure, Enthalpy
XPS	Quality	Pressure, Entropy
HPX	Enthalpy	Pressure, Quality
SPX	Entropy	Pressure, Quality

EXCERCISE 2.4 : PLANT SUMMARY

- Build a display with PI calculations

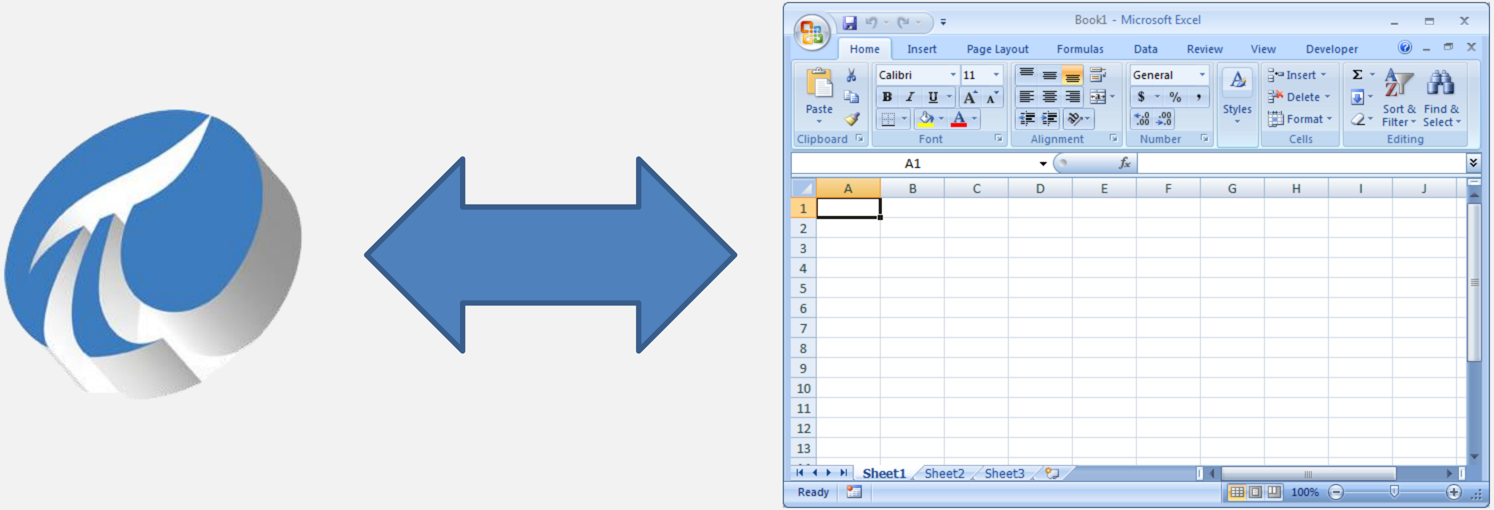




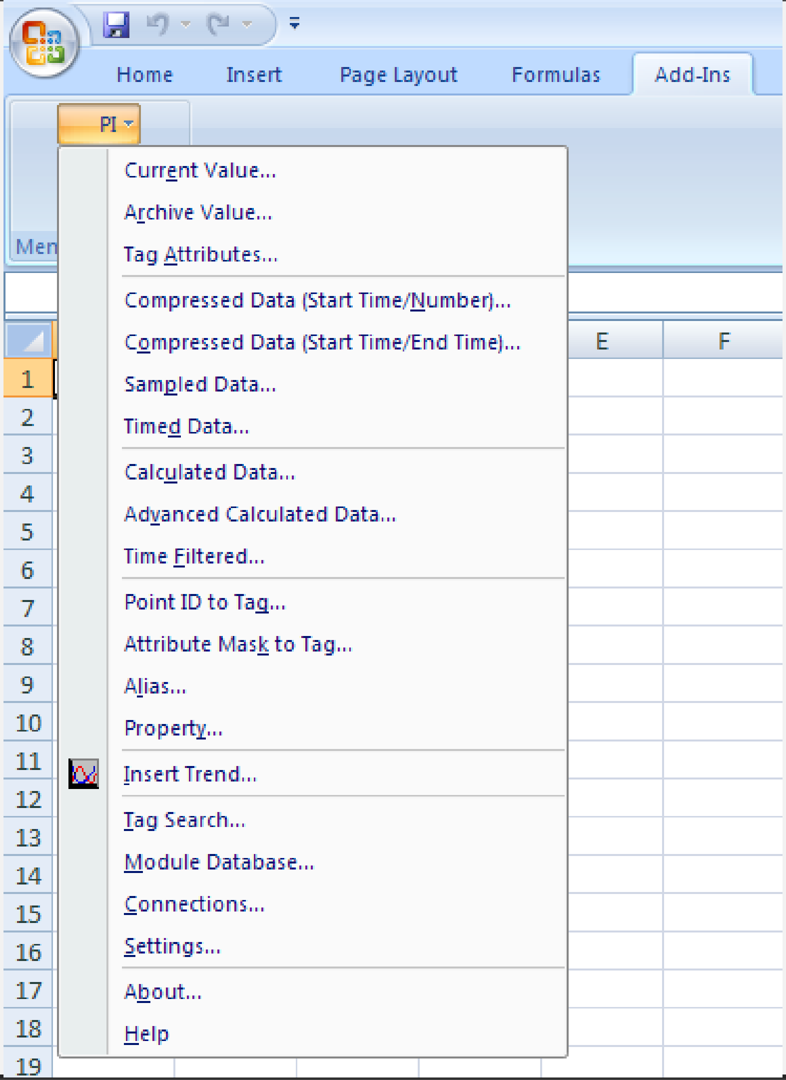
3. PI DataLink

WHAT IS PI DATALINK?

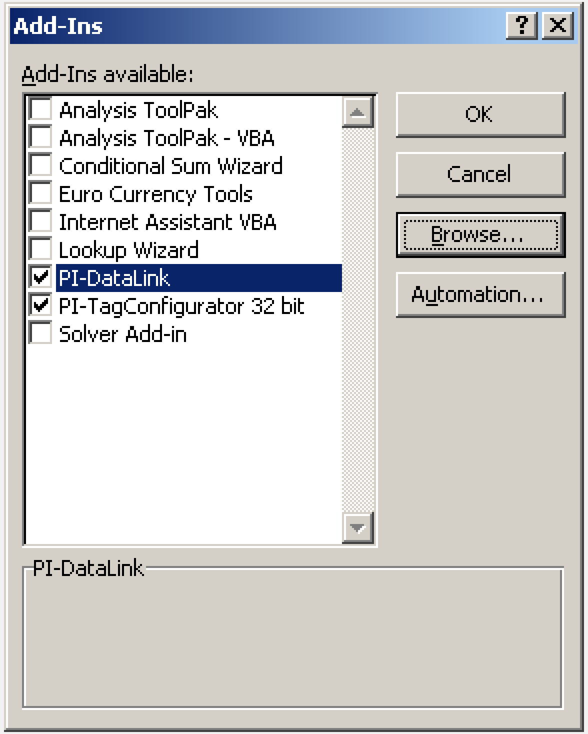
- PI DataLink provides a link between the PI Server and Microsoft Excel.



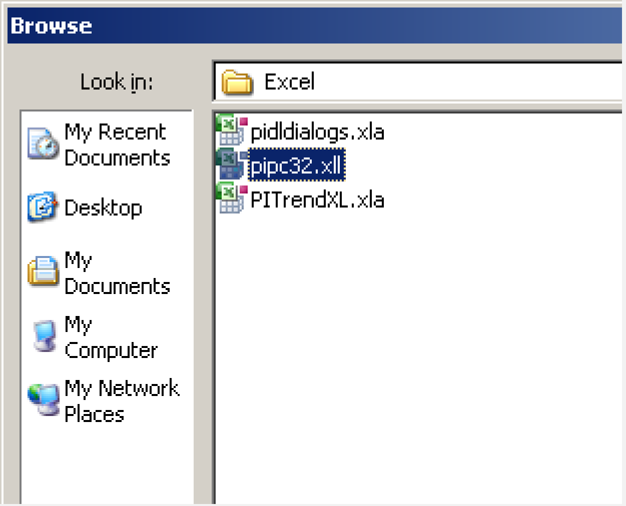
PI DATALINK MENU



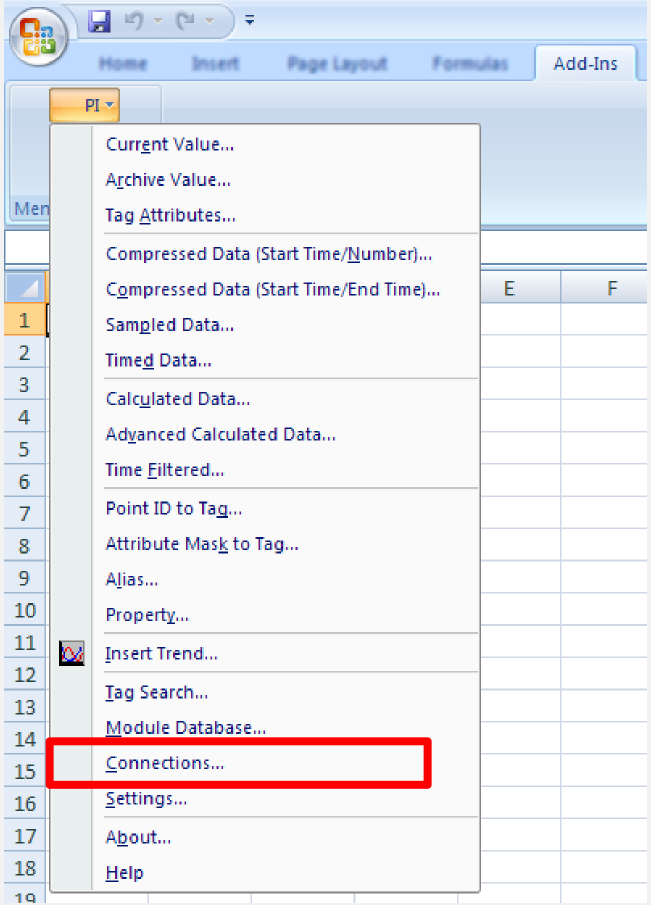
REGISTERING PI DATALINK ADD-IN

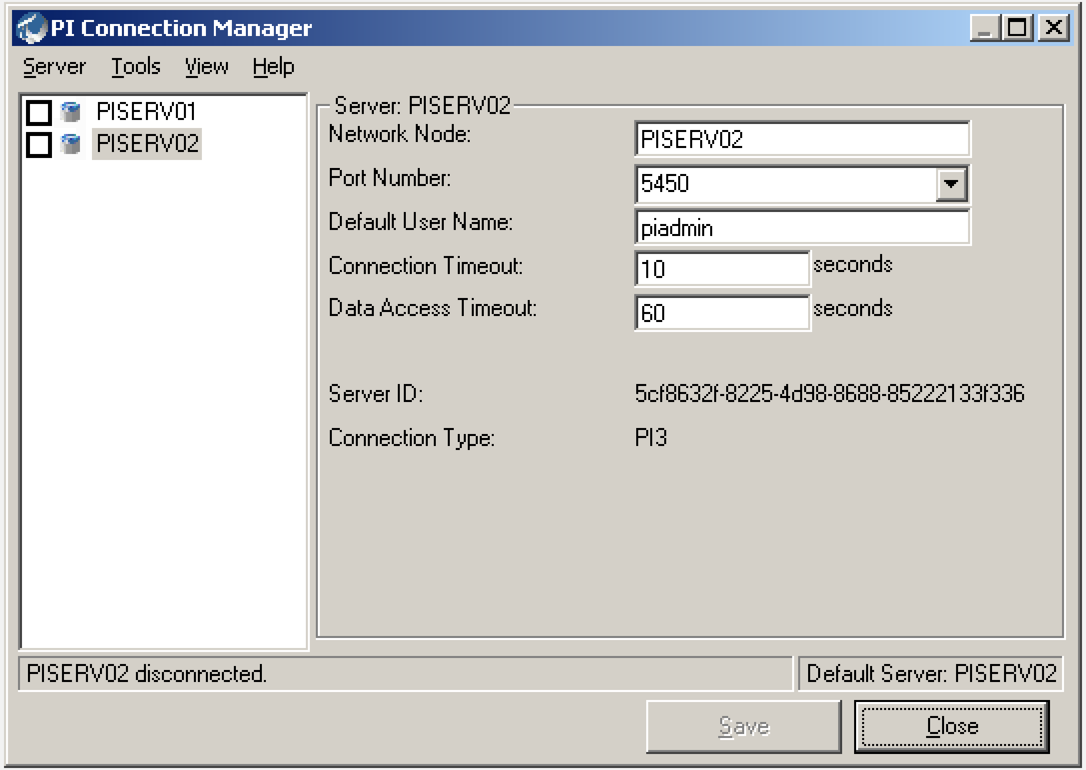


.\PIPC\Excel\pipc.xll

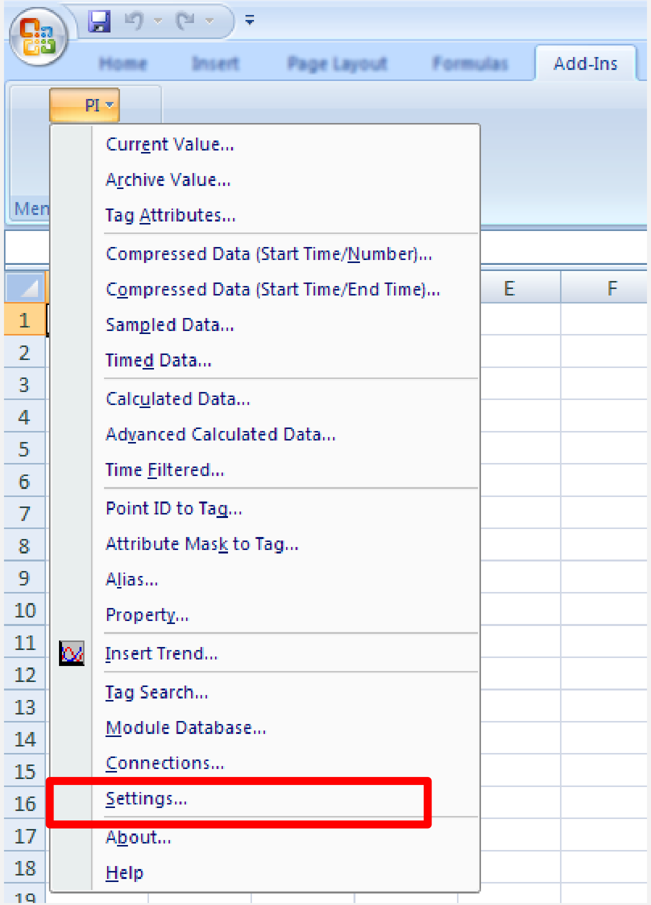


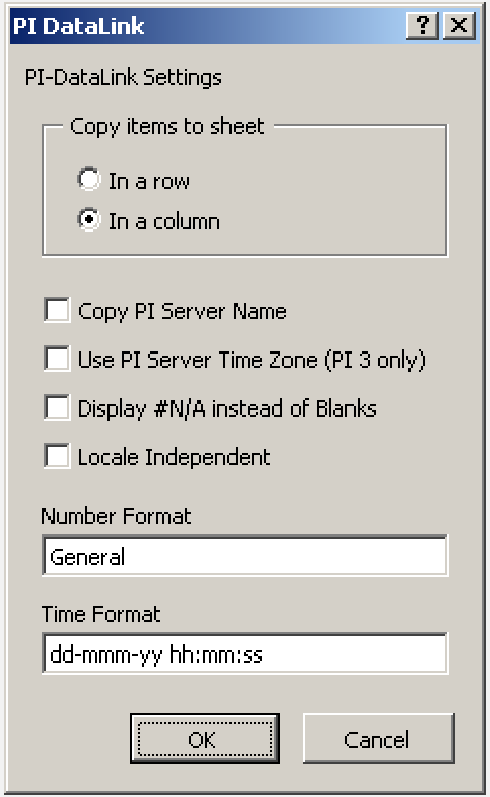
PI SERVER CONNECTION



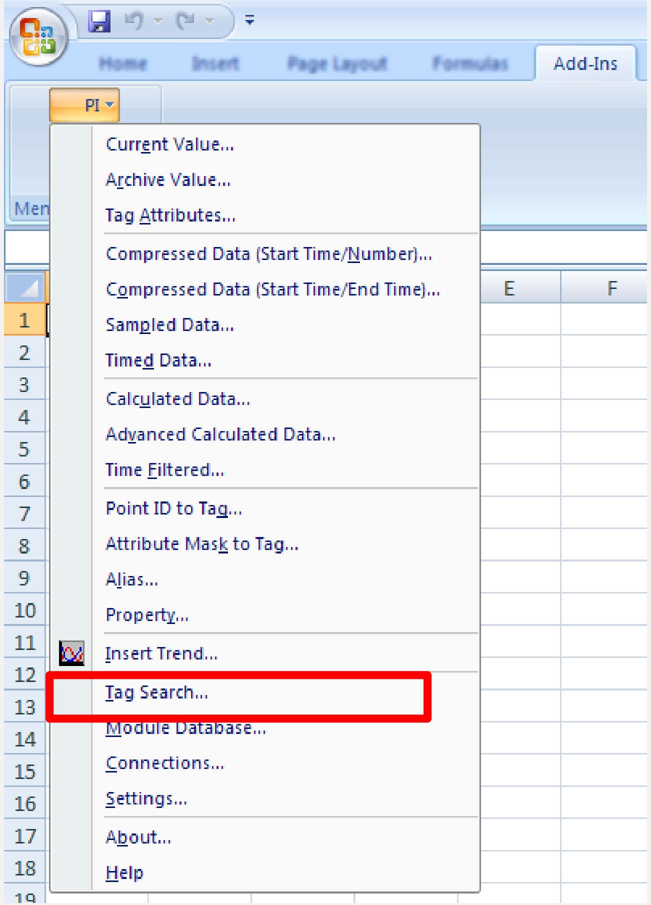


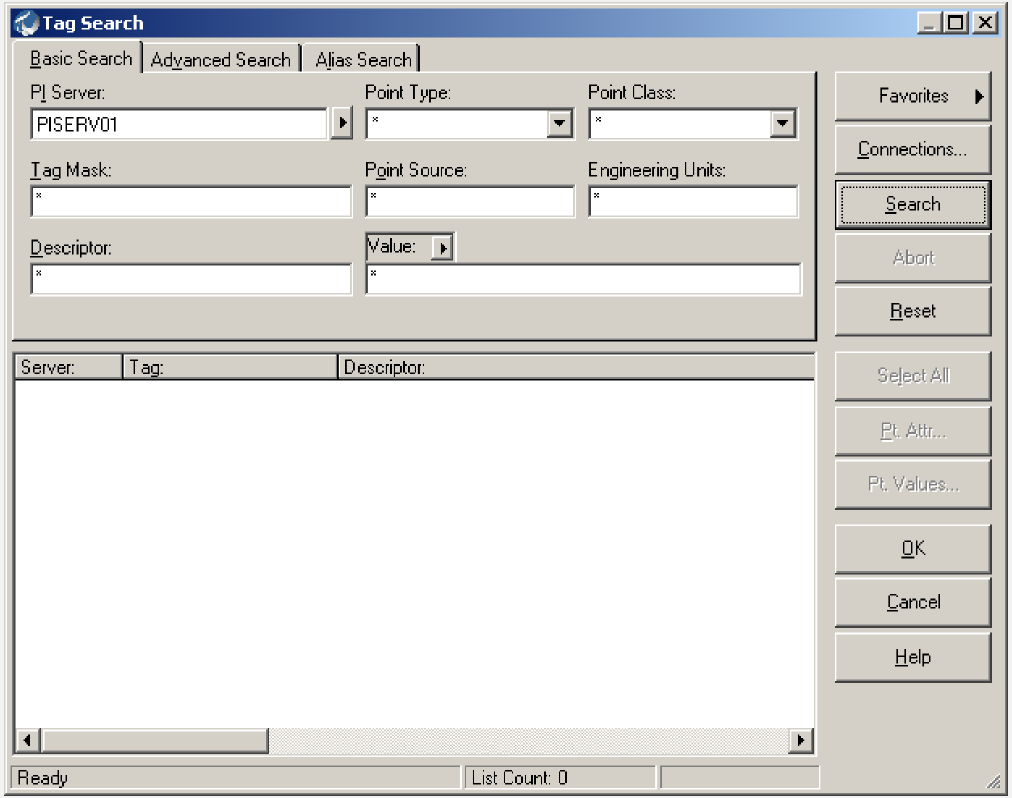
SETTINGS


Home
Insert
Page Layout
Formulas
Add-Ins
PI
Current Value...
Archive Value...
Tag Attributes...
Compressed Data (Start Time/Number)...
Compressed Data (Start Time/End Time)...
Sampled Data...
Timed Data...
Calculated Data...
Advanced Calculated Data...
Time Filtered...
Point ID to Tag...
Attribute Mask to Tag...
Alias...
Property...
Insert Trend...
Tag Search...
Module Database...
Connections...
Settings...
About...
Help

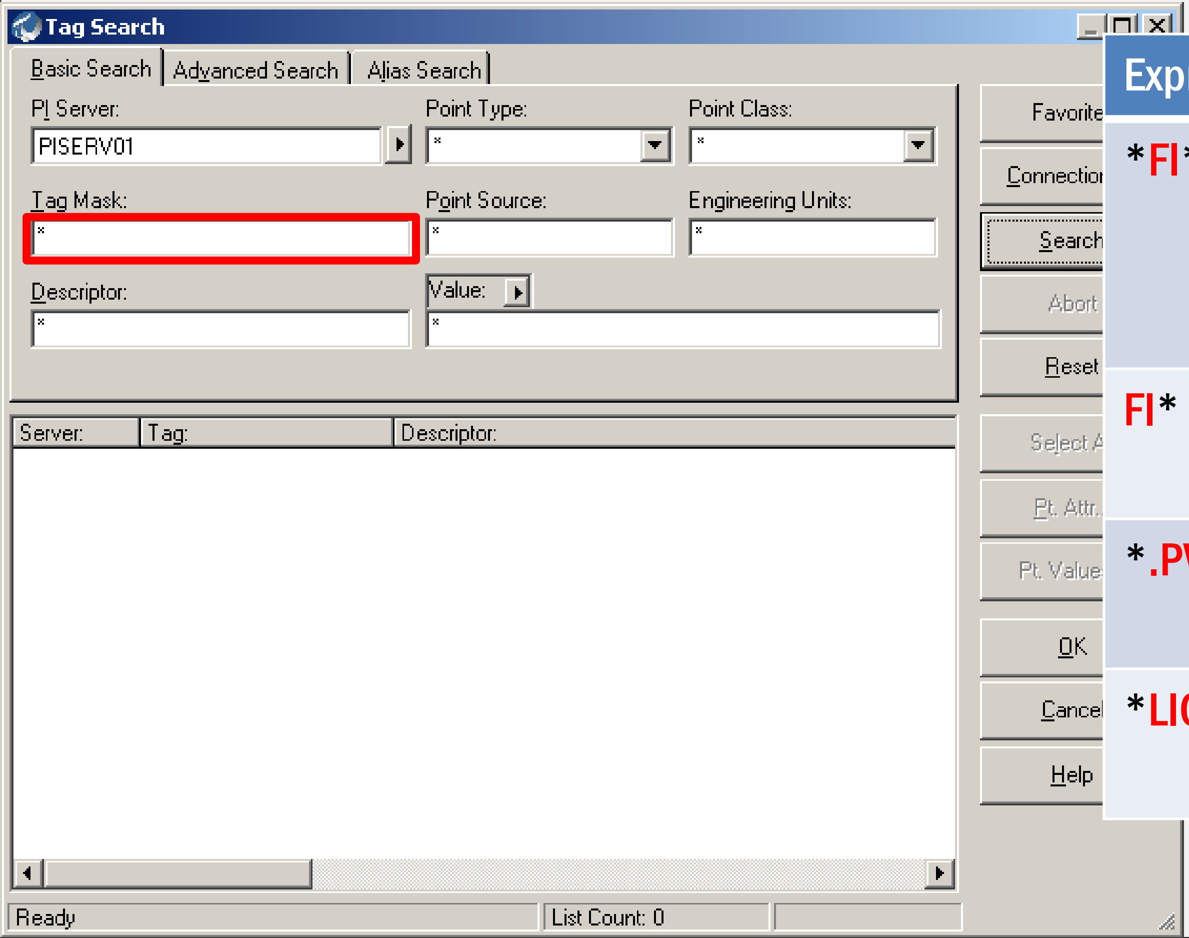

PI DataLink
PI-DataLink Settings
Copy items to sheet
In a row
In a column
Copy PI Server Name
Use PI Server Time Zone (PI 3 only)
Display #N/A instead of Blanks
Locale Independent
Number Format
General
Time Format
dd-mmm-yy hh:mm:ss
OK
Cancel

TAG SEARCH


Home
Insert
Page Layout
Formulas
Add-Ins
PI
Current Value...
Archive Value...
Tag Attributes...
Compressed Data (Start Time/Number)...
Compressed Data (Start Time/End Time)...
Sampled Data...
Timed Data...
Calculated Data...
Advanced Calculated Data...
Time Filtered...
Point ID to Tag...
Attribute Mask to Tag...
Alias...
Property...
Insert Trend...
Tag Search...
Module Database...
Connections...
Settings...
About...
Help
E
F


Tag Search
Basic Search
Advanced Search
Alias Search
PI Server:
PISERV01
Point Type:
Point Class:
Tag Mask:
Point Source:
Engineering Units:
Descriptor:
Value:
Favorites
Connections...
Search
Abort
Reset
Select All
Pt. Attr...
Pt. Values...
OK
Cancel
Help
Server:
Tag:
Descriptor:
Ready
List Count: 0

TAG SEARCH CRITERIA



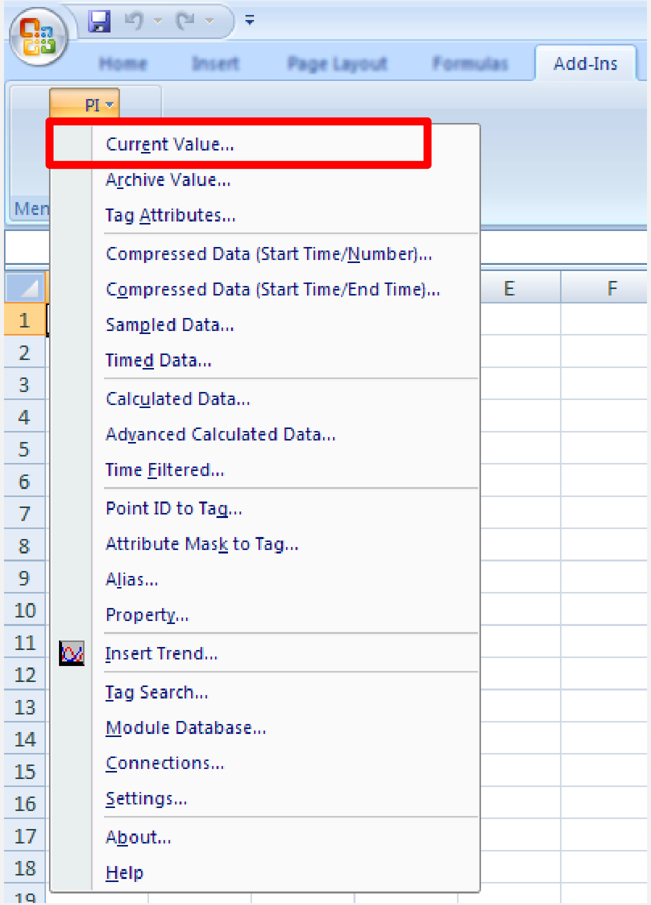
Expression	Meaning
FI	(Any) + FI + (Any) ▪ 10FI001.PV ▪ FI555.PV ▪ 555.FI
FI*	FI + (Any) ▪ FI123.PV
*.PV	(Any) + .PV ▪ 10TI003.PV
LIC.MV	(Any) + LIC + (Any) + .MV ▪ 10LIC01.MV

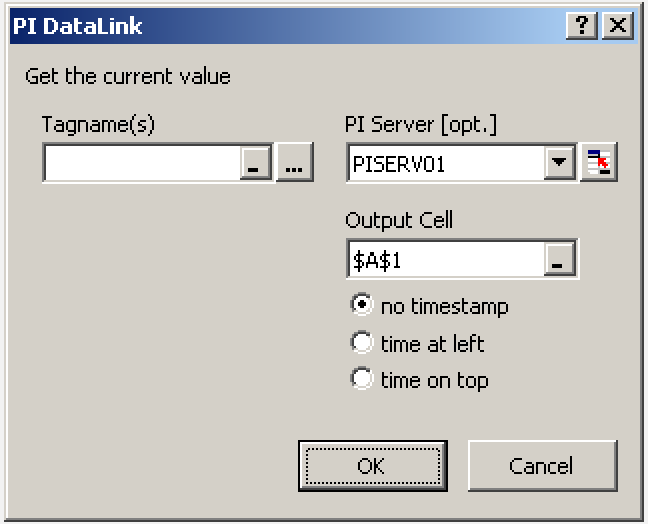
COMMON ARGUMENTS

Argument	Description	Example
PI Server	PI Server	PISERV01, PISERV02
Tagname	Tag Name or AF Attribute	10PI01.PV
Expression	PI Calculation	TagAvg('10PI01.PV','Y','T')
Timestamp Start Time & End Time	PI Time or Excel Time	*- 8h 15-Apr-15 12:13:14
Time Interval	Split Output to Interval	1h, 1d, 1w
Filter Expression	For Filter Data	'Tag' > 10
Output Cell	Output Cell Location	B2, C3

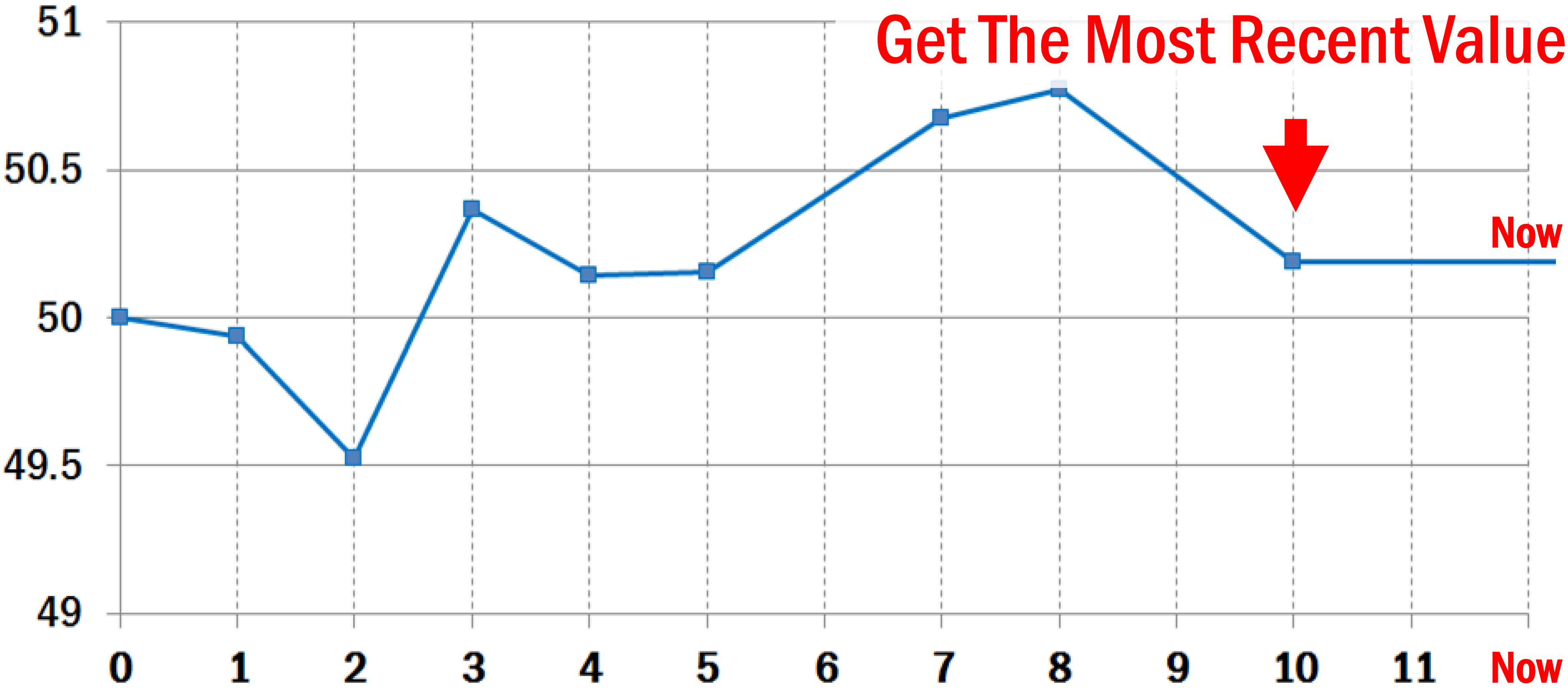
SAT Solutions

CURRENT VALUE

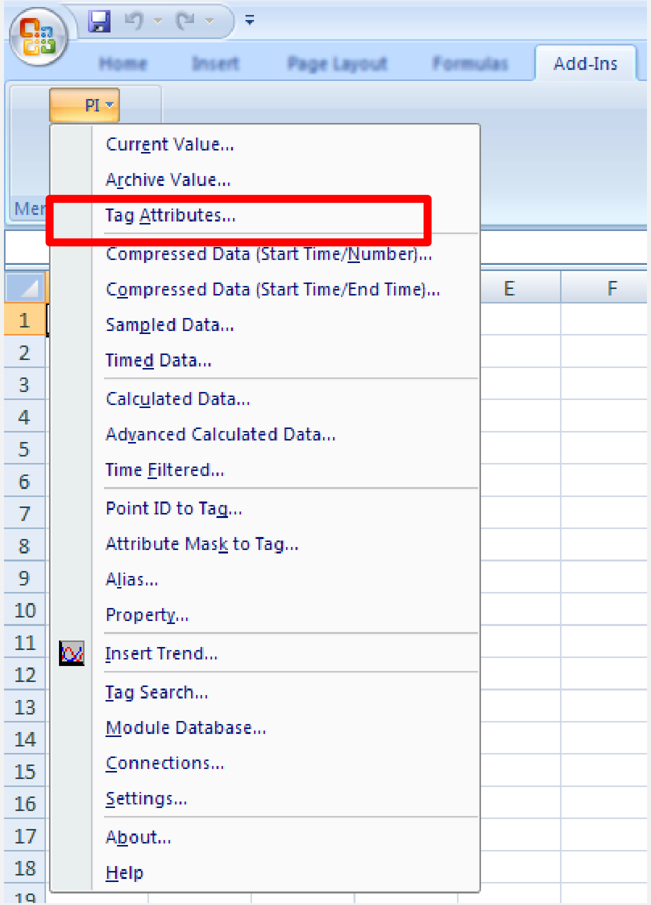


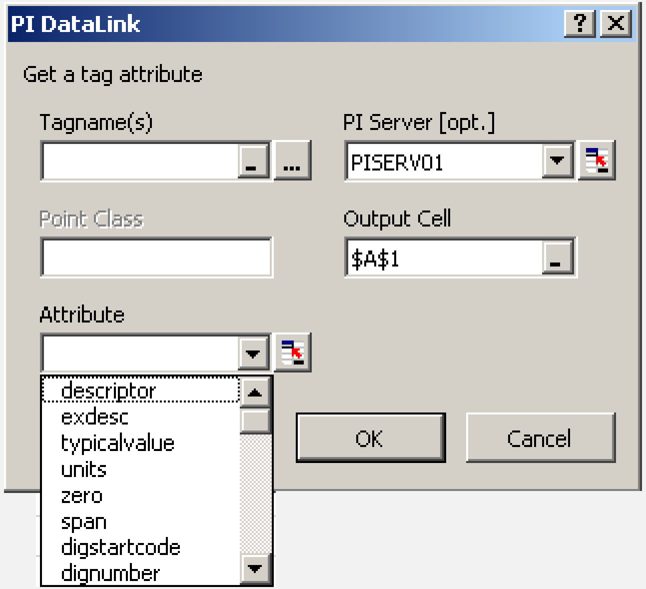


CURRENT VALUE



TAG ATTRIBUTES





EXERCISE 3.1 : REAL-TIME PLANT OPERATION

- Build a report with Current Value Function

EX 1 : Real-Time Plant Operation

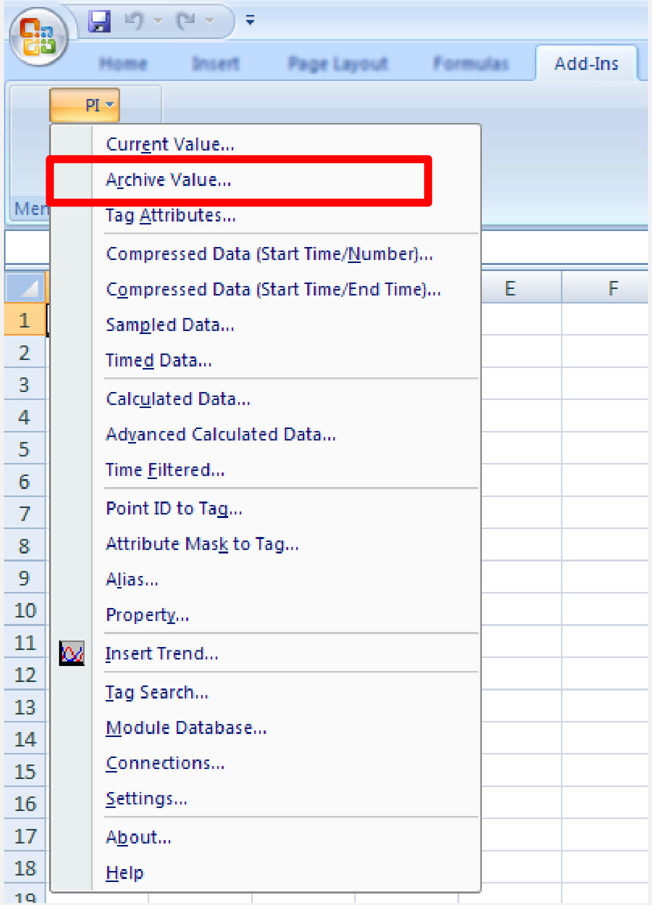
**** Use Current Value function to retrieve the Most Recent Value.*

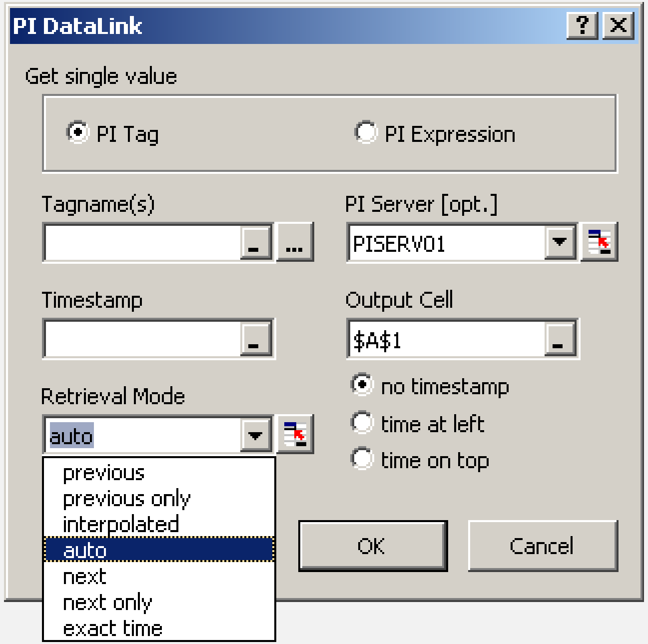
Tag	Timestamp	Value	Units	Descriptor

SAT Solutions

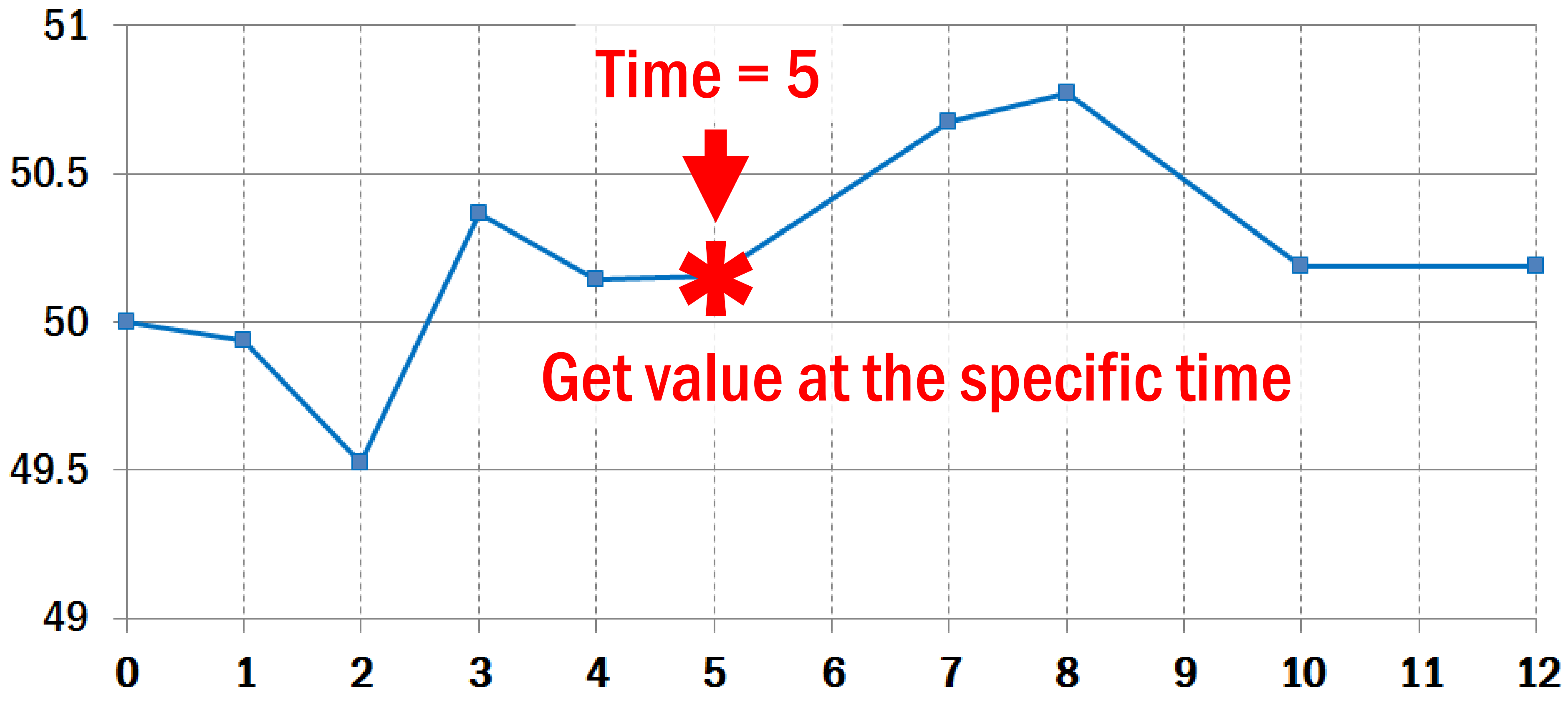
ARCHIVE VALUE

Get value at the specific time

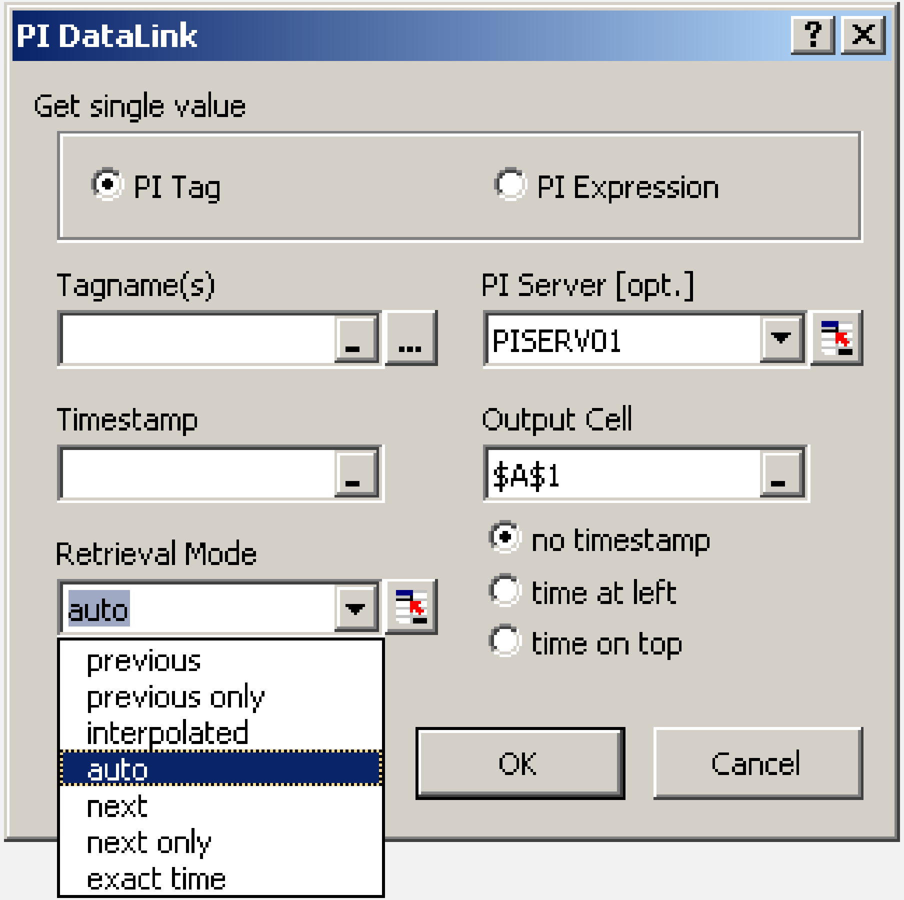




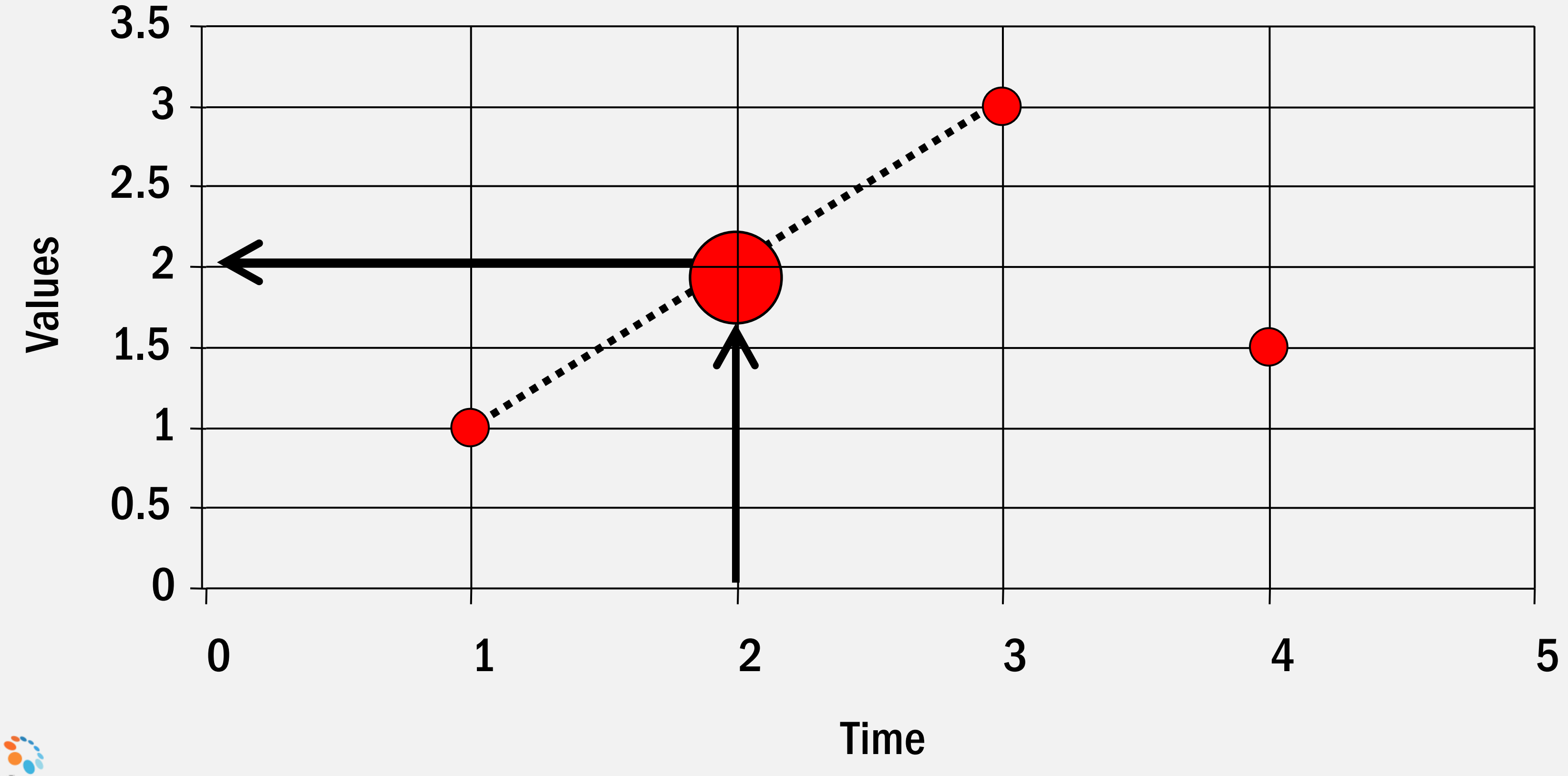
ARCHIVE VALUE



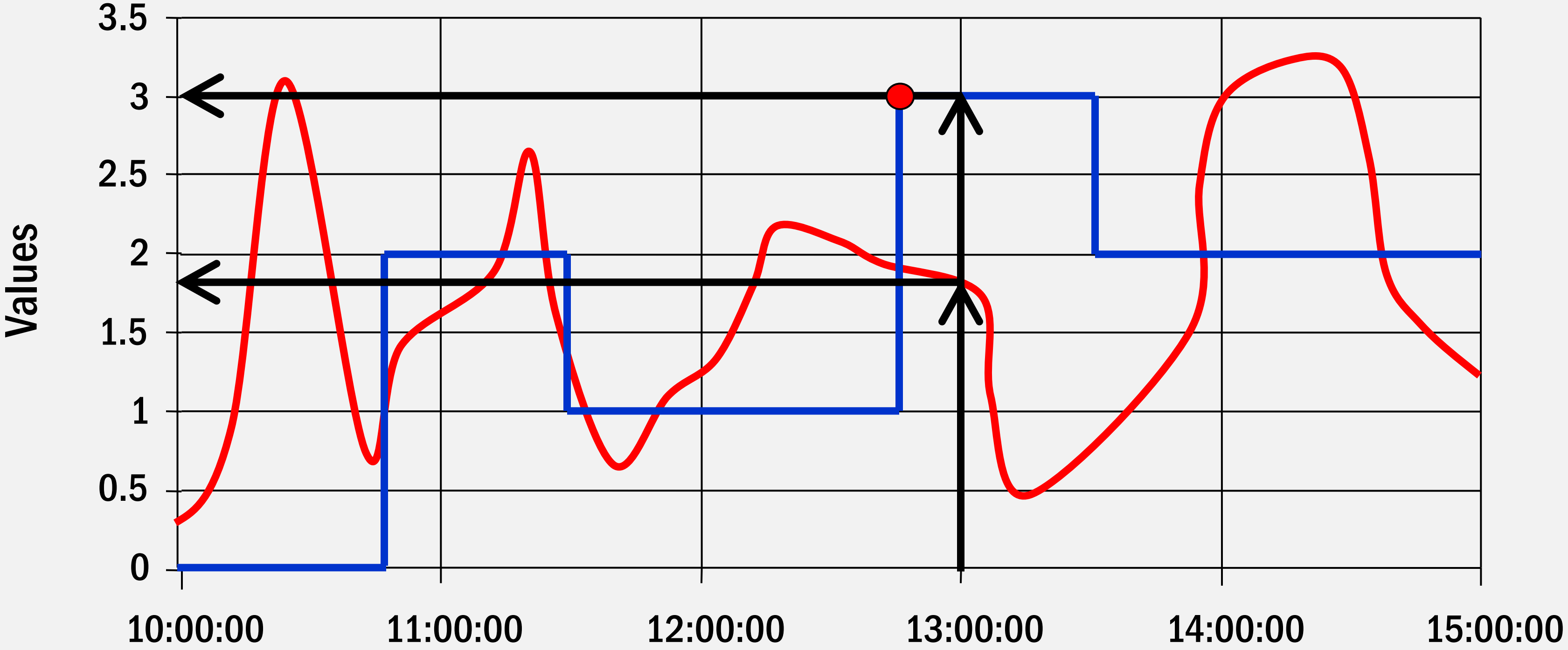
ARCHIVE VALUE



INTERPOLATED

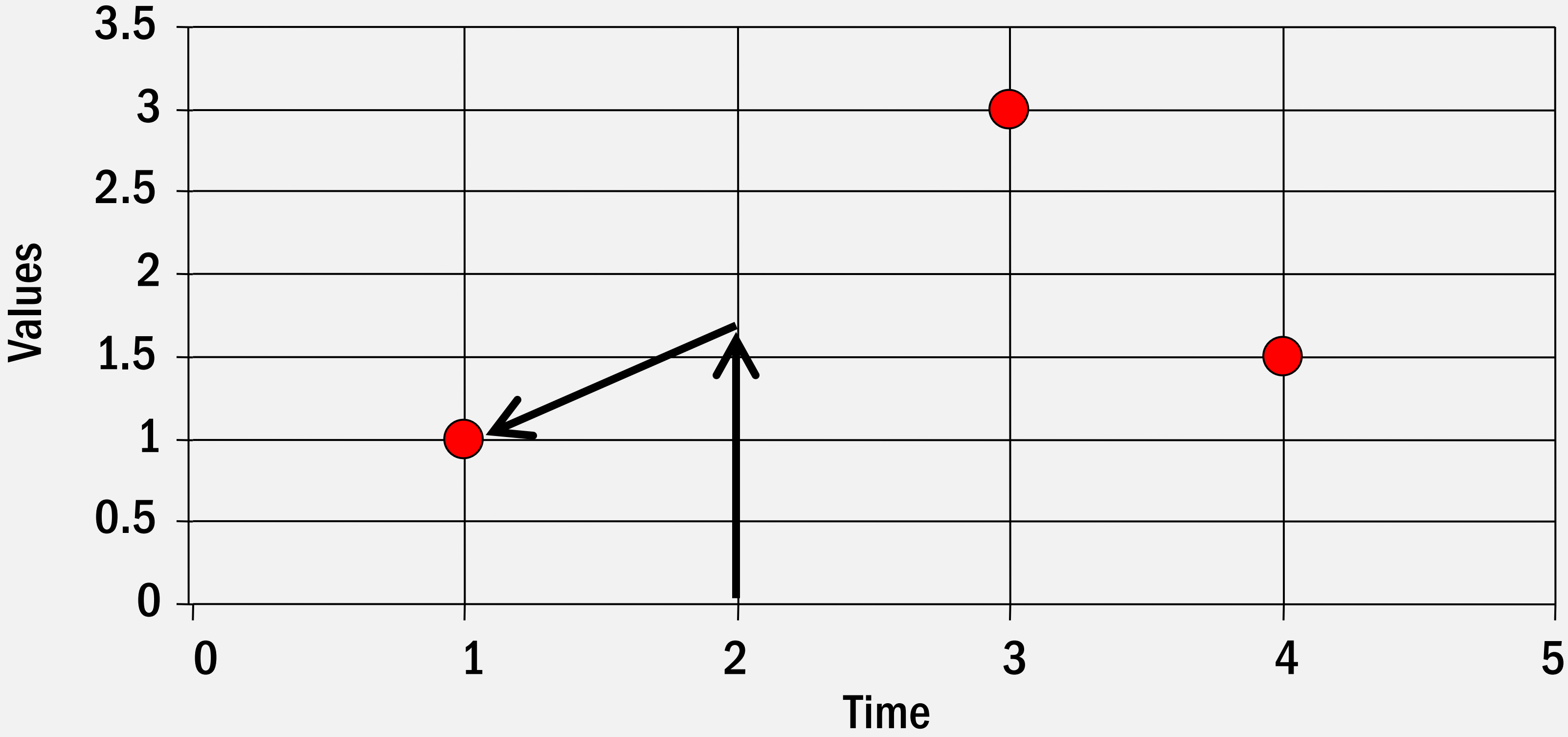


INTERPOLATED VS AUTO

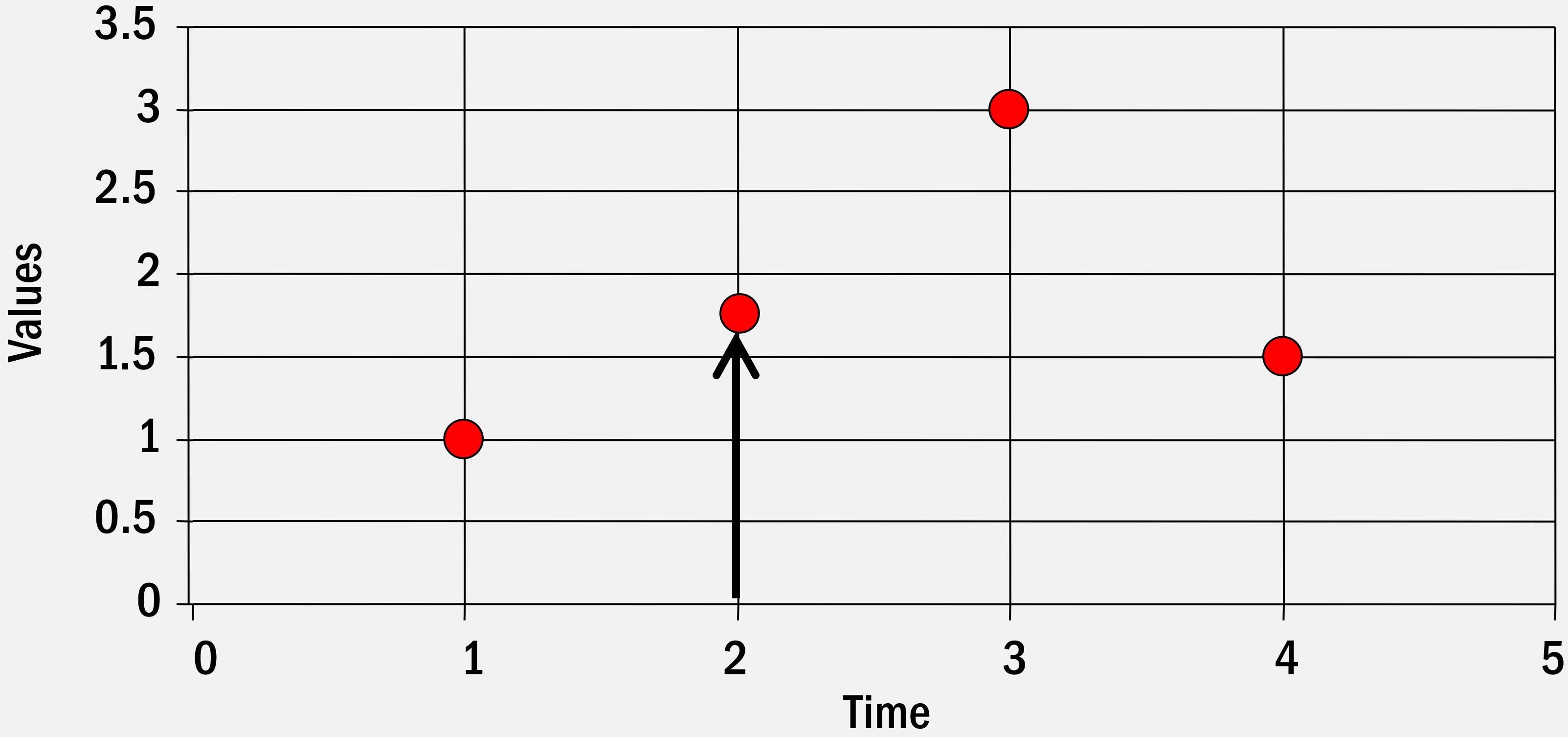


	Interpolated	Auto
Red Line (Analog)	13:00:00 - 1.81	13:00:00 - 1.81
Blue Line (Digital)	13:00:00 - 3	12:47:45 - 3

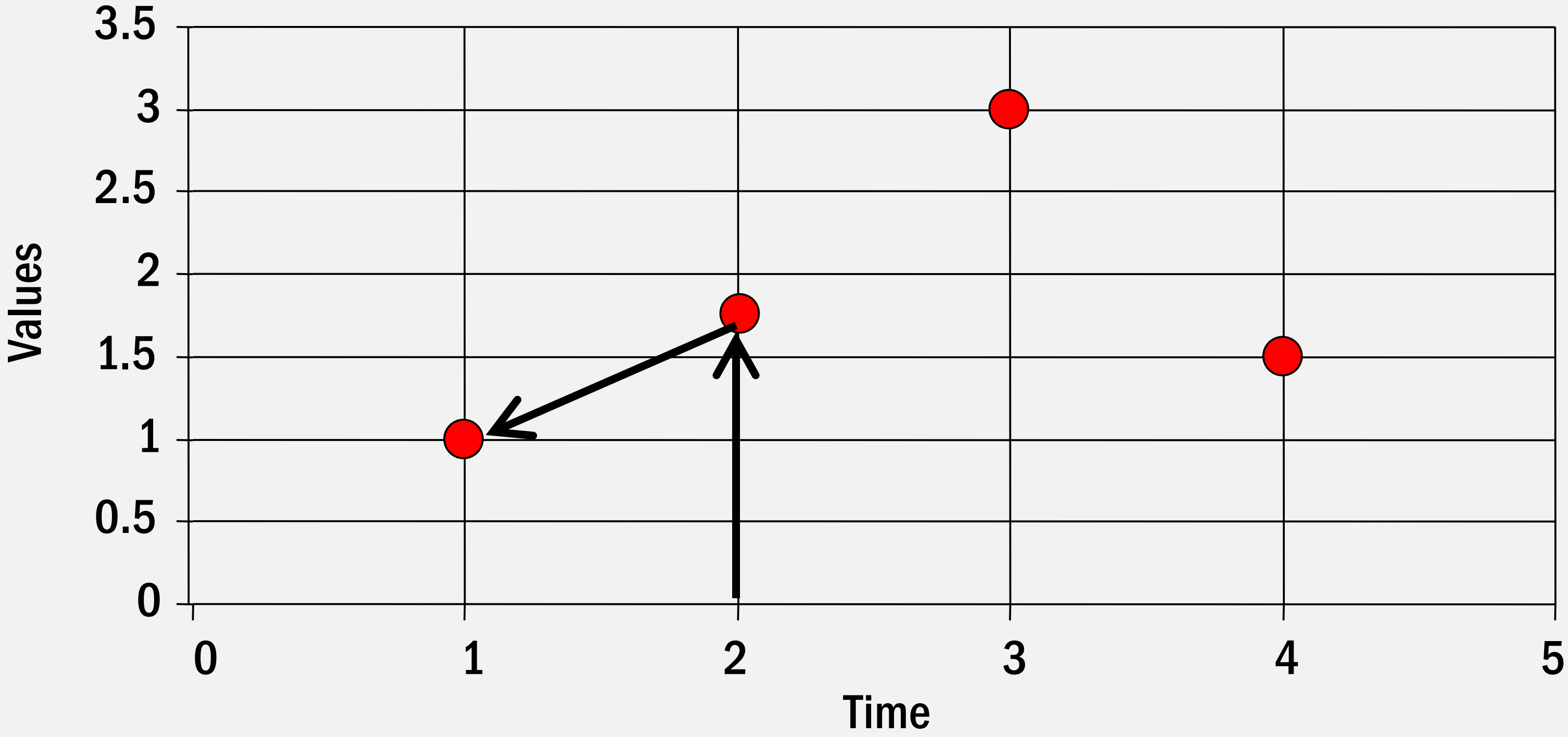
PREVIOUS & PREVIOUS ONLY (NO EVENT)



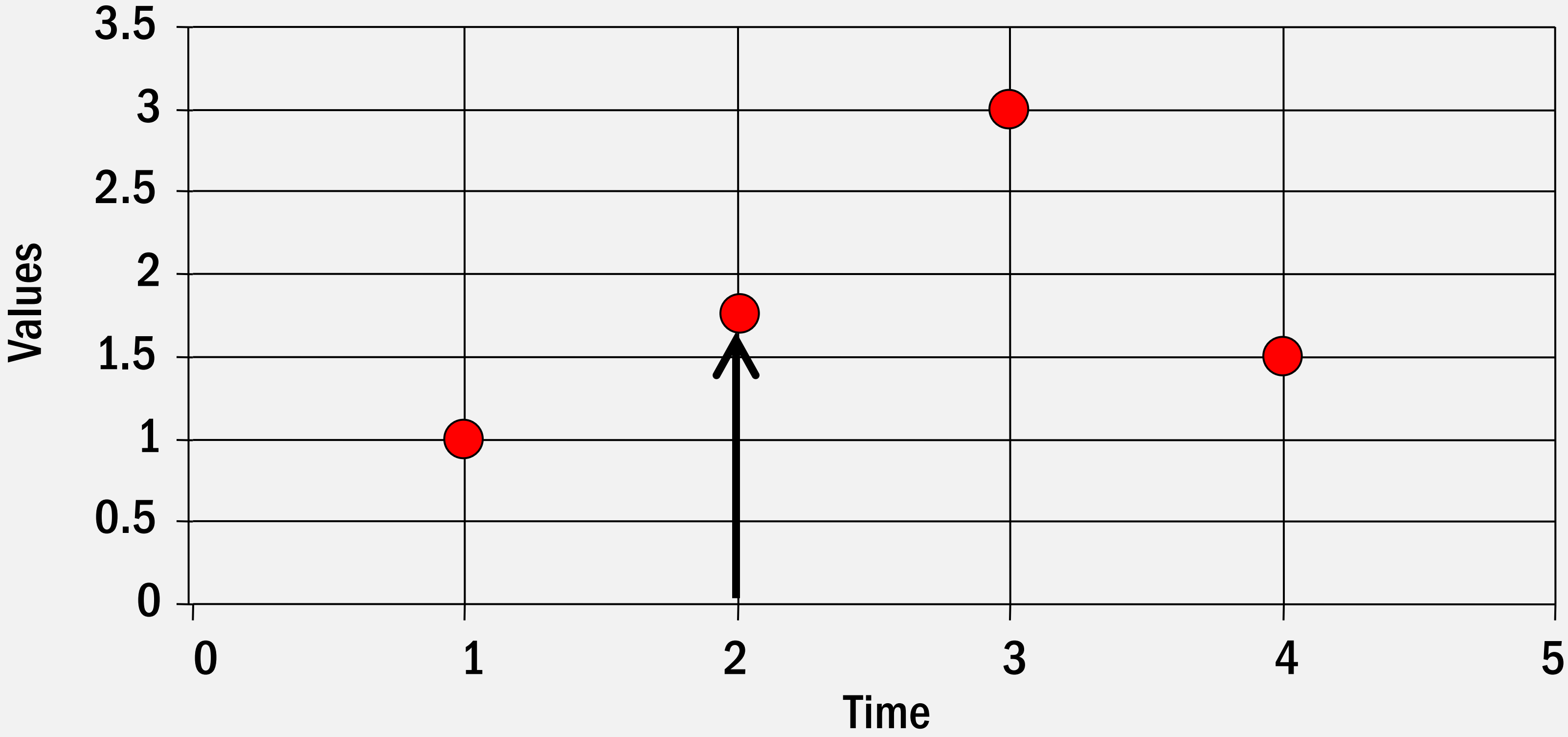
PREVIOUS (EVENT AVAILABLE)



PREVIOUS ONLY (EVENT AVAILABLE)



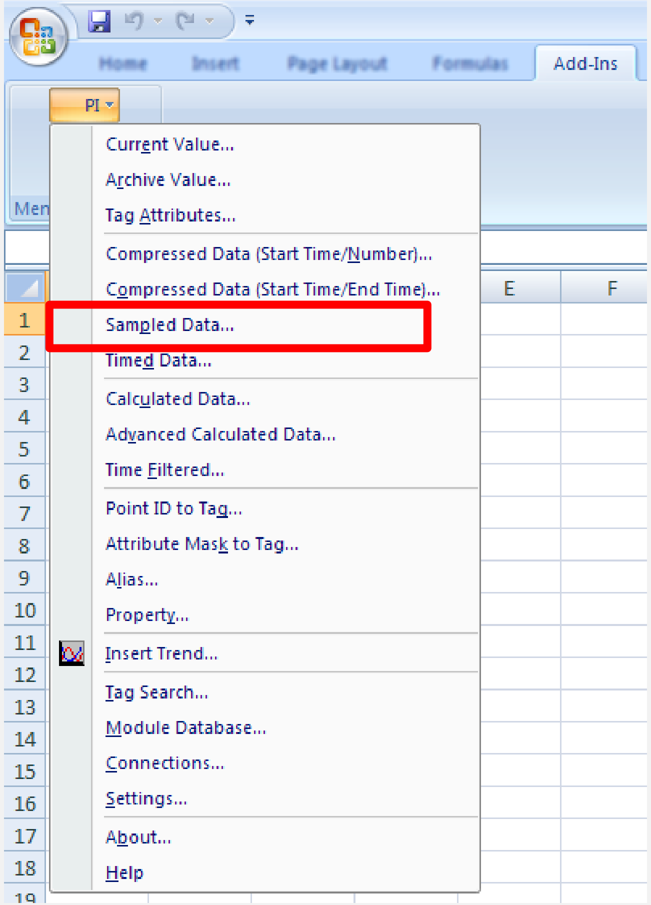
EXACT TIME (EVENT AVAILABLE)

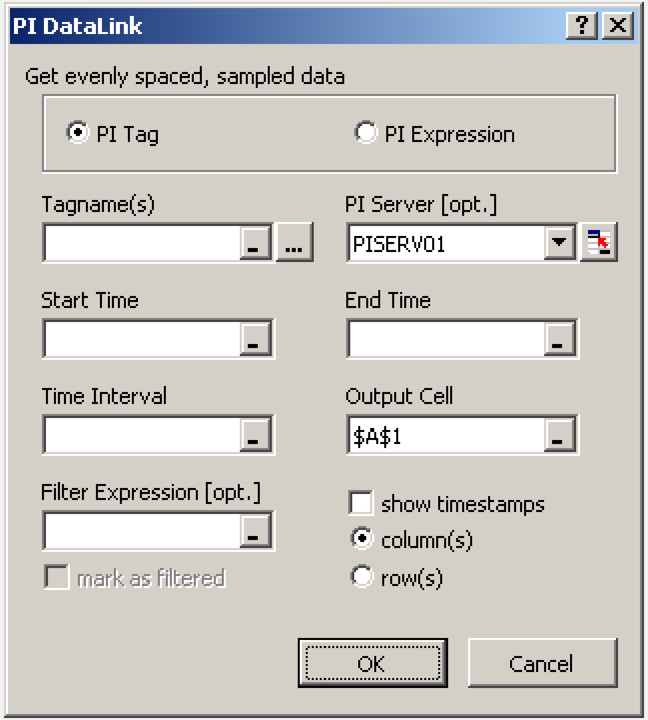


EXACT TIME (NO EVENT FOUND)

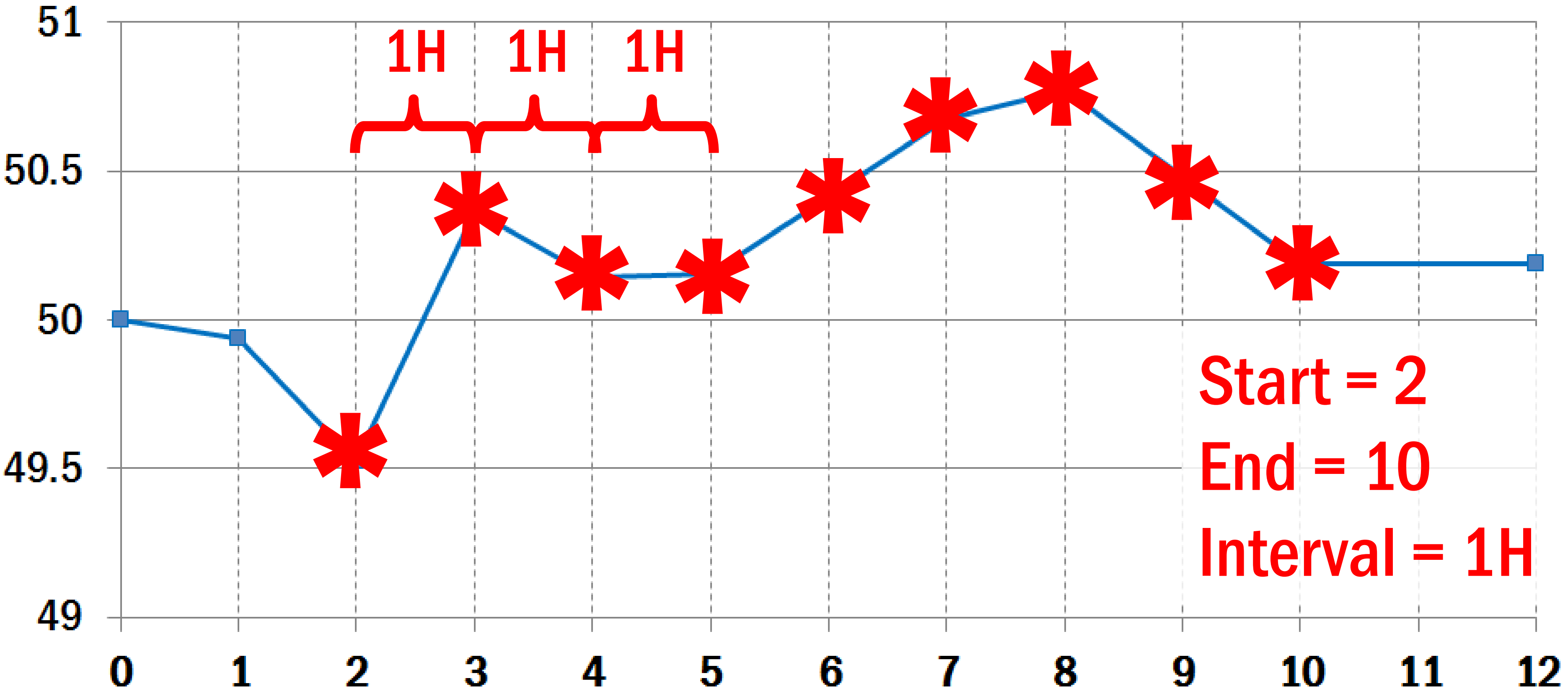
SAT Solutions

SAMPLED DATA

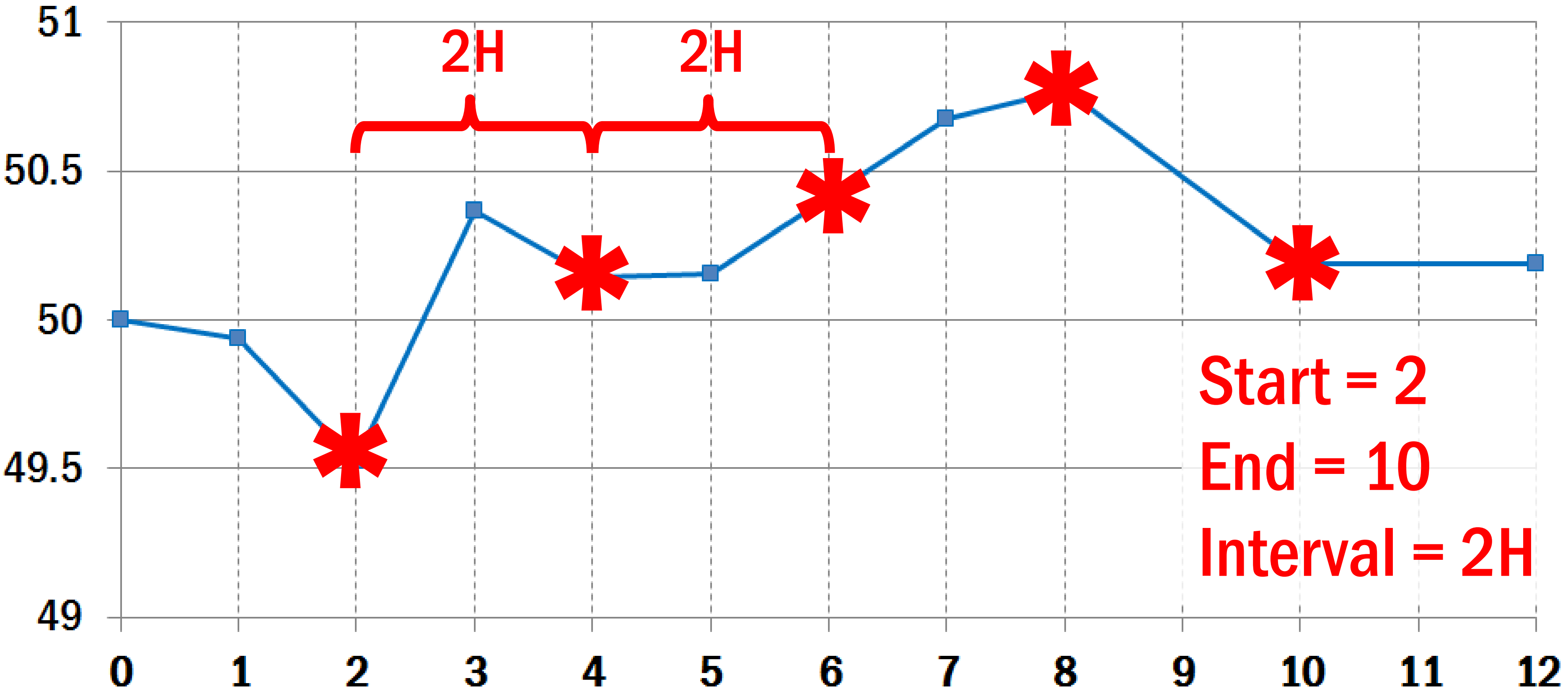




SAMPLED DATA



SAMPLED DATA



EXERCISE 3.2 – WATER LEVEL

- Build an Excel report with Archive Value and Sampled Data Function

EX 2 : Water Level

**** Use **Archive Value** to get a value at **a Specified Time**.*

**** Use **Sampled Data** function to get **Time-Interval Data**.*

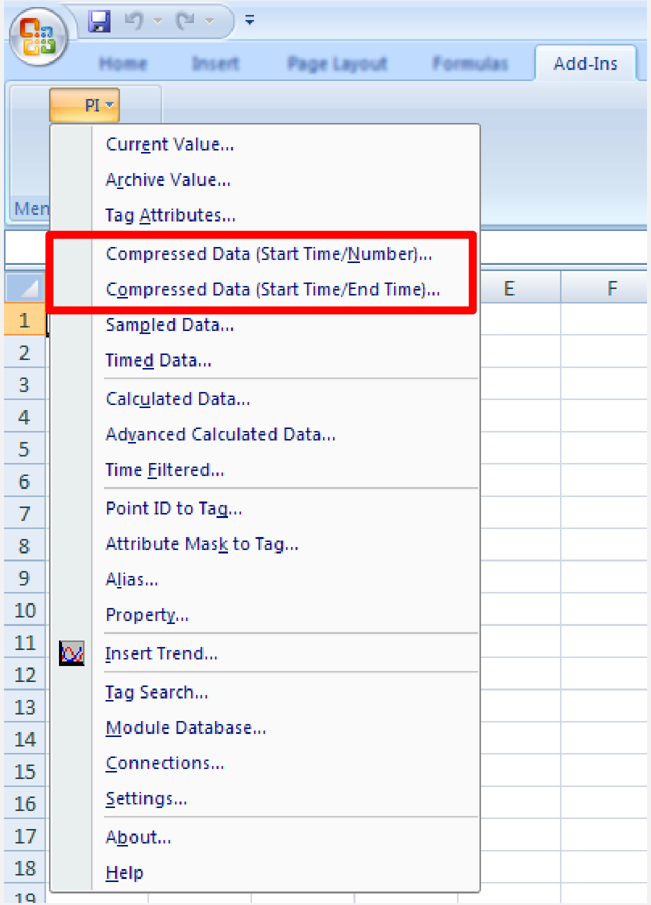
Start Time	
End Time	
Time Interval	

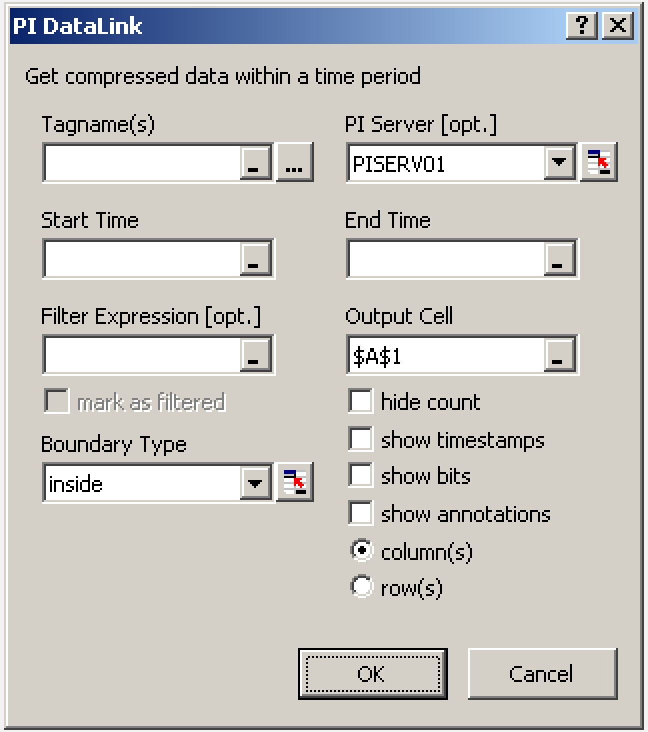
Tag	
Function	Archive Value
Timestamp	Value

Tag	
Function	Sampled Data
Timestamp	Value

SAT Solutions

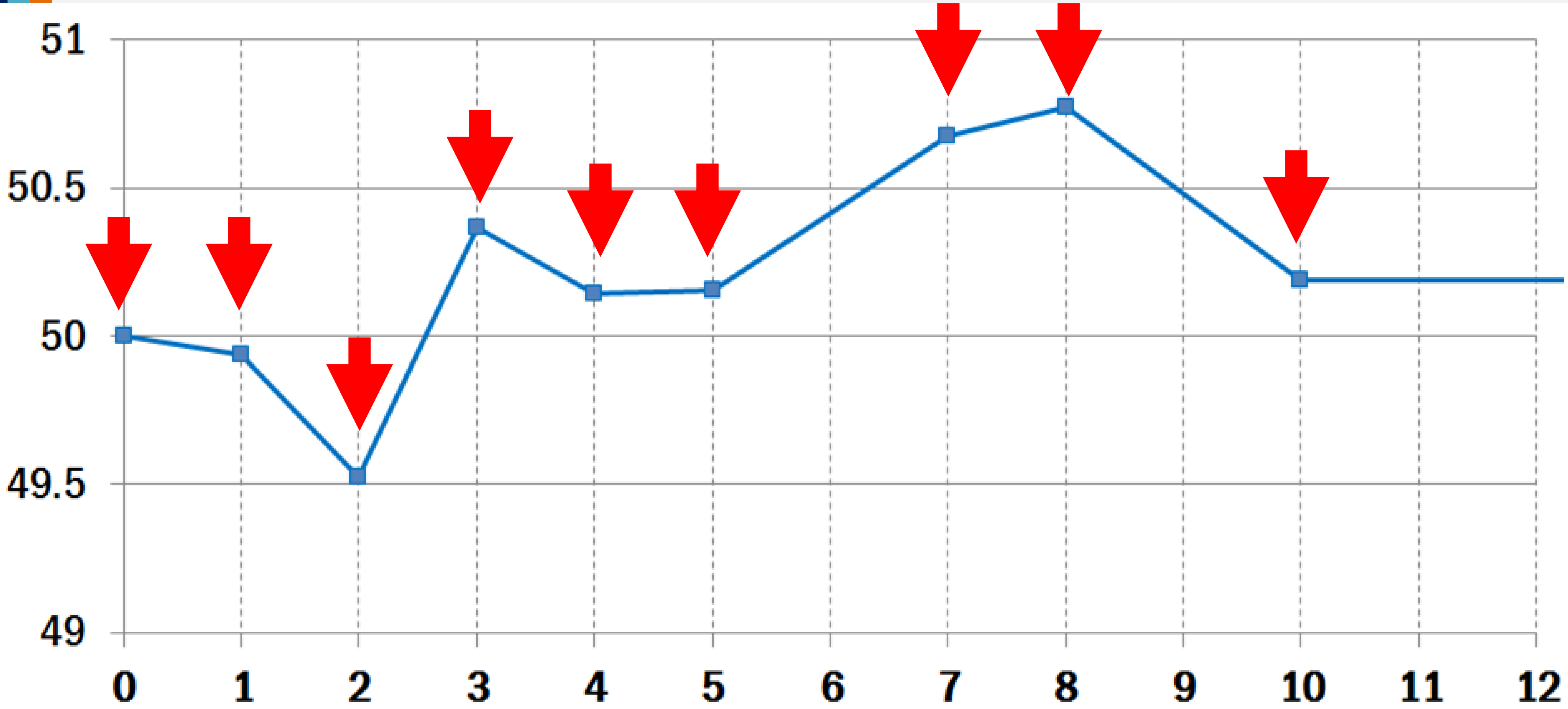
COMPRESSED DATA





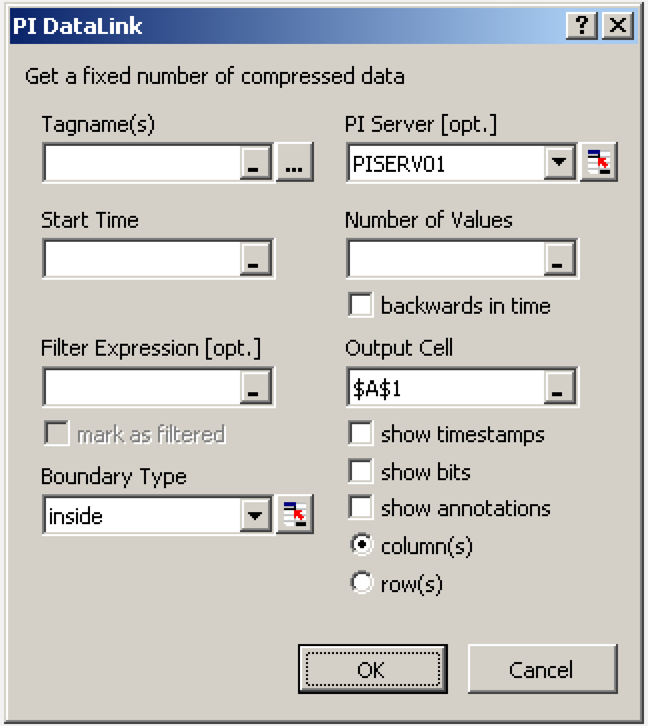
SAT Solutions

COMPRESSED DATA

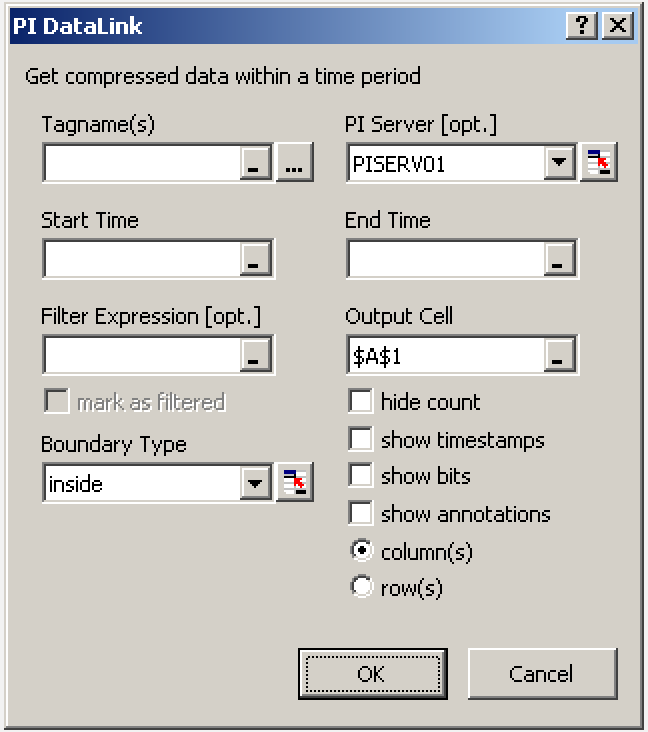


COMPRESSED DATA

No. of Values



Start/End Time



SAT Solutions

EXERCISE 3.3 – RAW EVENT DATA

- Build a Excel report with Compressed Data Function

EX 3 : Raw Event Data

**** Use Compressed Data function to retrieve raw event data by Time Range .*

**** Use Compressed Data function to retrieve raw event data by Number of Values .*

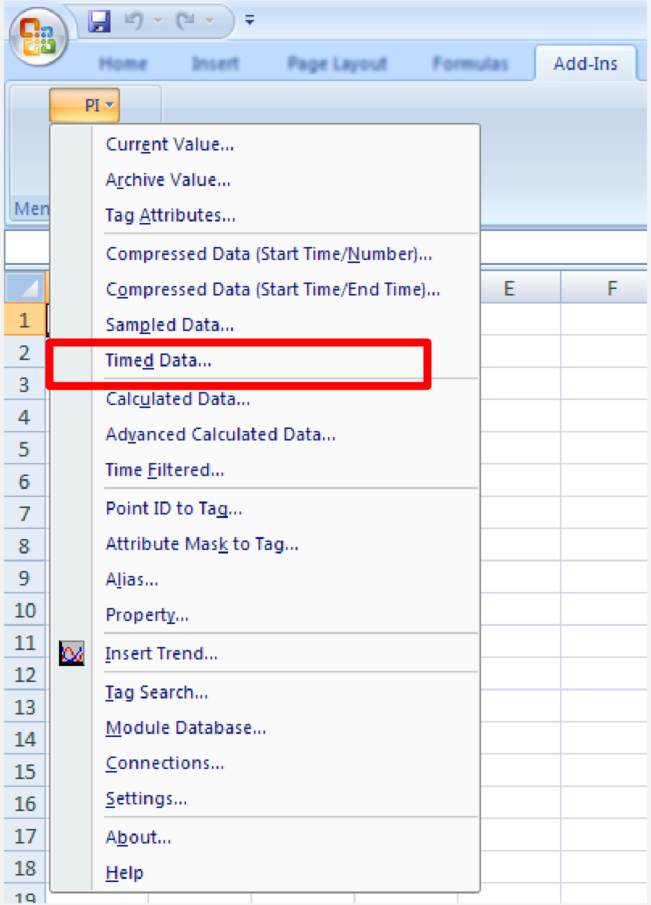
Time Range	
Start Time	
End Time	

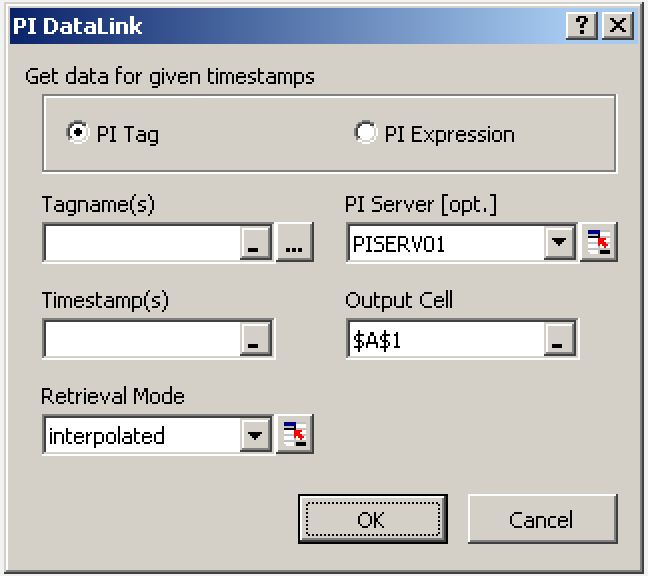
Number of Values	
Start Time	
Number of Values	

Tag	
Timestamp	Value

	Tag	
#	Timestamp	Value
1		
2		

TIMED DATA


Home
Insert
Page Layout
Formulas
Add-Ins
PI
Current Value...
Archive Value...
Tag Attributes...
Compressed Data (Start Time/Number)...
Compressed Data (Start Time/End Time)...
Sampled Data...
Timed Data...
Calculated Data...
Advanced Calculated Data...
Time Filtered...
Point ID to Tag...
Attribute Mask to Tag...
Alias...
Property...
Insert Trend...
Tag Search...
Module Database...
Connections...
Settings...
About...
Help


PI DataLink
Get data for given timestamps
PI Tag
PI Expression
Tagname(s)
PI Server [opt.]
PISERV01
Timestamp(s)
Output Cell
A1
Retrieval Mode
interpolated
OK
Cancel

EXERCISE 3.4 – DATA COMPARISON

- Build a Excel report with Timed Data Function

EX 4 : Data Comparison

For Reference Data

**** Use Timed Data function to get data based on Time References .*

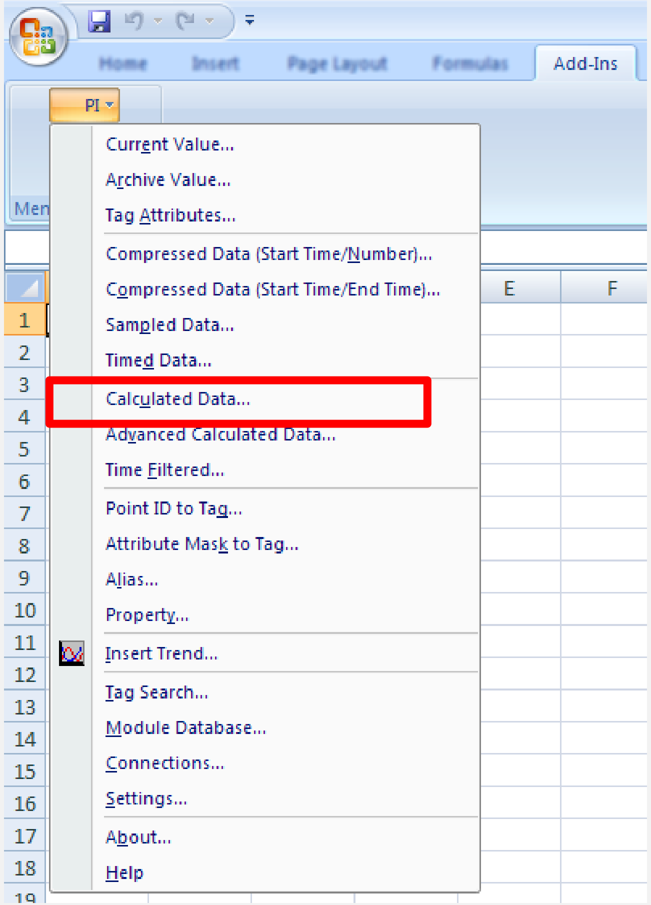
Time Range	
Start Time	
End Time	

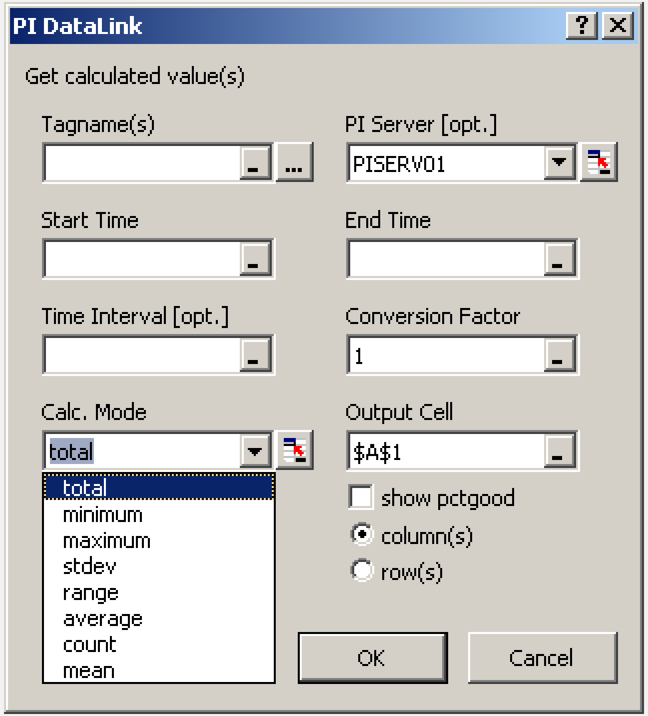
Tag	
Timestamp	Value

Value	Value

SAT Solutions

CALCULATED DATA





EXERCISE 3.5 – DAILY REPORT

- Build a Excel report with Calculated Data Function

EX 5 : Daily Report

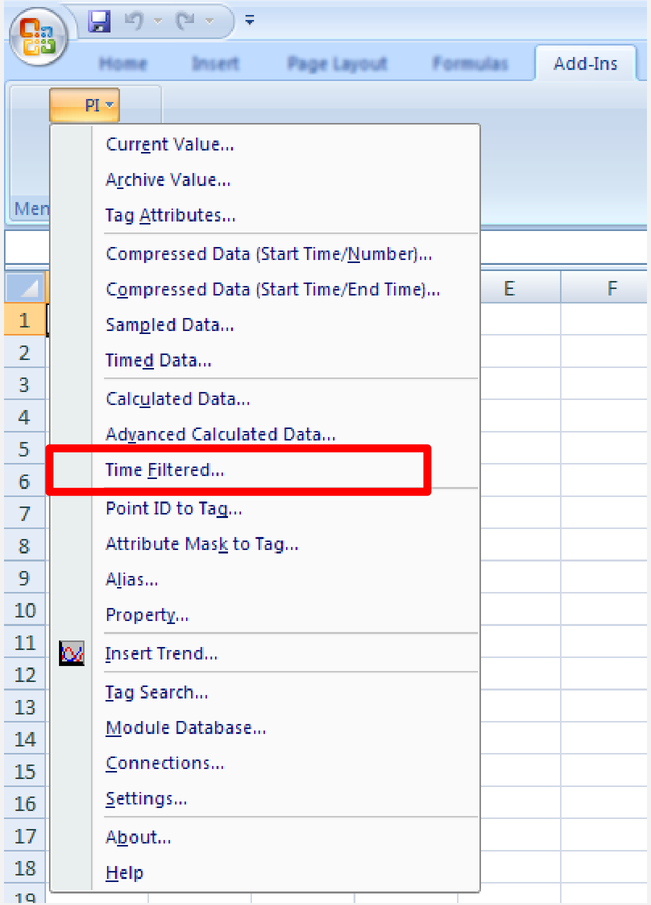
*** Use *Calculated Data* to calculate the *Total, Average, Minimum, Maximum, Standard Deviation, and Range* .

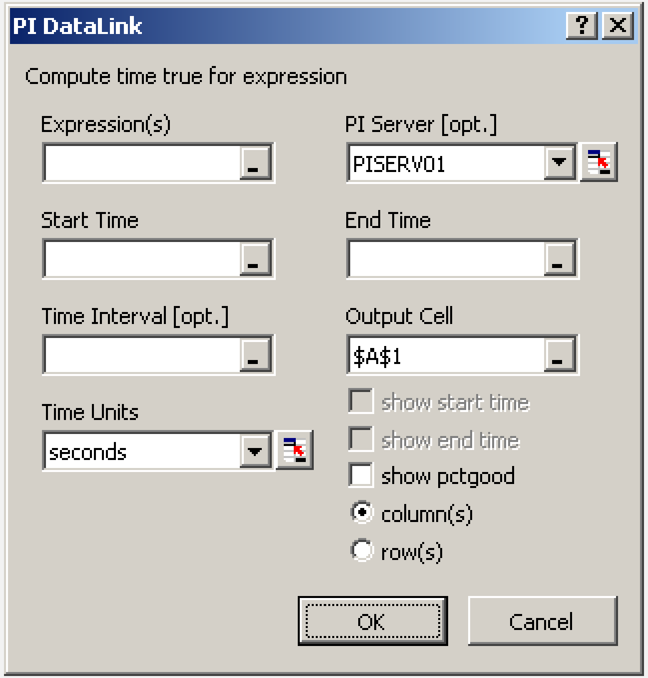
Start Time	
End Time	
Time Interval	

Tag					
Units					
Calc Mode					
Conversion Factor					
Start Time	End Time				

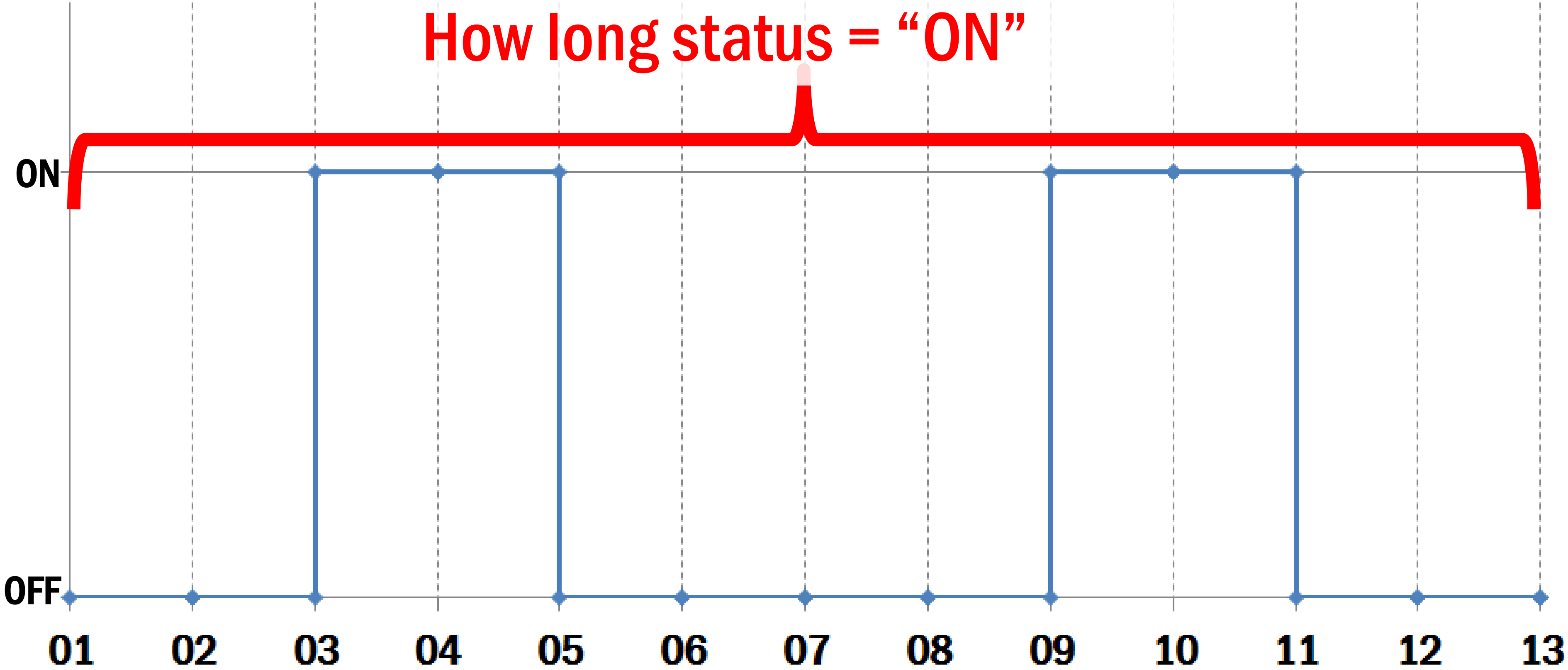
SAT Solutions

TIME FILTERED





TIME FILTERED



EXERCISE 3.6 – RUNNING HOURS

- Build a Excel report with Time Filtered Function

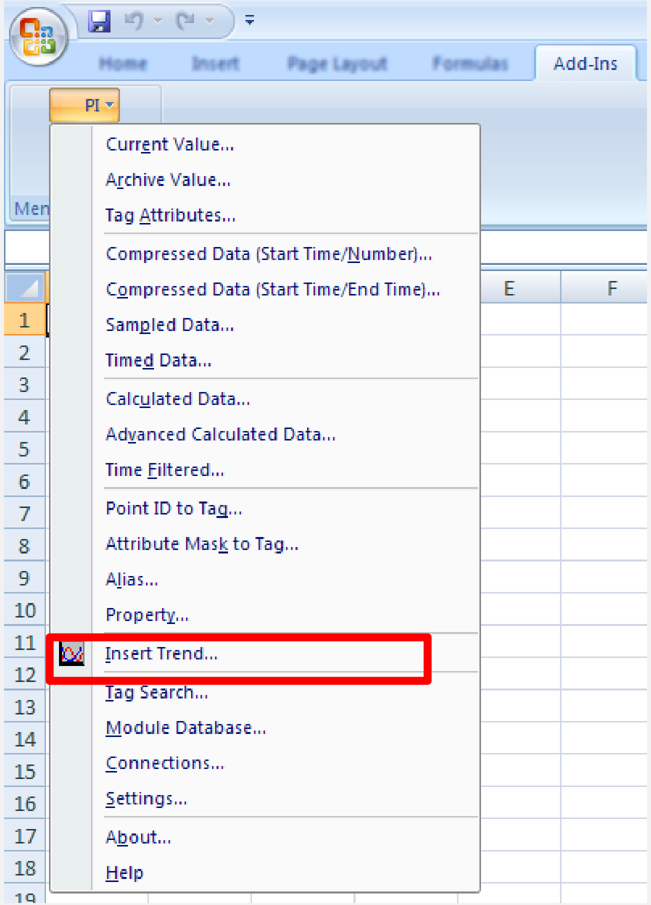
EX 6 : Running Hours

*** Use *Time Filtered* to get *The Amount of Time Where an Expression is True* .

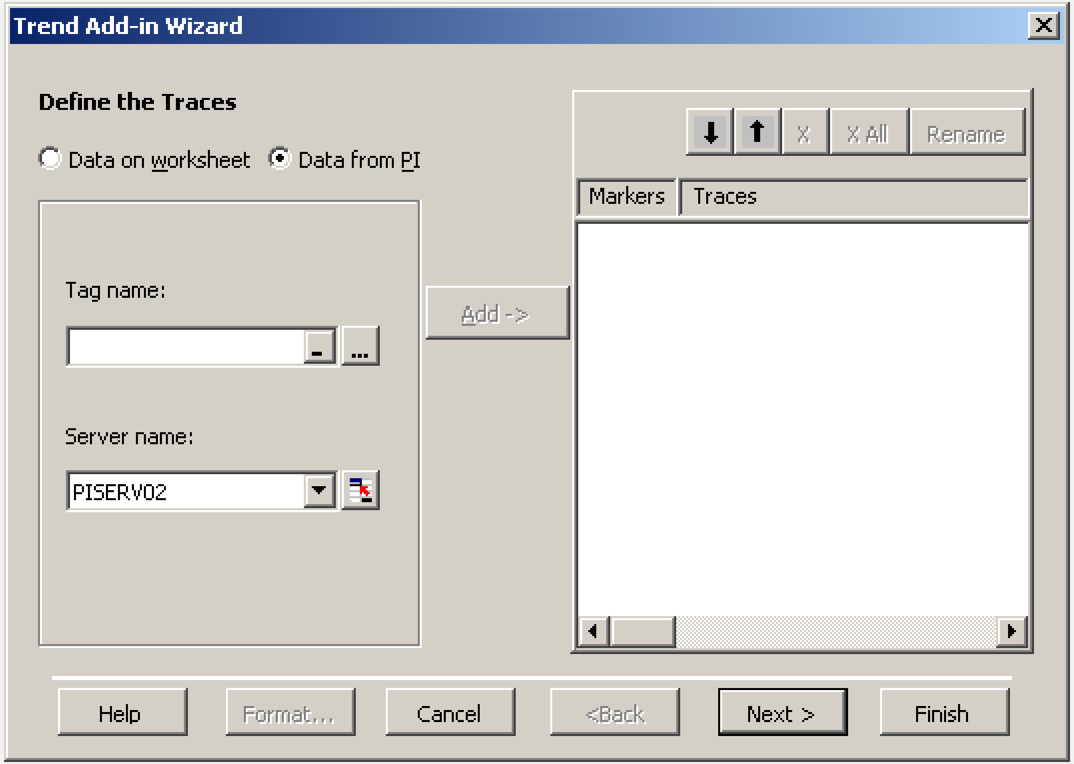
Start Time	
End Time	
Time Interval	

Expression				
Time Units				
Start Time	End Time			

TREND

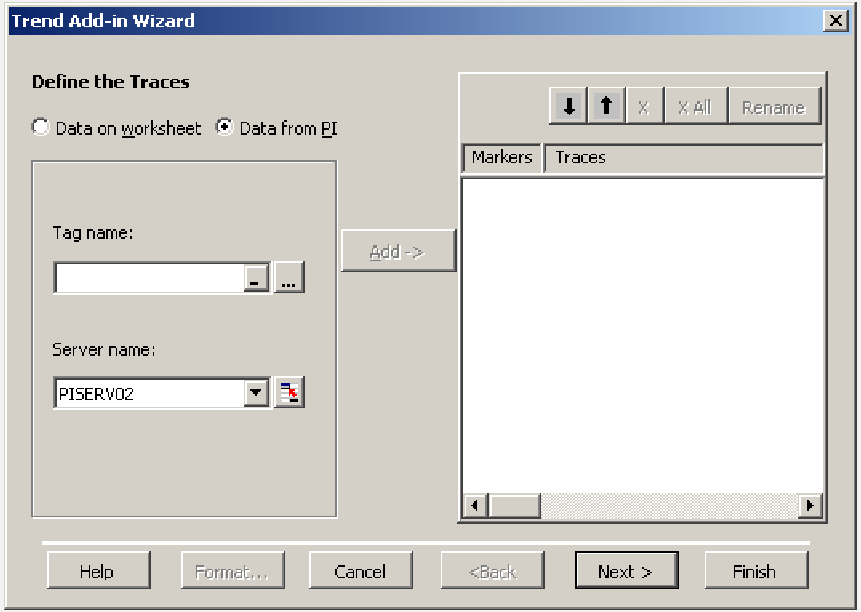


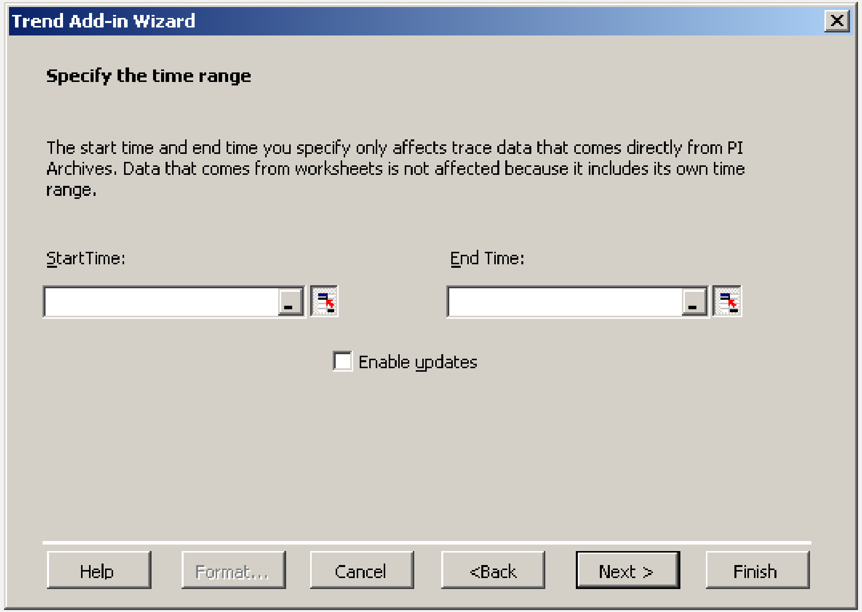
Home
Insert
Page Layout
Formulas
Add-Ins
PI
Current Value...
Archive Value...
Tag Attributes...
Compressed Data (Start Time/Number)...
Compressed Data (Start Time/End Time)...
Sampled Data...
Timed Data...
Calculated Data...
Advanced Calculated Data...
Time Filtered...
Point ID to Tag...
Attribute Mask to Tag...
Alias...
Property...
Insert Trend...
Tag Search...
Module Database...
Connections...
Settings...
About...
Help
E
F



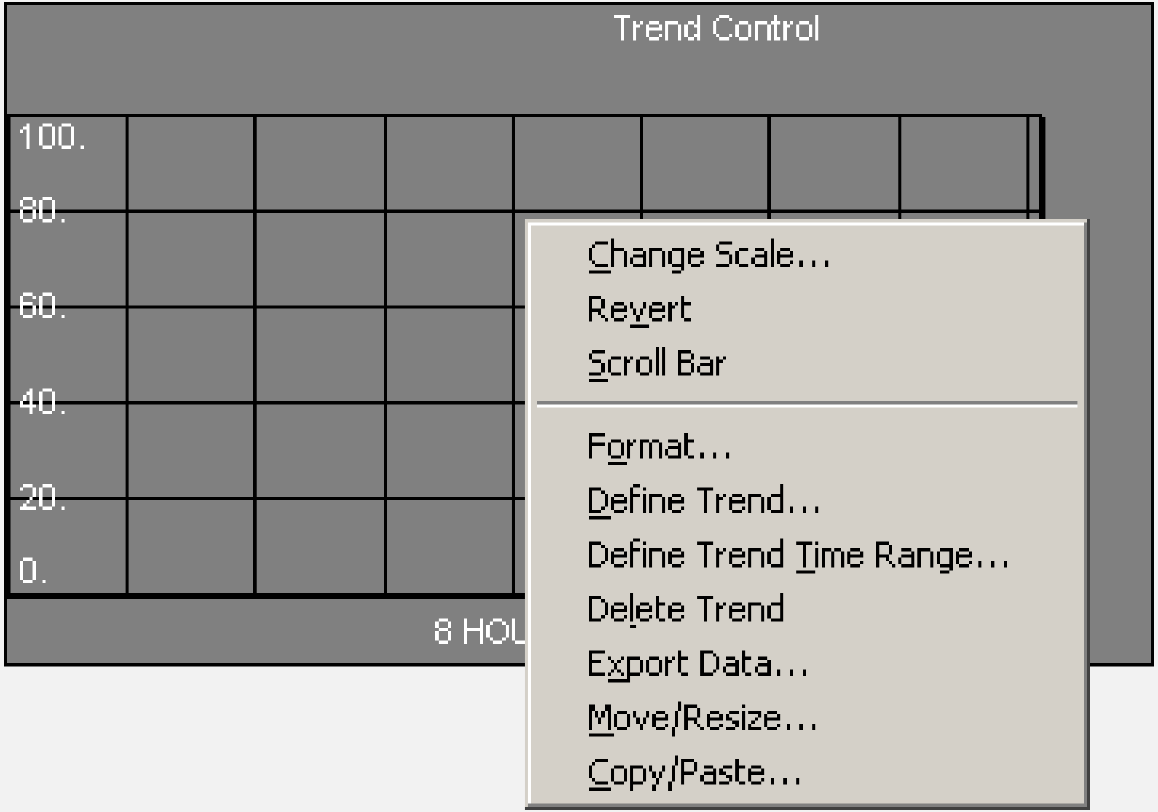
Trend Add-in Wizard
Define the Traces
Data on worksheet
Data from PI
Tag name:
Server name:
PISERV02
Add ->
X
X All
Rename
Markers
Traces
Help
Format...
Cancel
<Back
Next >
Finish

TAG & TIME SELECTION

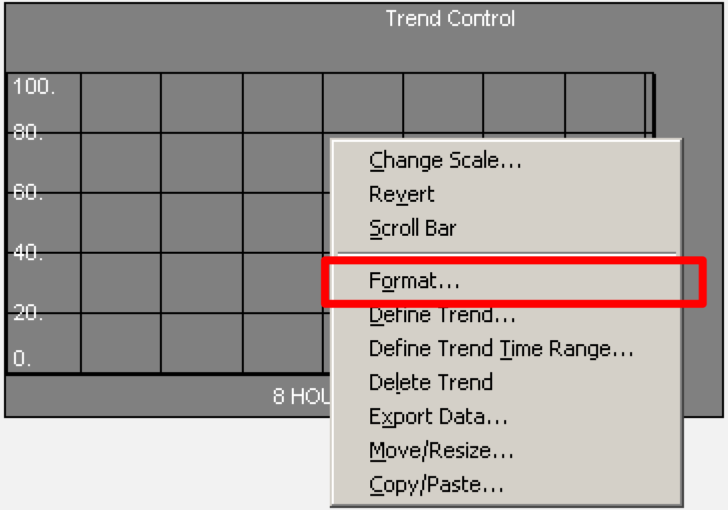


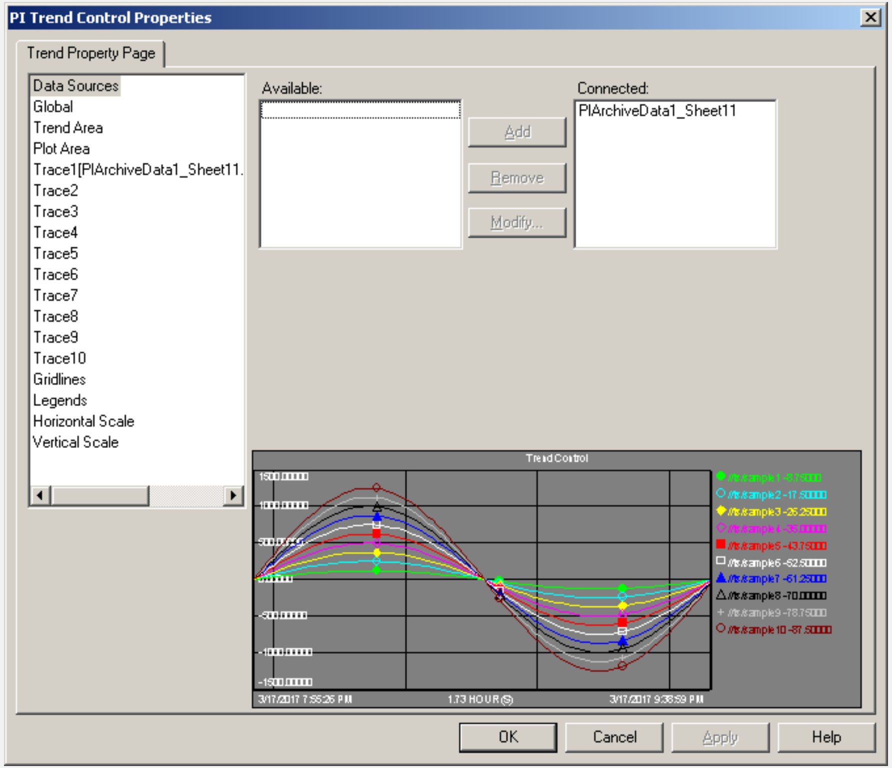


RIGHT CLICK OPTIONS



TREND FORMAT





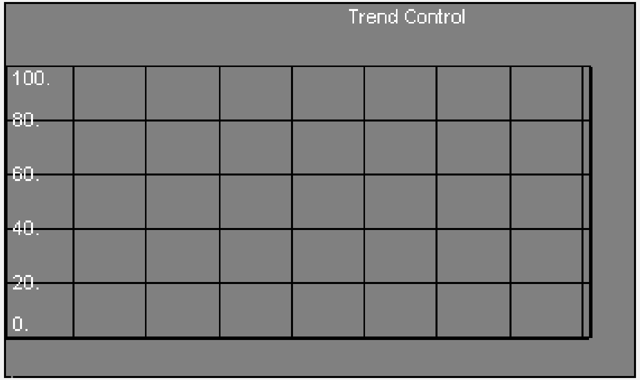
EXERCISE 3.7 - TREND

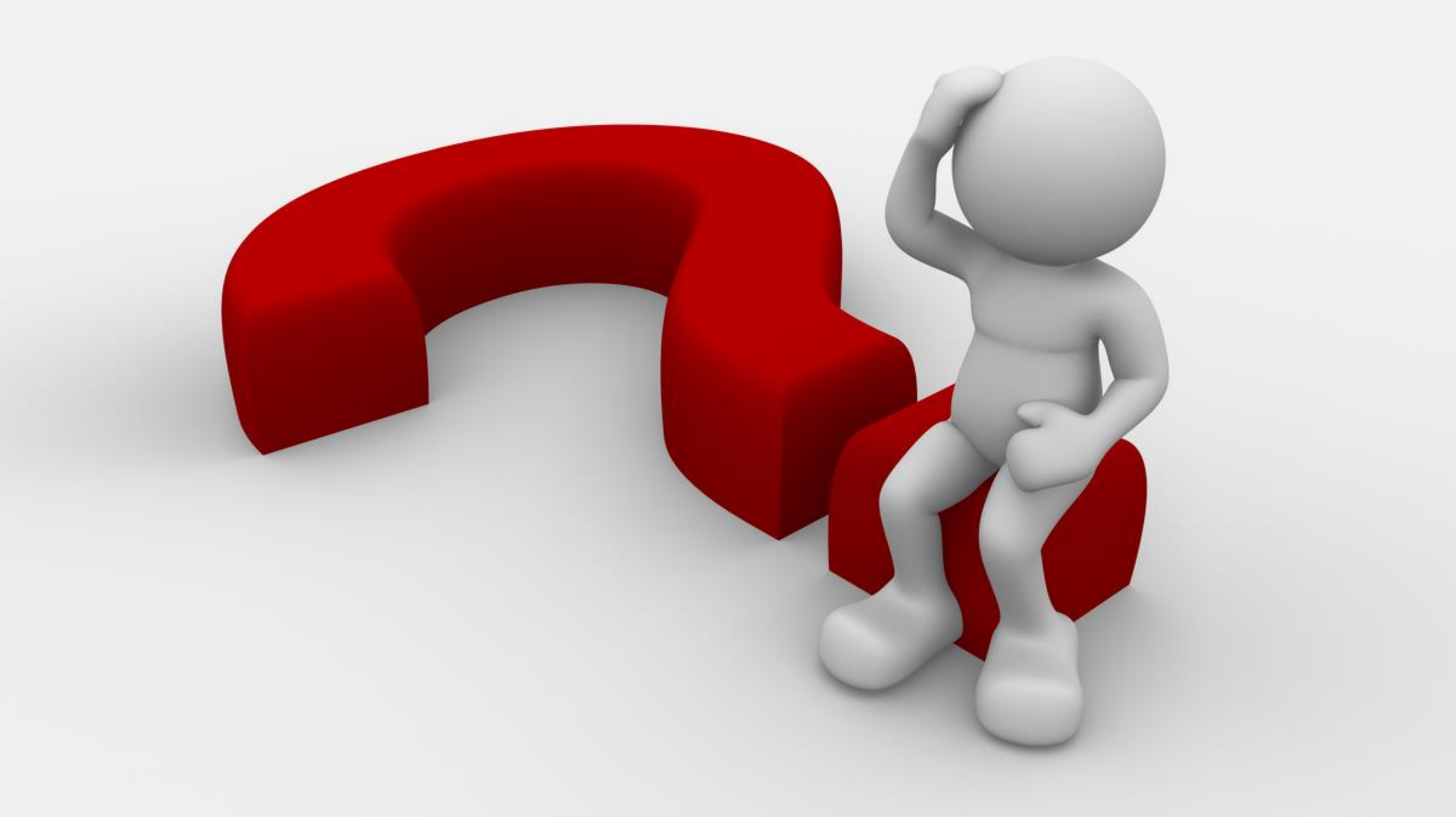
- Insert a **PI Trend** to an Excel Sheet

EX 7 : Trend

**** Use Trend to display data in Trend format.*

Start Time	
End Time	
Tag1	
Tag2	
Tag3	
Tag4	
Tag5	





thank you
danke
謝謝
ngiyabonga
teşekkür ederim
gracias
merci
obrigado
bedankt
dziękuję
sukriya
kop khun krap
go raibh maith agat
mochchakkeram
arigatô
dakujem
grazie
terima kasih
감사합니다
ありがとう
спасибо
рахмат
Баярлалаа
mersi
kia ora
barka
welalin
tack
spas
dank je
misaotra
paldies
grazzi
mahalo
tapadh leat
vinaka
blagodaram
спасибі
matondo
faafetai lava
kiitos
dankie
nanni
nandri
dhanyavad
mauruuru
hvala
gràçie
bayarlalaa
köszönöm
enkosi
akun
dankon
aċiū
хвала
asante
manana
obrigada
tenki
murakoze
djiere dieuf
tau
дякую
mamnun
chokrane
chnorakaloutioun
gratias ago
gràcies
sulpáy
taiku
sobodi
děkuji
sagolun
mèsi
didi madloba
najis tuke
kam sah hamnida
rahmat
tanemirt
rahmet
ευχαριστώ
diolch
dhanyavadagalu
takk
shukriya
mercé
мерси
trugarez
xièxie
তোমাকে ধন্যবাদ